Board guide

How the chairman, the treasurer and the electrician work in the system, and the rules it calculates by: tariffs, readings, charges, payments, losses, water — with worked examples.

Reader	Chairman, treasurer, electrician, other board members
System state	Phases 1–4 (payments, documents, delivery by e-mail and Telegram): part I — working in the system, part II — how the system calculates
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1 The system, roles and signing in

Nadel keeps the cooperative's records: who the members are, which plots they own, what is charged and at which tariffs, how much electricity and water was used. Payments are entered at the cash desk and allocated to charges; invoices, certificates and receipts are printed as PDF; the map comes next.

Who sees what

Everyone signs in with their own account. What a person sees and may do depends on their **roles**, and roles are made of **permissions** (chapter 3). The cooperative has three base roles:

Role	May do
Chairman	Everything: members, roles, plots, owners, tariffs, meters, readings, charges, payments, import, settings, audit log. The chairman's rights cannot be reduced, and a cooperative always keeps at least one chairman.
Treasurer	Views members, plots, tariffs, meters, charges and the audit log; enters payments, splits them over plots, reallocates them and allocates prepayment. Cannot reverse a payment - the chairman does that.
Member	Own data only: own plots, their tariffs, charges and payments, own meters, submitting readings.

The chairman can create custom roles. For example the role "Electrician" has only two permissions, "View meters" and "Enter readings"; a person with it sees only "Readings" and "Meters" in the menu.

Signing in

1. Open the system's address and press "Sign in".
2. Enter your email and password.
3. The system finds you among the members by email. If your member card has no email or a different one, you see "Account not linked": ask the chairman to correct the email on your card.

NOTE

For now the sign-in account (email and password) is created by the system administrator. Self sign-up and email invitations come later.

Language and theme

At the top right: the language switch **УКР / ENG** and the theme **System / Light / Dark**. The choice is remembered in this browser. The interface always shows exactly one language: with English selected, every name, hint and message is English.

At the top left is your cooperative's name, below it the menu. On a phone the menu hides behind the "Menu" button.

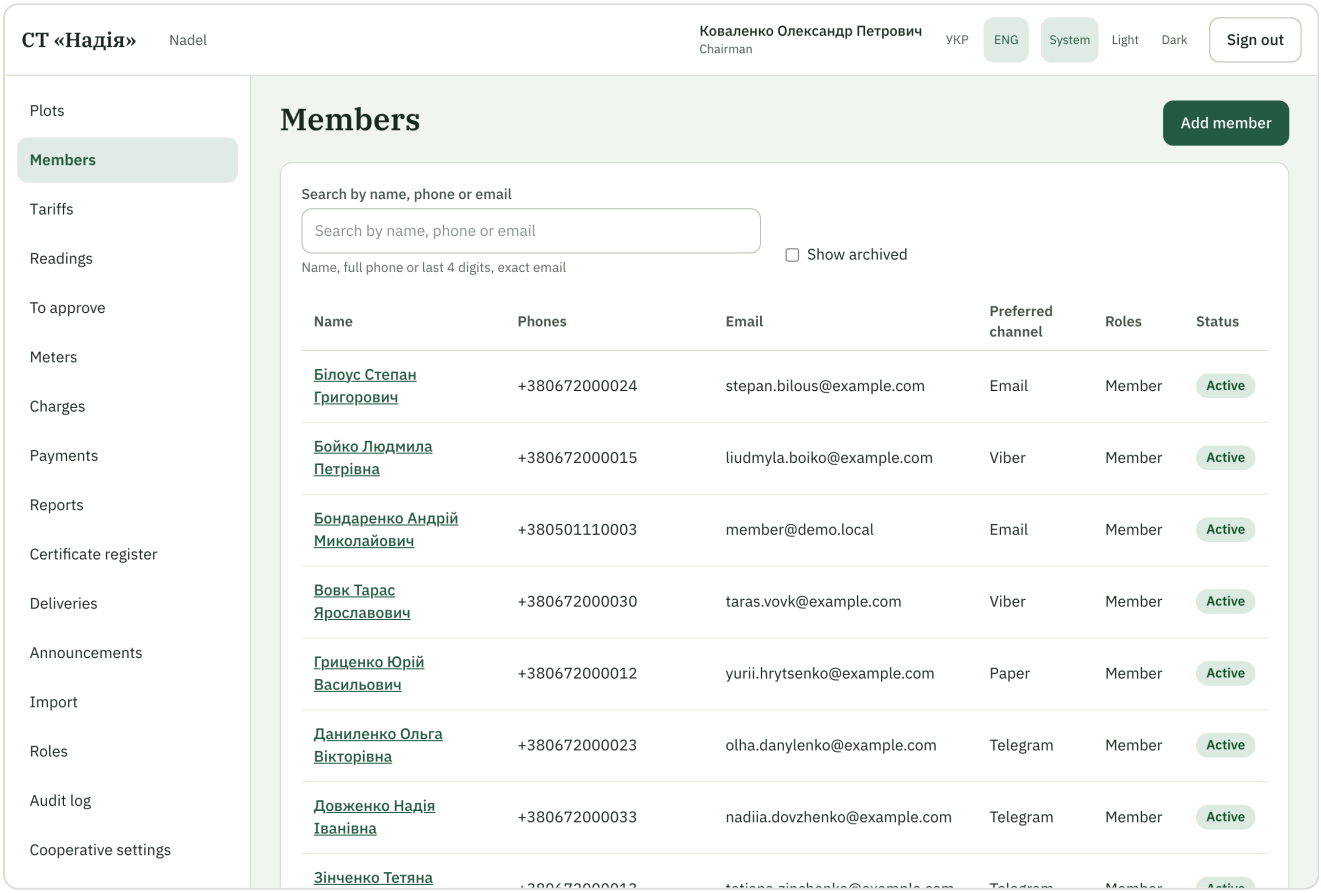
2 Members

"Members" lists every member. The search field says what it understands.

Searching

- By **name**: any part of the last or first name, in any letter case.
- By **phone**: the whole number written in any usual way (`+380 50 123 45 67`, `0501234567`, `380501234567`), or exactly its **last 4 digits**.
- By **email**: the whole address, in any letter case.

A part of an email (for example only what comes before the @) or digits from the middle of a phone find nothing. If a member is not found, try the surname or the last 4 digits of the phone.



The members list

Adding a member

1. Press "Add member".
2. Fill in the last and first name (required), patronymic, phones, email, postal address.
3. Choose how invoices are delivered: email, Telegram, Viber or paper. The matching contact is required.

4. Press "Save". A phone typed as `0XXXXXXXXX` is stored as `+380XXXXXXXXX`.

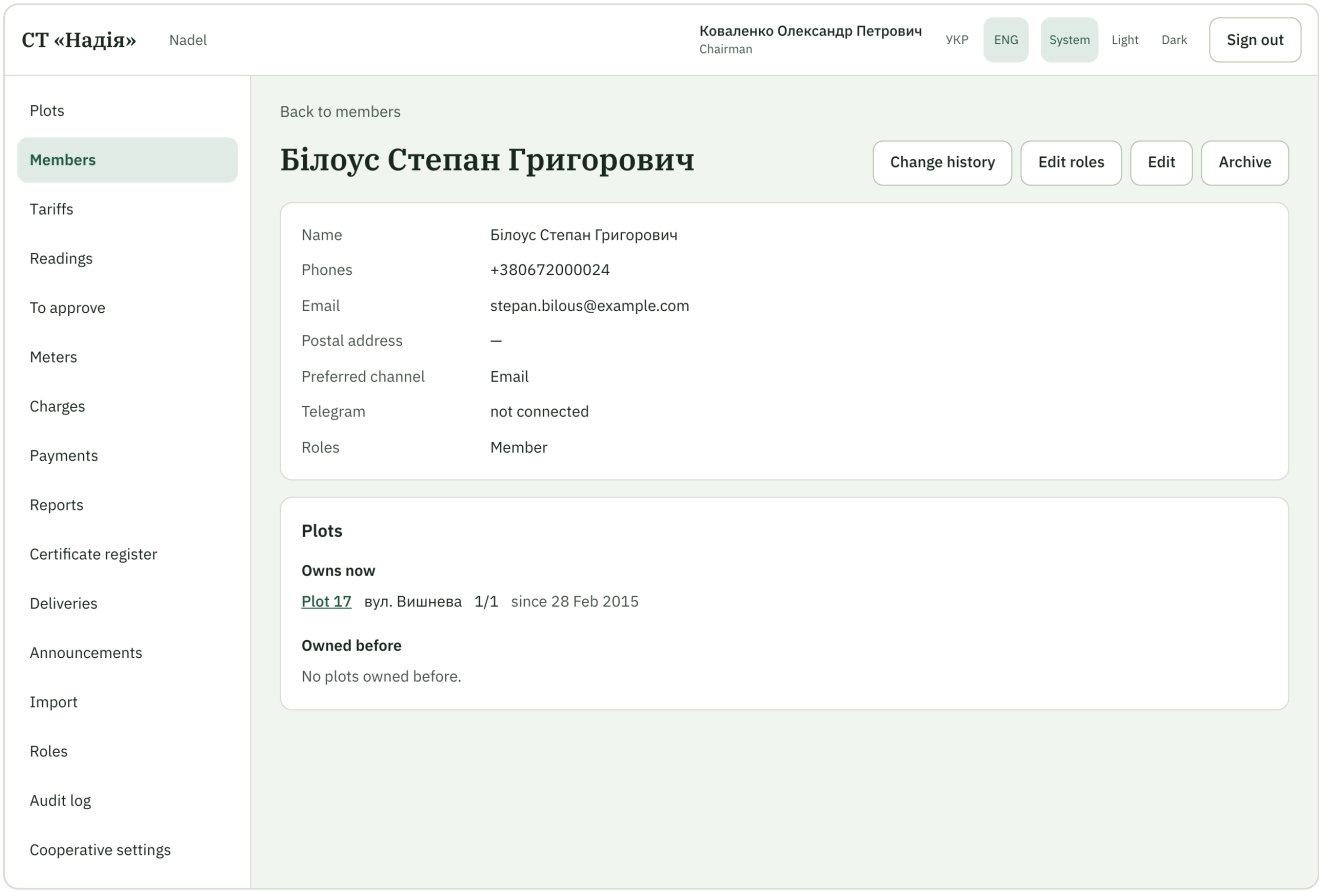
NOTES AND PERSONAL DATA

Notes are for working information about a member: how best to reach them, arrangements, relatives' contacts. Only people with "View members" see them. The system has no fields for tax numbers (РНОКПП), passport data or dates of birth — the cooperative does not need them for its records.

Phones, email, postal address, Telegram, Viber and notes are stored encrypted, so a copy of the database does not reveal them. The system decrypts them for people who may see them, so nothing changes in daily work, except the search rules above. Names stay readable.

The member card

Contacts, roles, owned plots and notes. Notes are visible only to people with "View members".



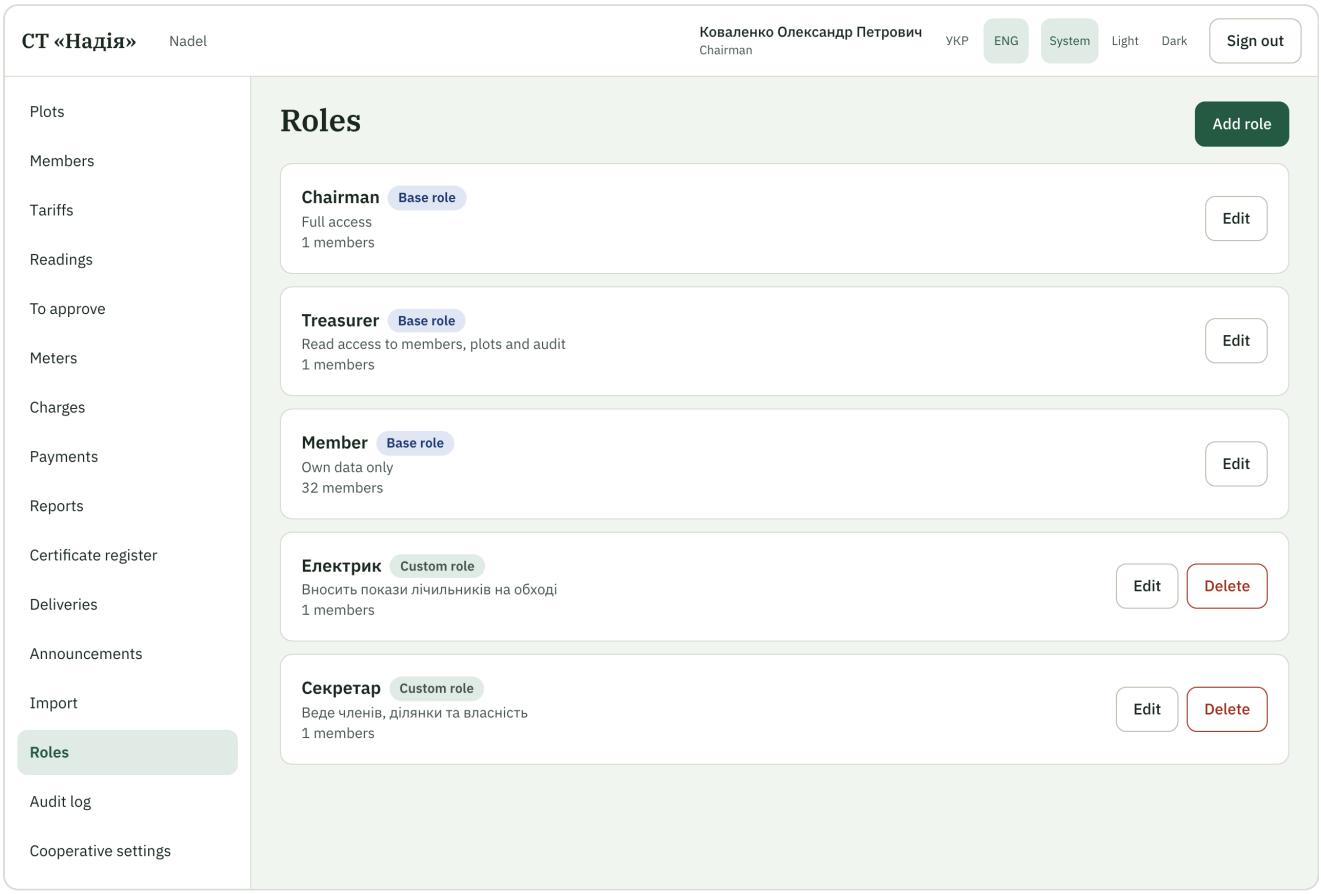
A member card

Archiving

A member is archived, not deleted: hidden from the list (visible with "Show archived"), unable to sign in, history kept. "Restore" brings them back. A current plot owner cannot be archived until the plot changes hands, nor the last chairman until another is appointed.

3 Roles and permissions

In "Roles" the chairman (or anyone with "Manage roles") decides who may do what.



Base and custom roles

- **Base roles** (Chairman, Treasurer, Member) cannot be deleted but can be renamed; the new name and description then show in every language. "Restore standard name" undoes it.
- **Custom roles**: "Add role", a name, a description and any set of permissions. A person may hold several roles; permissions add up.
- A role still assigned to someone cannot be deleted.

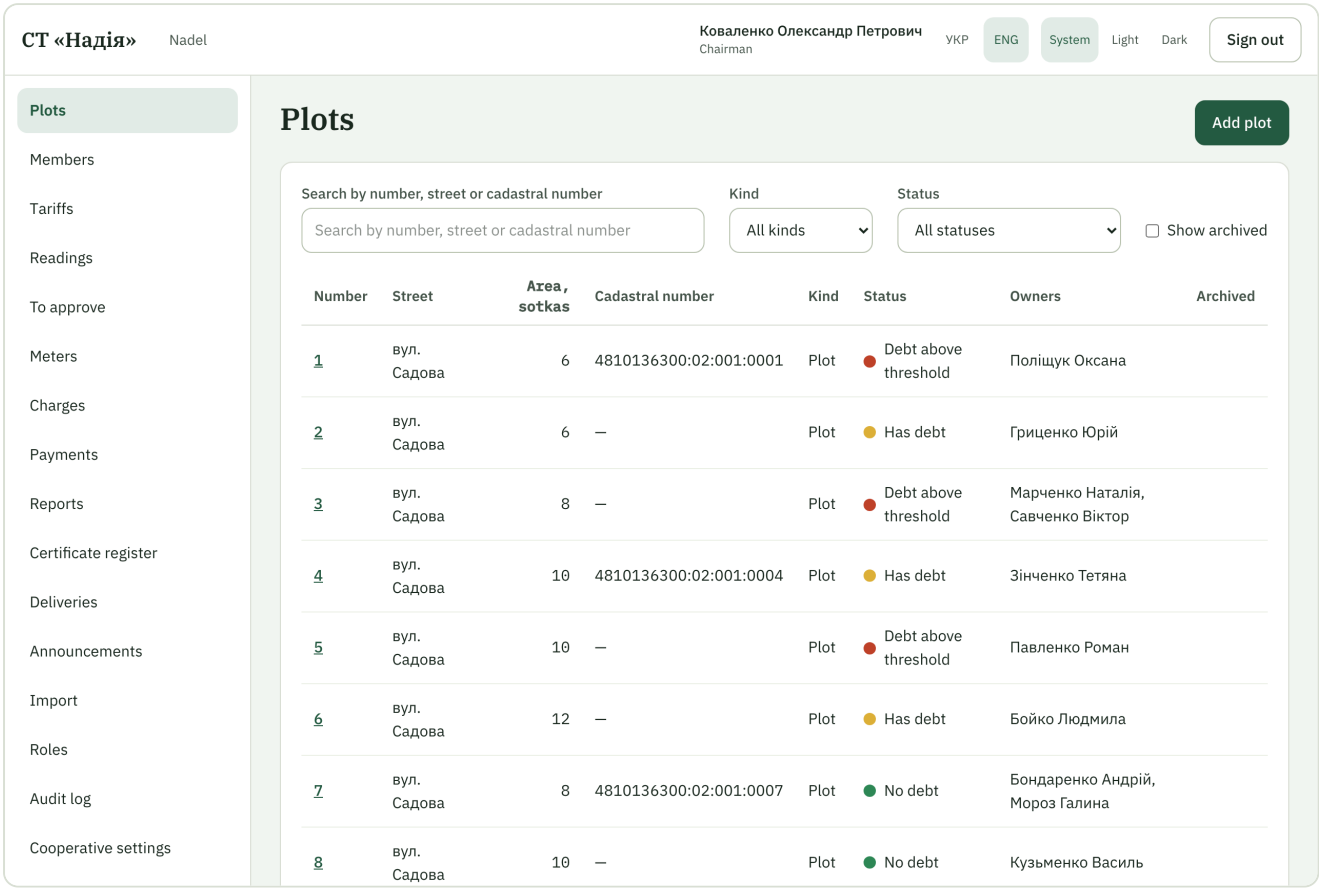
Permissions

Permission	Allows
View / edit members	See all members and contacts; add, change, archive
Manage roles	Create roles, change permissions, assign roles
View / edit plots	See every plot in detail; add, change, archive
Edit ownership	Add, change and correct owners

Permission	Allows
Run import	Upload data from Excel
Manage settings	Change the cooperative settings
View / manage tariffs	See tariffs; create charge types, tariffs and versions
View / manage meters	See the register and the tree; register, replace, remove meters
Enter readings	Enter, correct and delete readings (the electrician's round)
Approve readings	Approve or reject readings submitted by members
View audit log	Read the audit log
View reports	Open and download reports, generate files from templates (the "Debtors" report also needs "View members")
Manage report templates	Upload, replace, download and archive report templates
View payments	See payments, their allocation and the "Payments" report
Enter payments	Enter cash and bank payments, split them over plots, reallocate, allocate prepayment
Reverse payments	Reverse a payment with a reason (the chairman only by default)

4 Plots

"Plots" is the register of every plot, with search by number, street or cadastral number and filters by kind and status.



The plot register

What is kept about a plot

Number — Required, unique in the cooperative: `12`, `12a`, `R-1`. Natural order: 2 before 10, 12a after 12.

Kind — **Plot** or **Common land** (roads, shared areas). Common land has no owners, tariffs or plot meters.

Area — In sotkas or m², entered in either unit (1 sotka = 100 m²).

Cadastral number — Optional, format `NNNNNNNNNN:NN:NNN:NNNN`.

Building — Whether a building stands on the plot. Reference only; it never affects charges.

Notes — Visible only with "View plots".

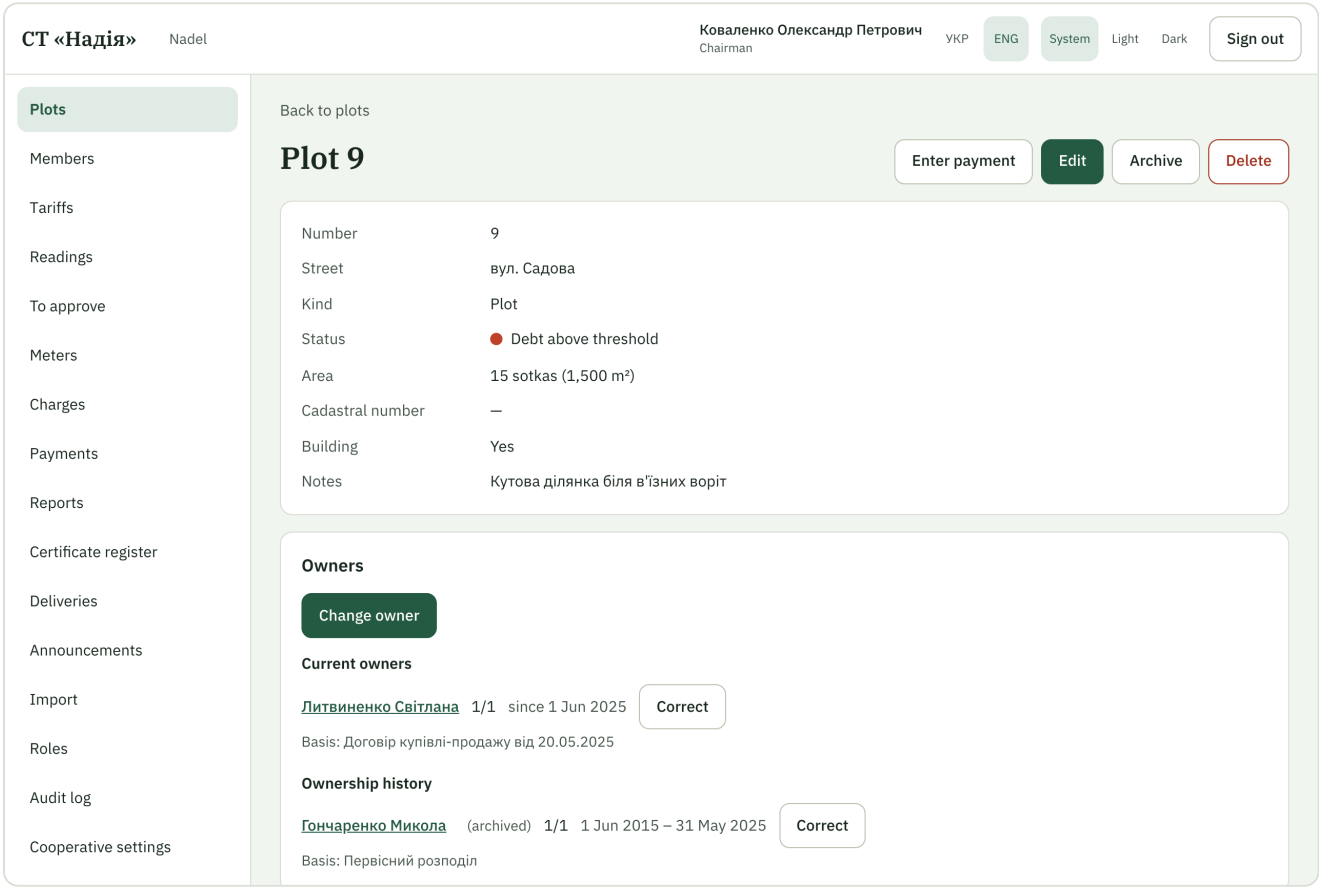
Plot status (colour)

Colour	Meaning
Green	No debt (also when there is a prepayment)
Yellow	Has a debt
Red	Debt over the threshold (amount or age set by the chairman)
Grey	A plot without an owner, or common land

The colour follows the debt after payments: a charge that has been paid is not a debt (chapters 20 and 25).

The plot card

All the plot's data, owners with history, tariffs for a chosen month, meters.



Plot 9: owners, tariffs, meters

"Archive" hides a plot from the list and keeps its history. Only a plot created by mistake that never had an owner can be deleted. A plot with a lasting water connection (or an active meter) cannot be archived or turned into common land: the system refuses and asks to end the connection on the plot card first.

5 Plot owners

Owners are kept on the plot card. A plot can have several co-owners with shares (`1/2`, `1/3`); the shares always add up to one. One member may own several plots, each with its own account.

Change of owner (sale, inheritance)

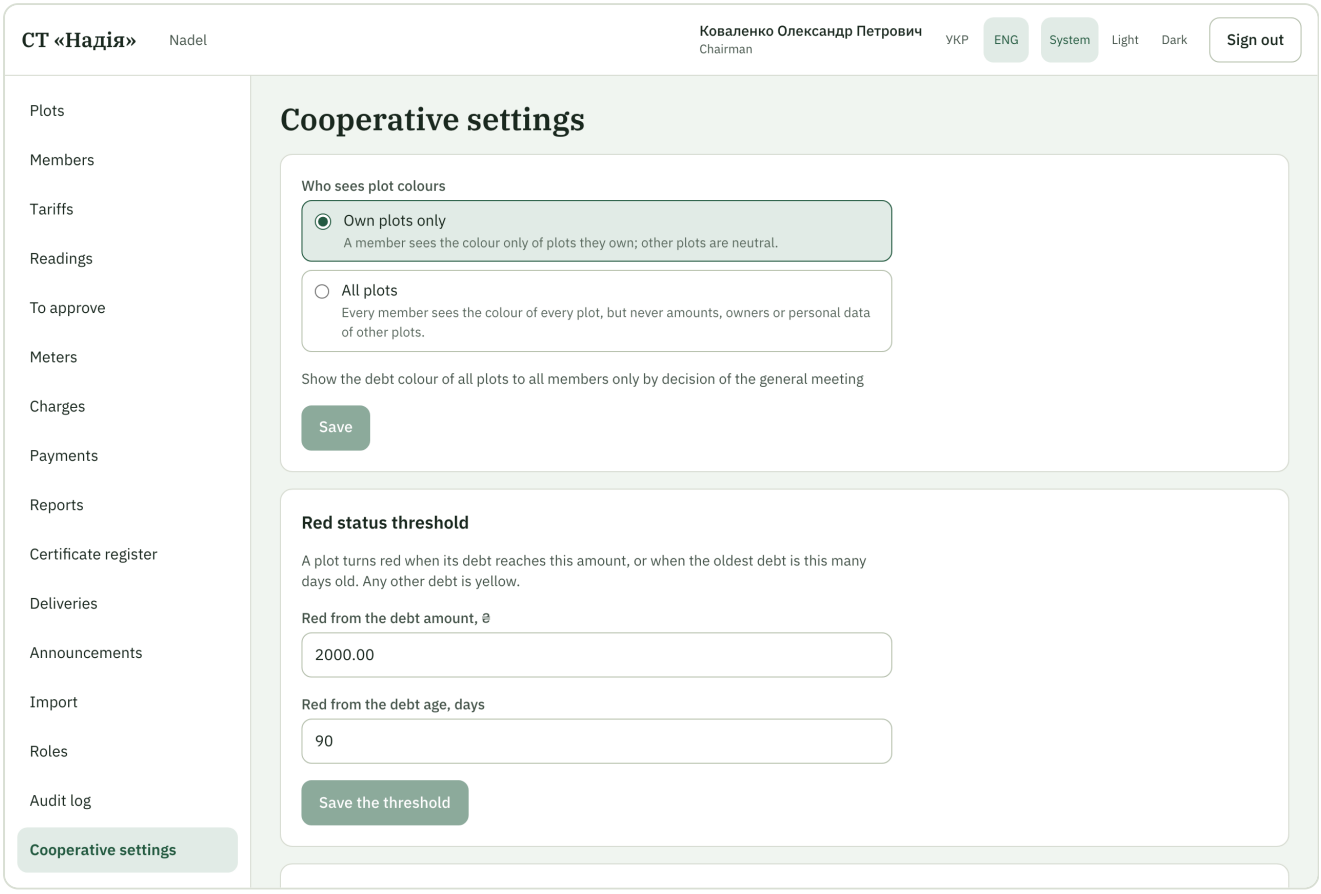
1. On the plot card press "Change owner".
2. Enter the date and the new owners with shares. If a new owner is not a member yet, use "Add member" right in the form.
3. Enter the basis (purchase contract, inheritance, court decision) and save.

The previous owners stay in the history up to the day before. **A debt belongs to the plot, not the person:** after a sale it stays on the plot.

Correcting a mistake

A wrong share, date or person in a record is fixed with "Correct" next to the record. The old values stay in the audit log. Ownership records are never deleted.

6 Cooperative settings



Cooperative settings

Who sees plot colours

- **Own plots only** (default): a member sees the colour of their own plots only; other plots are neutral.
- **All plots**: everyone sees every plot's colour, but never the amounts, owners or personal data of other plots.

> **WHY "OWN PLOTS ONLY" IS THE DEFAULT**
>
> Neighbours know whose plot is whose, so the colour effectively names the debtor. Switch to "All plots" only after a general meeting decision. The change is written to the audit log.

Prepayment

The **"Allocate prepayment automatically"** switch (on by default; changed by whoever has "Manage settings", read-only for the others) decides whether a plot's prepayment goes to new charges by itself. See chapter 13.

Meter tree

The meter tree is opened from here too — see chapter 9.

Charging settings

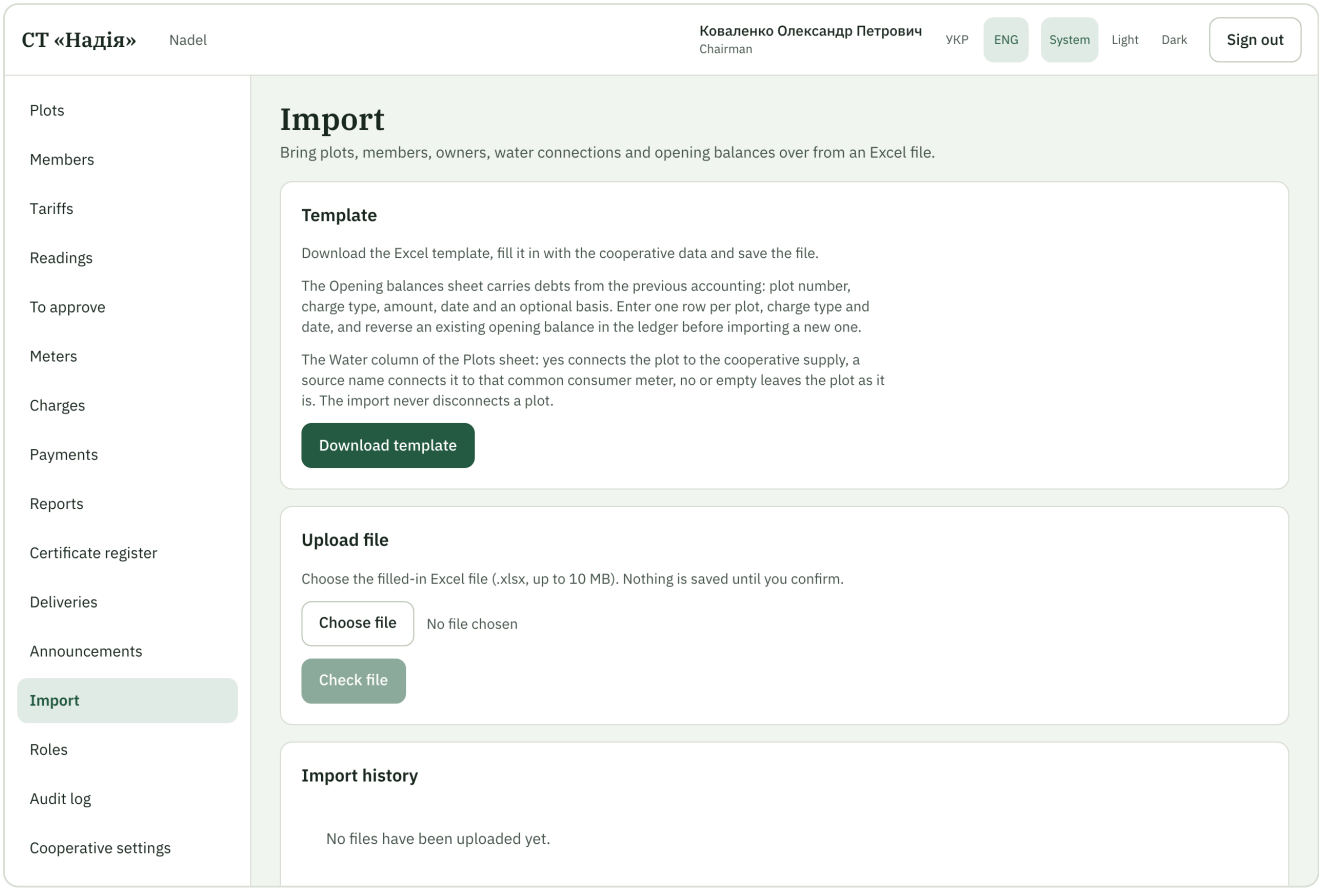
Below the colour choice: the red status threshold, how losses are charged, the split of common consumers and the water supply of plots. They are described in chapters 20–22.

Requisites

The "Requisites" block: official name, legal address, bank, IBAN; the EDRPOU code is read-only. Report templates use them — see chapter 24.

7 Excel import

Instead of typing the cooperative in by hand, plots, members, owners and meters can be loaded from one Excel file.



The import page

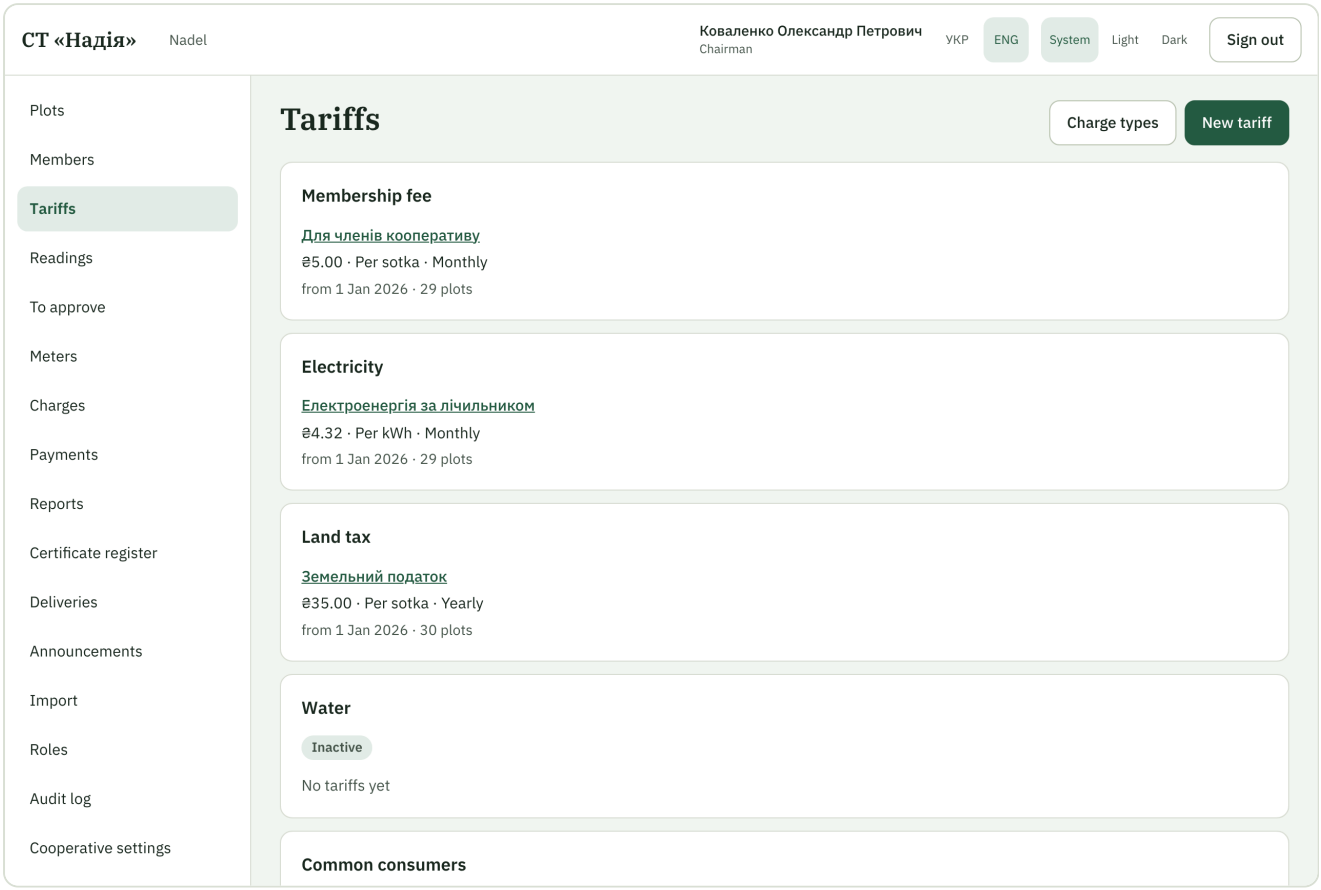
1. "Download template" gives a file in the interface language with the sheets Plots, Members, Ownership and Meters, each with a hint and an example row.
2. Fill in the sheets. Dates as Excel dates or `DD.MM.YYYY`, area in sotkas or m², shares as `1/2`.
3. Upload the file. The system checks everything **without saving** and shows how many records will be added, updated or left unchanged, and every problem with its sheet, row and column.
4. With no problems, confirm. Everything is saved together: the whole file or nothing.

- Importing the same file again duplicates nothing: plots are matched by number, members by email (or name and phone).
- An import cannot change an owner; use "Change owner" (chapter 5).
- Limits: `.xlsx` up to 10 MB, up to 2,000 plots and 3,000 members. A check is valid for 30 minutes.
- Sheet and column names are accepted in Ukrainian and English, even mixed.

Two additions: the **Opening balances** sheet (plot, charge type, amount, date, basis; one row per plot and charge type; amounts with at most two decimals) and the **Water** column of the Plots sheet ("yes" connects to the cooperative supply, any other text names a well; "no" or empty changes nothing). Importing the same file again books nothing twice.

8 Tariffs

Here the chairman describes what the cooperative charges for and how much. The charges themselves come later (chapter 20), but every plot card already shows what the plot will be charged.



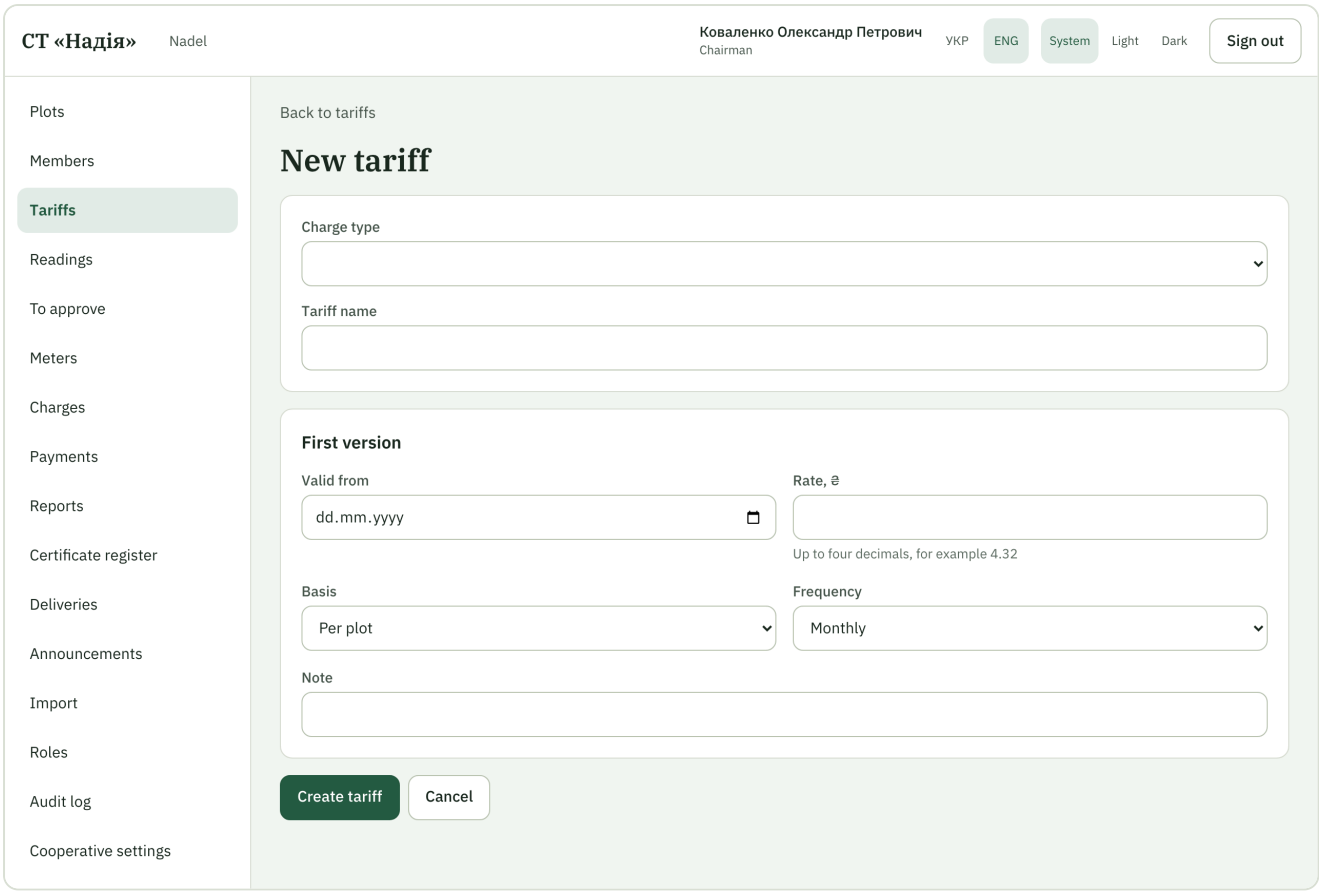
Tariffs by charge type

Charge types

The "Charge types" button. The cooperative starts with membership fee, electricity, land tax and water (inactive until there are water meters). Add your own, such as "Road repair", rename, reorder, archive.

A new tariff

1. "New tariff" → choose the charge type and name it, for example "For members".
2. Fill in the first version: from which date it applies, the basis, the frequency and the rate.
3. Save, then on the tariff card choose the plots it applies to.



The new tariff form

Basis	Example
Per plot	300 ₴ per plot per quarter
Per sotka	Membership fee 50 ₴ per sotka per year
Per kWh	Electricity 4.32 ₴ per kWh, by meter readings
Per m³	Water, by water meter readings
Fixed one-time amount	Special-purpose fee 1,200 ₴ with a due date

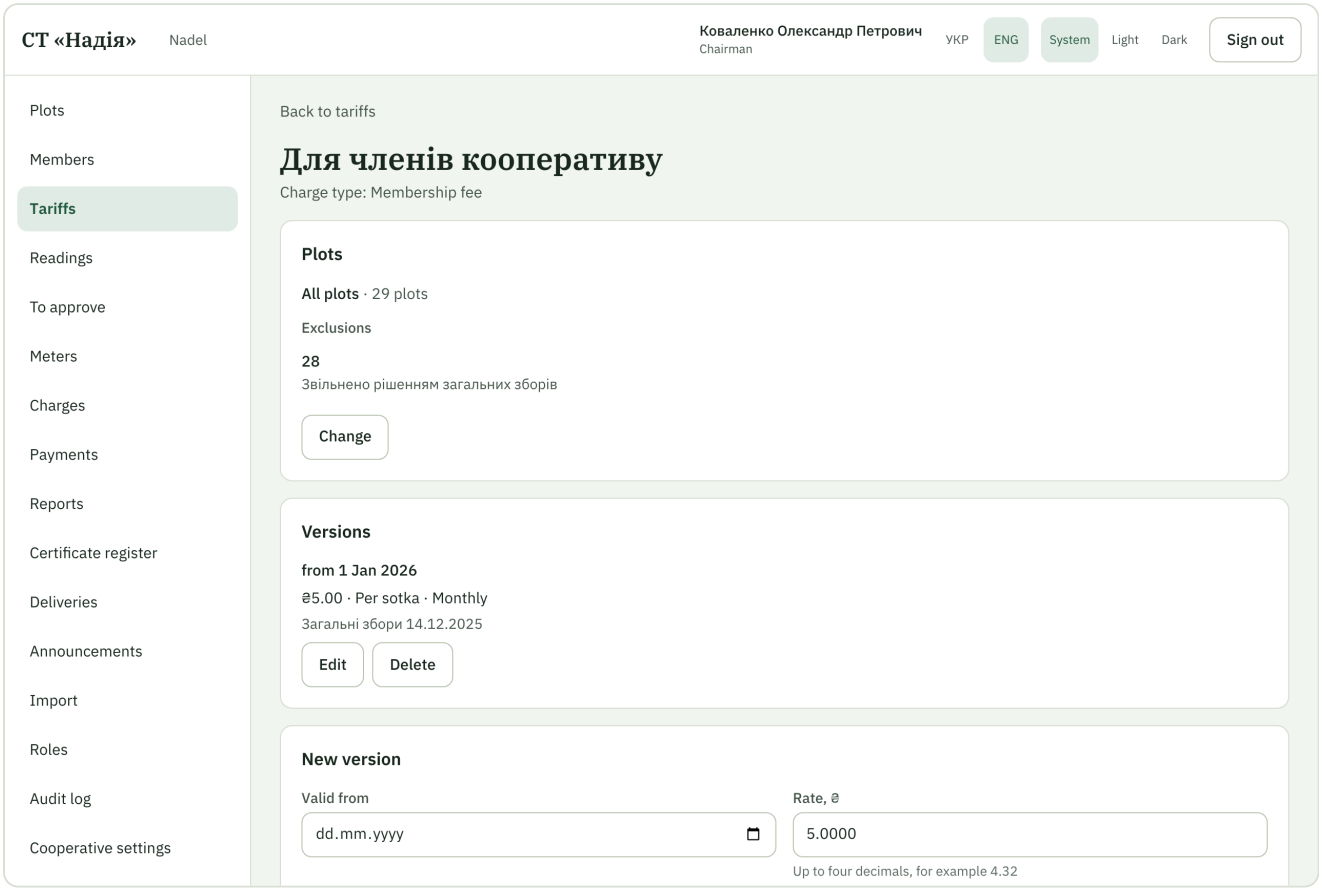
Frequency: monthly, quarterly, yearly or one-off. The cooperative bills monthly, so a yearly tariff is divided by 12 and a quarterly one by 3.

> **EXAMPLE**
>
> Membership fee 50 ₴ per sotka per year, a 6-sotka plot: 50 × 6 / 12 = **25.00 ₴ per month**.

A rate may have up to four decimals (for example 1.4532 ₴ per m² for land tax); amounts are rounded to kopecks only at the end. A rate is shown without trailing zeros: 50.00 ₴, not 50.0000 ₴.

Tariff versions

A tariff is changed not by editing but with a **new version** and the date it starts ("New version"). The version history is on the tariff card. A version starting in the past shows a warning that past charges will be recalculated.



A tariff card: versions and plots

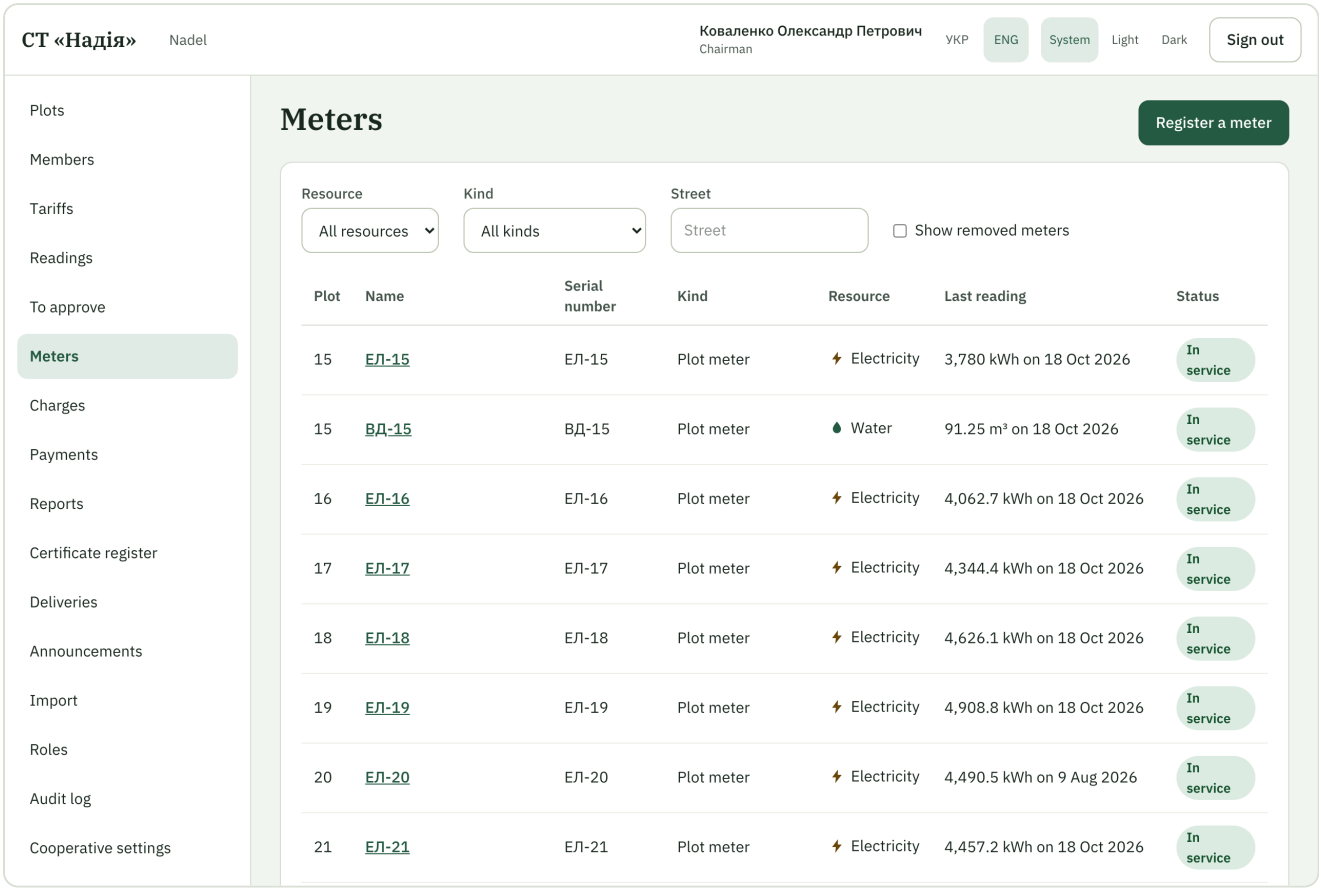
A mistake in a version can be fixed ("Edit") or the version deleted **while no charge uses it**. The only version of a tariff cannot be deleted; archive the tariff instead.

Which plots it applies to

- **All plots** (never common land), with **exclusions** if needed, each with a mandatory reason such as "exempted by the general meeting".
- **Selected plots**: an explicit list.
- Per-kWh and per-m³ tariffs apply only to plots with a meter of that resource. A plot without one shows "no meter".

9 Meters and the meter tree

"Meters" is the register of every meter in the cooperative. Each one has a resource icon: a **lightning bolt** for electricity, a **drop** for water.



The meter register

Kinds of meters

Plot meter	Belongs to a plot. A plot has at most one active meter per resource.
Substation	The meter at the cooperative's transformer: the root of the tree.
Quarter meter	The meter of a group of plots (a street or a block).
Common consumer	A well pump, street lighting, the office.

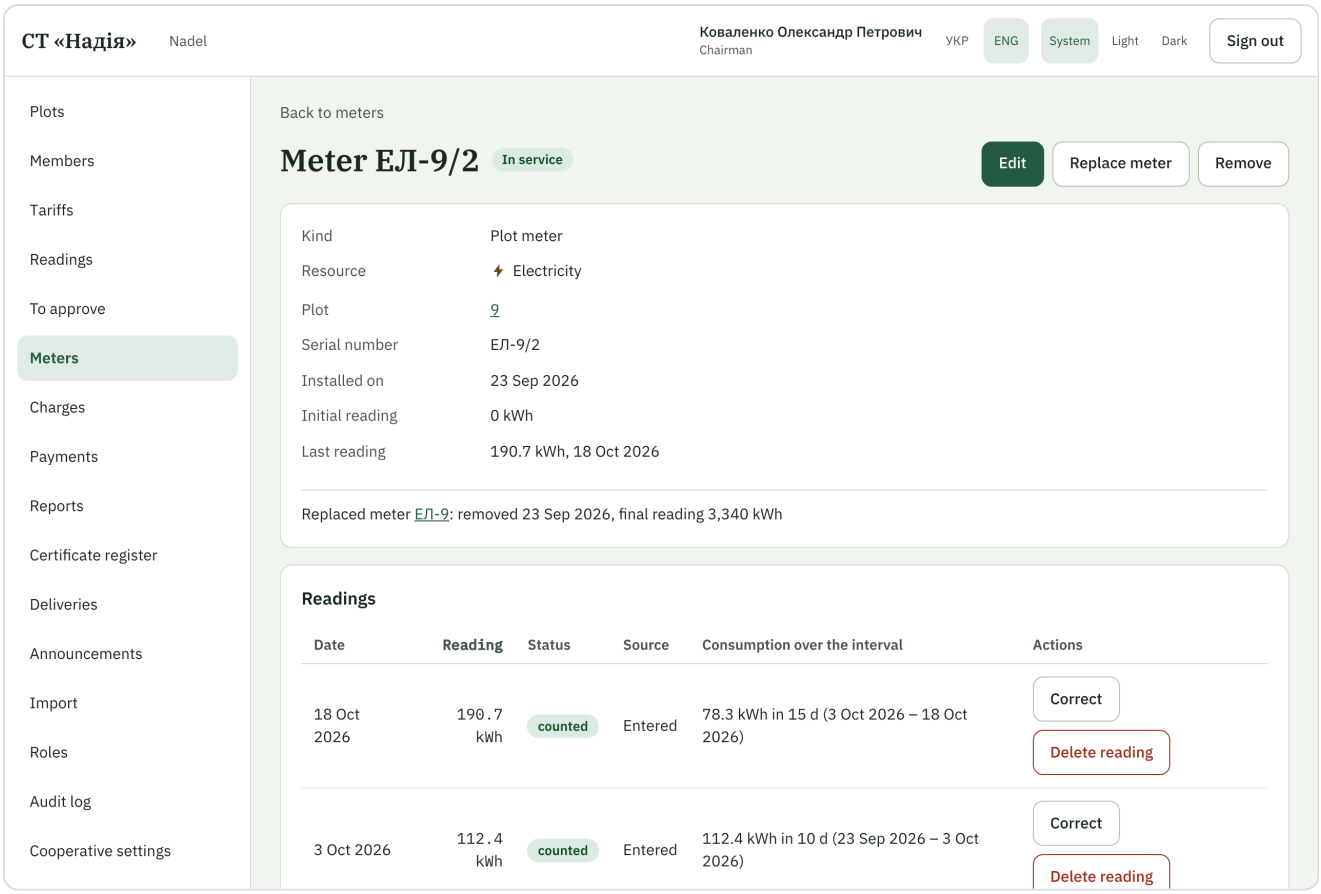
A cooperative may have any number of substations, quarter meters and wells — the system sets no limit.

Register, replace, remove

- **Register a meter**: kind, resource, serial number, plot (or a name for other kinds), installation date, initial reading.
- **Replace meter** is one action: the date, the final reading of the old meter, the serial number and initial reading of the new one. Both keep their history, and their cards link to each other ("Replaced

a previous meter…", "Replaced by the meter ЕЛ-9/2").

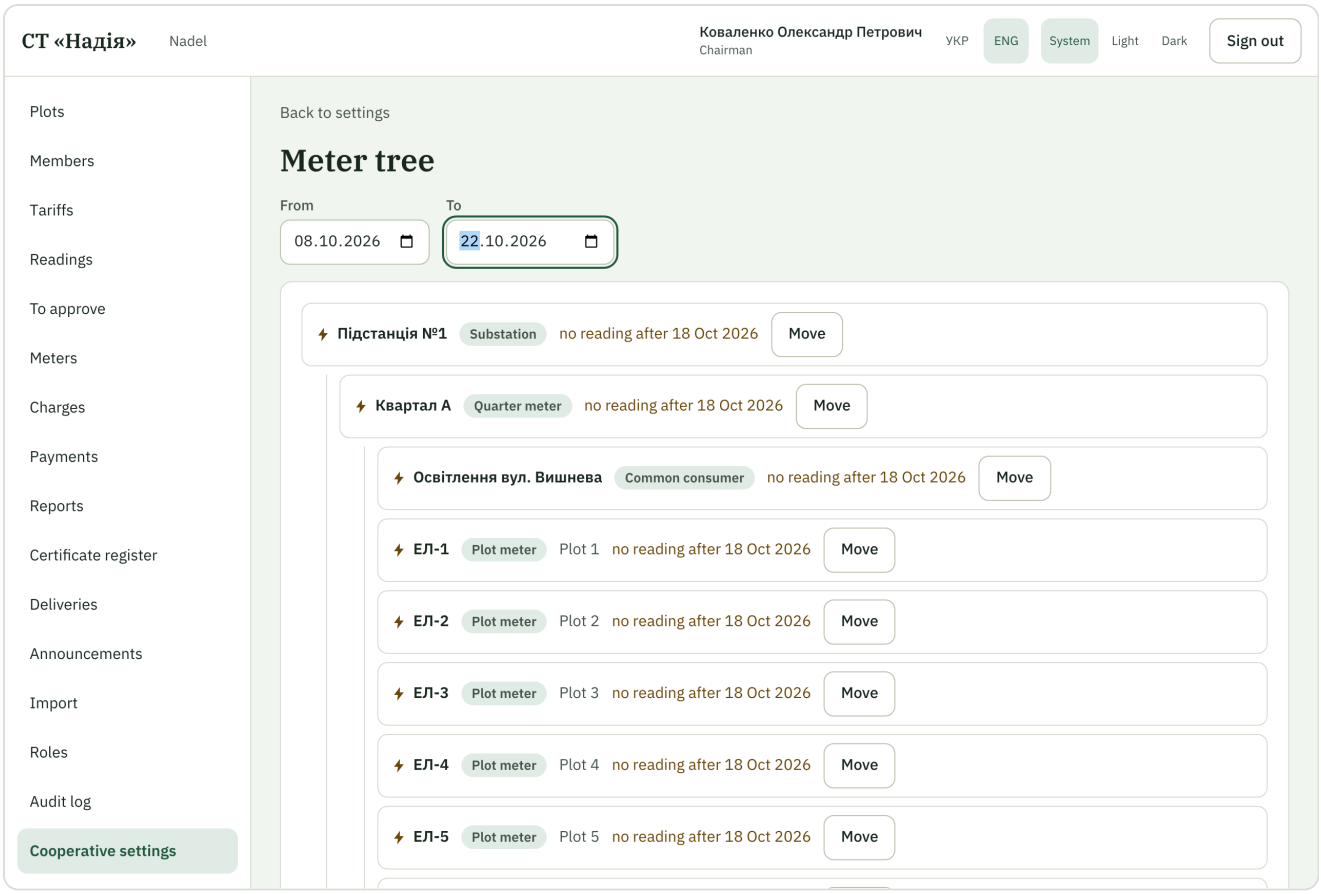
- **A counter wrapping past zero** (99 999 → 00 012) is a replacement of the meter by itself: "Fill in as a wrap-around" in the replace form.
- **Remove**: the meter stops working on the removal date; its readings are kept. Only a meter registered by mistake, without readings, can be deleted.



A meter card with its readings

The meter tree

"Cooperative settings" → "Meter tree". The chairman builds the tree: substations, below them quarter meters, common consumers and plot meters. "Move" puts a meter under another node **from a given date**; earlier periods use the old tree.



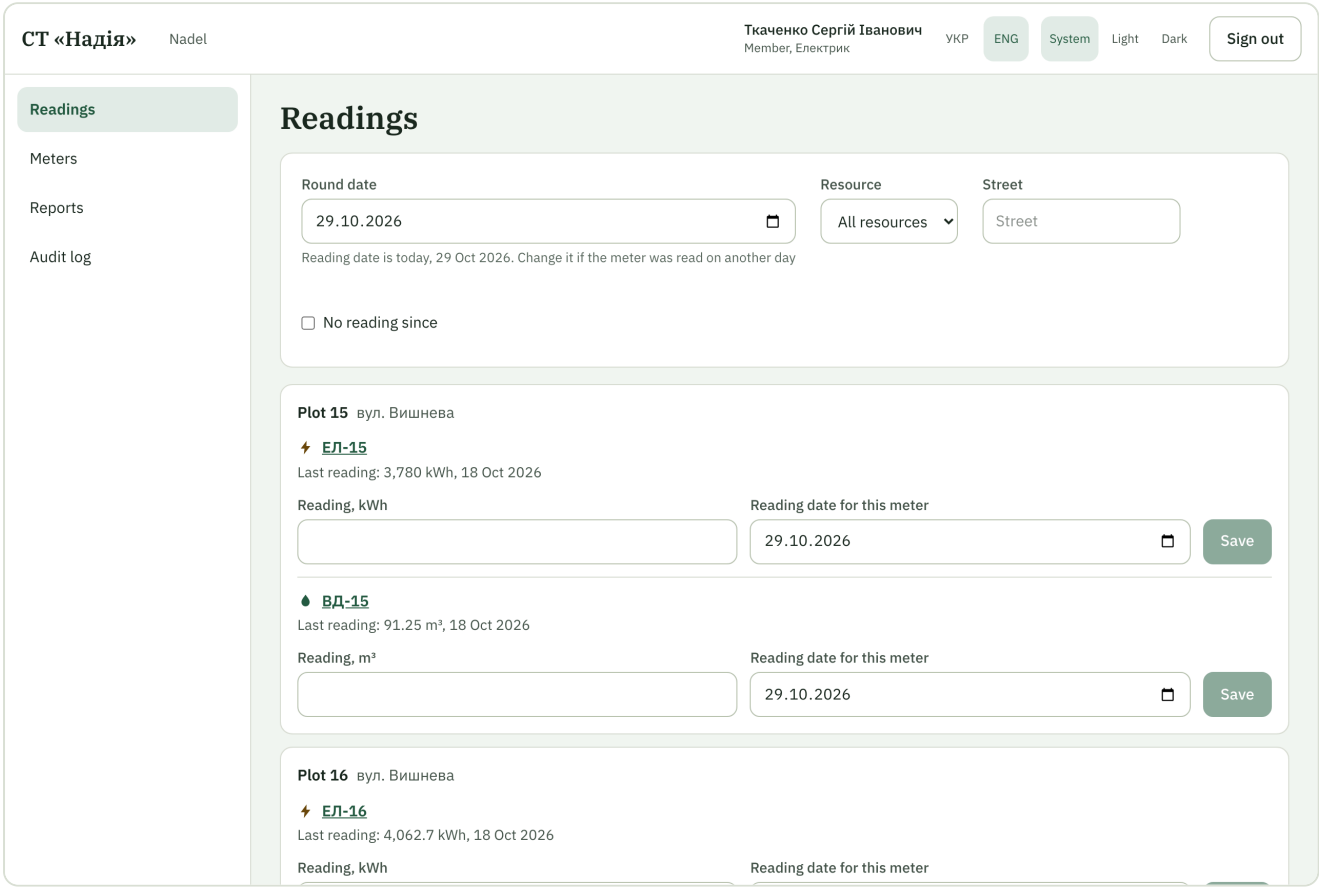
The meter tree with consumption for a range

Choose a range "from … to …" and every meter shows its consumption for it. With no reading on a range boundary, the system estimates the value in proportion to the days between the neighbouring readings and marks it "calculated". With no reading after the range end yet it says "no reading after …".

10 Readings: the round and approvals

Readings are not tied to months: the electrician may come after a month, after three or after a few days. **A reading is a meter's value on a date.** Consumption is counted between neighbouring readings.

The electrician's round

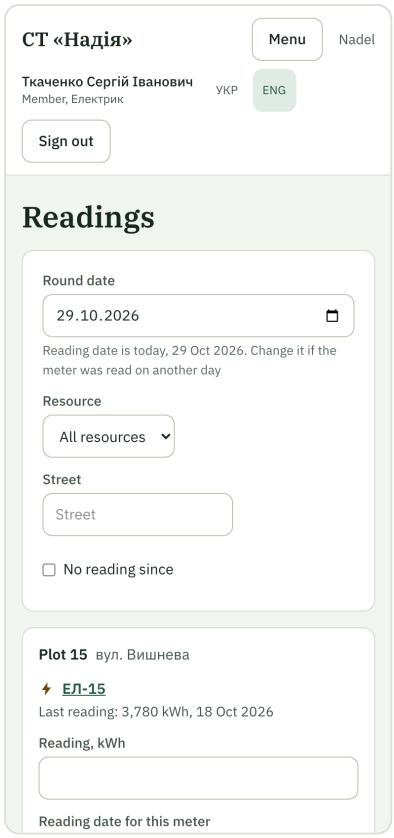


"Readings": one card per plot, a row per meter

1. Open "Readings". "Round date" already holds today's date with the hint "Reading date is today... Change it if the meter was read on another day". A single meter's date can be changed in its row.
2. Filter by resource (lightning / drop) if needed — the choice is remembered — and by street.
3. Each meter shows its last reading and date. Type the new reading and press "Save" in that row. The interval consumption appears at once.

- A reading lower than the previous one is refused: "The reading is lower than the previous one or higher than the next one".
- When consumption is much higher than usual, the reading is saved with the warning "Check the reading".
- "No reading since" lists meters that have not been read for a long time.

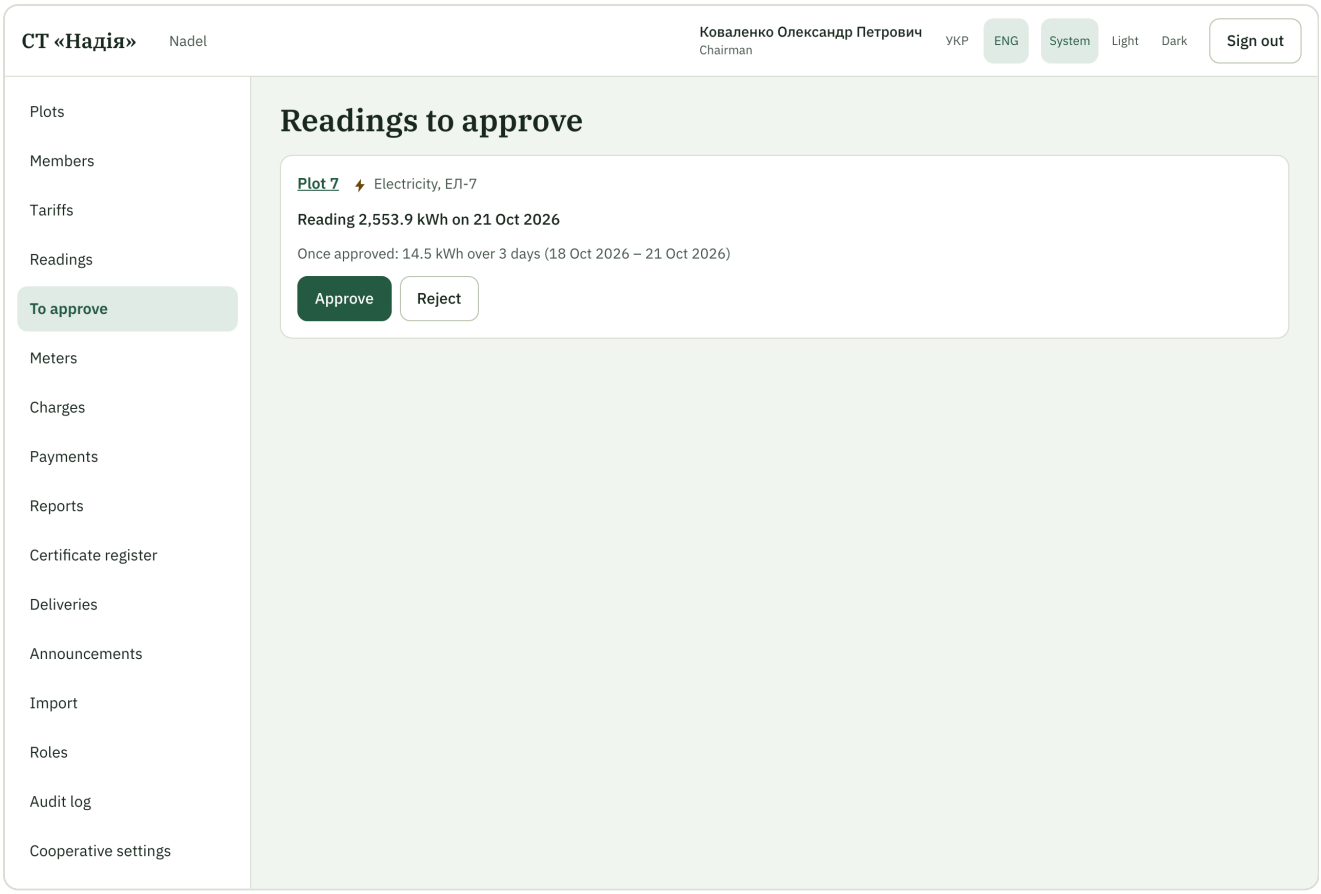
- A second reading for the same date replaces the first; the old one stays in the history marked "superseded".



The same screen on a phone

Readings from members

A member can submit the reading of their own meter from their account. It waits in "To approve": "Approve" or "Reject" with a reason the owner sees. A reading that is not approved is never used.



Readings to approve

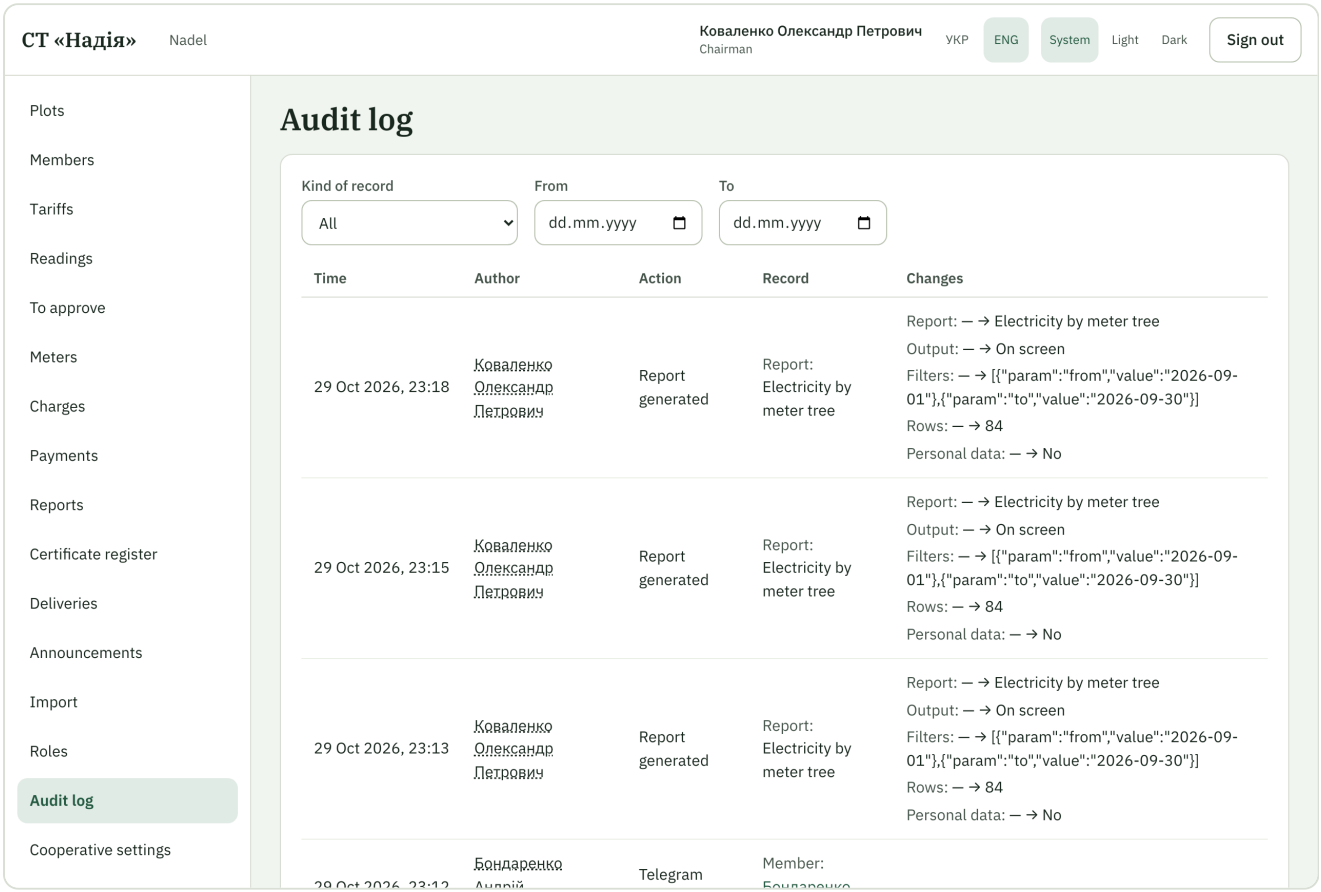
Corrections

On the meter card every reading can be corrected or deleted **while no charge uses it**. Once charged, a correction will go through recalculation (chapter 23).

11 Audit log

Every board action is recorded: who, when, what changed — with old and new values. Only the fields that really changed are shown; a change of a role's permissions lists the added ones ("+") and the removed ones ("–"). Filters by person, action, object and date.

Old and new values of a member's phones, email, address, Telegram, Viber and notes are shown only to someone who may also view members. Anyone else who is allowed to read the audit log sees that the field changed, with the word «приховано» ("hidden") in place of the values. Other fields, such as a name, are shown as usual.

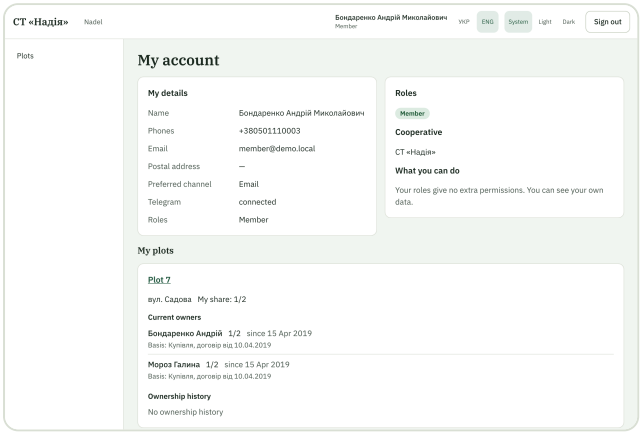


The audit log

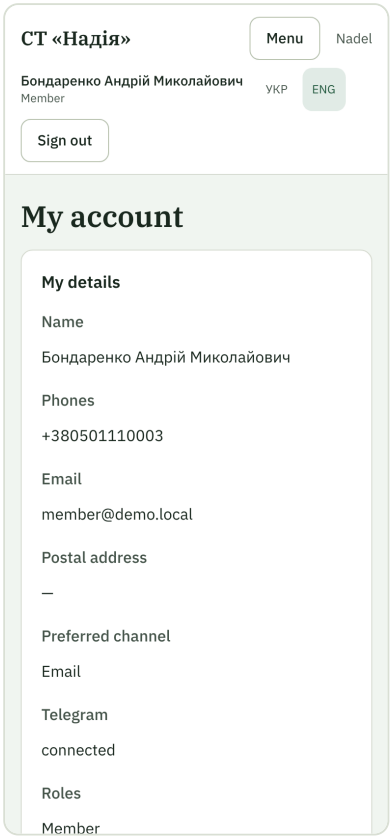
Log entries are never changed or deleted. Data loaded by demo scripts does not appear in the log — only actions of people in the system.

12 The member's account

After signing in a member sees "My account": own plots, what they will be charged by the tariffs, own meters with their readings and "Submit a reading". The payments of their plots are in "My charges".



My account

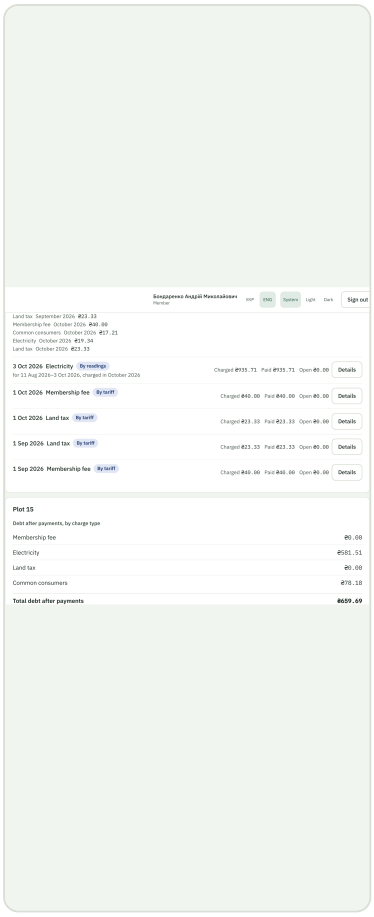


On a phone

A member never sees other plots' amounts, owners or personal data. Other plots' colours only when the chairman chose "All plots" (chapter 6).

My charges

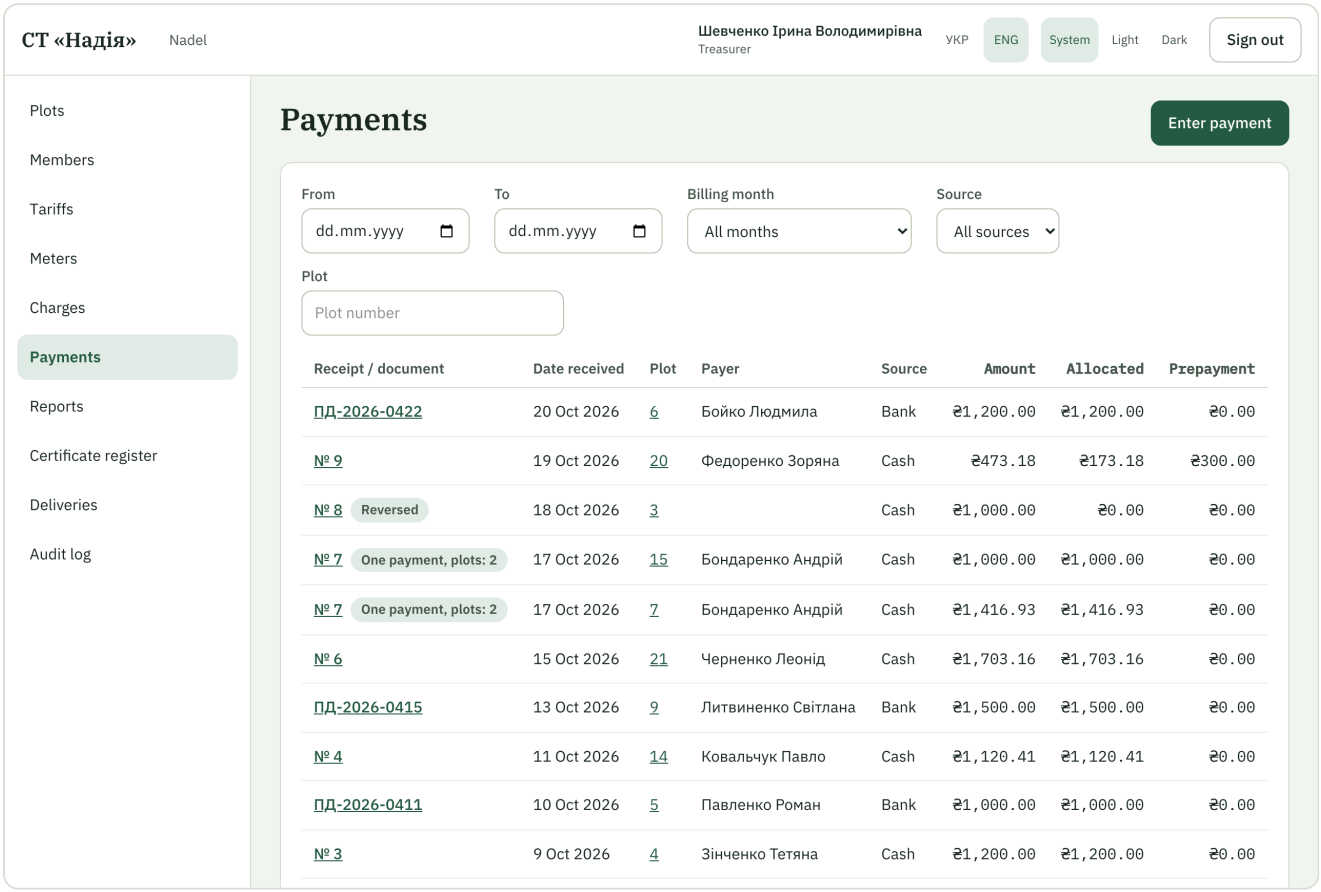
For each of their plots the member sees the debt per charge type and every charge with its calculation, read-only. See chapter 20. The payments of the plot are there too: date, receipt or bank document number, amount and what it paid, reversed payments with a mark, the prepayment and the debt after payments. A member sees neither who entered a payment nor the note about the payer, and never the payments of other plots.



"My charges" in the member account

13 Payments and cash desk works

The treasurer enters money received - cash at the cooperative's cash desk or a bank transfer - and the system allocates it to the plot's debts. From then on the debt, the plot colour and the reports reflect payments. The "Payments" menu is for those with "View payments"; payments are entered by those with "Enter payments" (chairman and treasurer); only the chairman reverses.

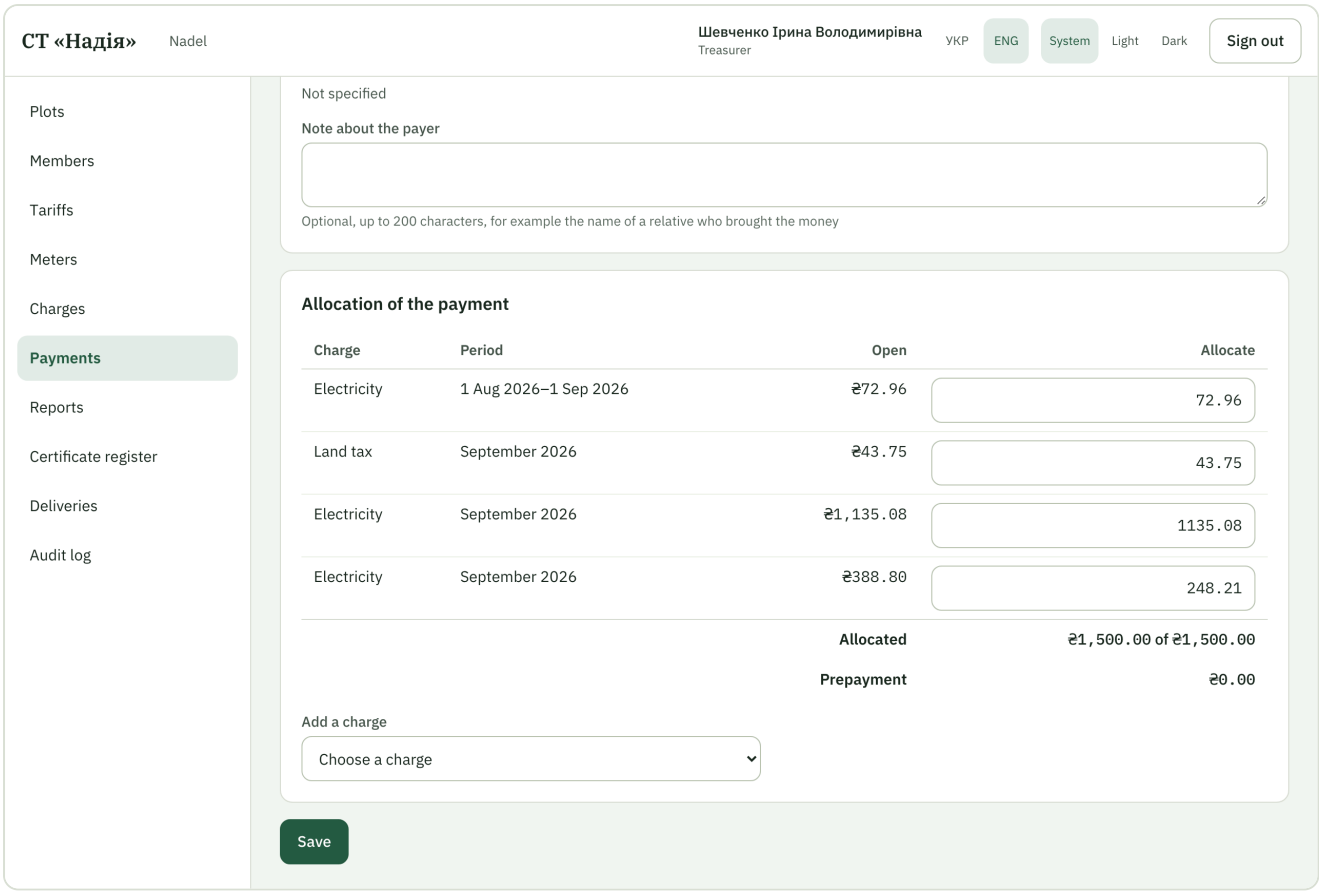


The list of payments

The list shows payments from newest to oldest: receipt or bank document number, date received, plot, payer, source, amount, how much is allocated and how much is left as prepayment. Filters: "From", "To", "Source", "Plot", "Billing month". Reversed payments stay in the list marked "Reversed".

Enter a payment

Press "Enter payment" in the "Payments" list or on a plot card (the plot is then already chosen).



The payment form: the allocation proposal

Field	What to enter
Plot	The plot number. Common land and an archived plot take no payments
Source	"Cash" or "Bank"
Date received	Today by default. Another date is allowed with the warning "The date is not today"; a future date is not
Amount	Above zero, at most two decimals
Payer	Optional: the plot's owners first, then any member
Note about the payer	Optional, up to 200 characters, for example the name of a relative who brought the money. It is personal data, so it is stored encrypted and shown only to someone who also has "View members"
Bank document number	Bank payments only, optional, up to 64 characters

The allocation proposal

As soon as the plot and the amount are known, the form shows the **proposal**: a table of the plot's unpaid charges (charge, period, open amount, "Allocate") and a "Prepayment" line - what is left over the debts. The system pays the oldest debt first; for debts of the same day, in the order of the cooperative's charge types. Examples are in chapter 25.

- You can change the amount on any line, or add another open charge of the plot with "Add a charge". The treasurer chooses the lines; the system places the free remainder itself (see below).
- While automatic allocation is on, a note under the "Prepayment" line says "The remainder will be placed automatically on the oldest debts", and the preview shows the resulting lines, the automatic ones marked. If a line is wrong, the table stays on screen with a message at that line and "Allocated X of Y" below; you can save as soon as there are no problems.
- The total may not exceed the payment ("More is allocated than the payment amount"), and a line may not exceed what is open on its charge ("The amount exceeds what is open on the charge"). Then nothing is saved.
- If someone changed the plot's account between the preview and saving, you see "The data changed after the preview. Review it again before saving." - look at the allocation once more.

Save and the receipt number

"Save" records the payment. For cash the screen shows **"Receipt № 42"** large and **"Another payment"**; for a bank payment it shows the saved payment. Receipt numbers run through the whole cooperative with no gaps and no repeats, even when two treasurers enter cash at the same moment; a number is never changed or used twice. The receipt can be printed (chapter 14); it is an internal receipt, not a fiscal one.

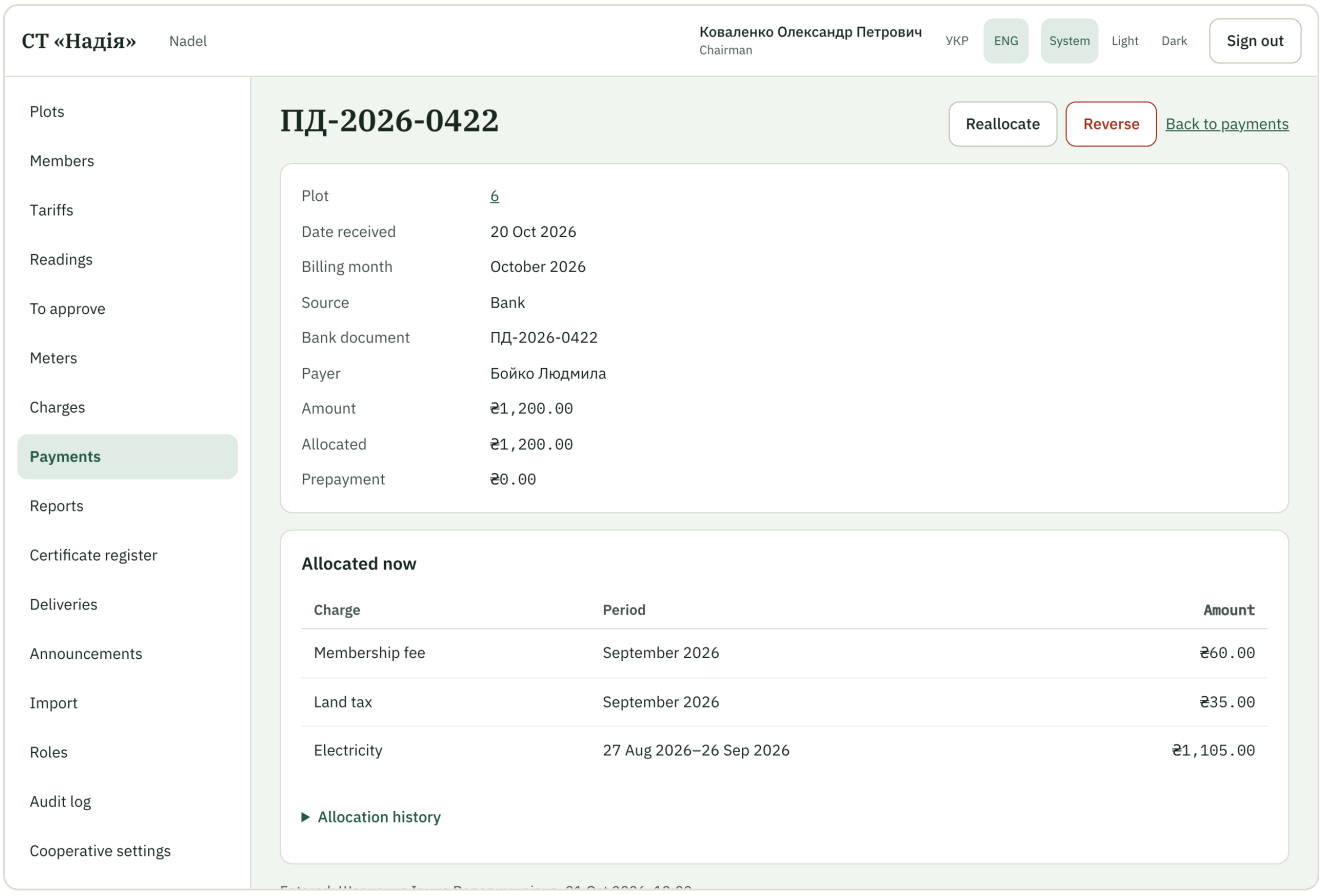
> **A PAYMENT FOR A PAST MONTH**
>
> If the money arrived on 31 January and you enter it on 3 February while January is closed, the payment keeps the date 31.01 but counts in February; January's totals do not change.

One payment for several plots of one payer

If a member brought one sum for several of their plots, choose **"Several plots of one payer"** in the form, then the payer and the amount. The form lists the payer's current plots with their debts and proposes the parts **in proportion to the debts**. Kopecks that do not divide go to the plots with the largest remainders (a tie goes to the plot listed first); an amount above the whole debt goes to the first plot as prepayment. Examples are in chapter 25.

- You can change any part, but together they must equal the payment ("The parts do not add up to the payment amount"). Each part has its own allocation proposal, which can be changed too.
- Each part is a payment of its own plot; all parts belong to one group. A cash group has **one** receipt number. On each plot's account the part is shown with that number and a link to the other parts.

The payment card



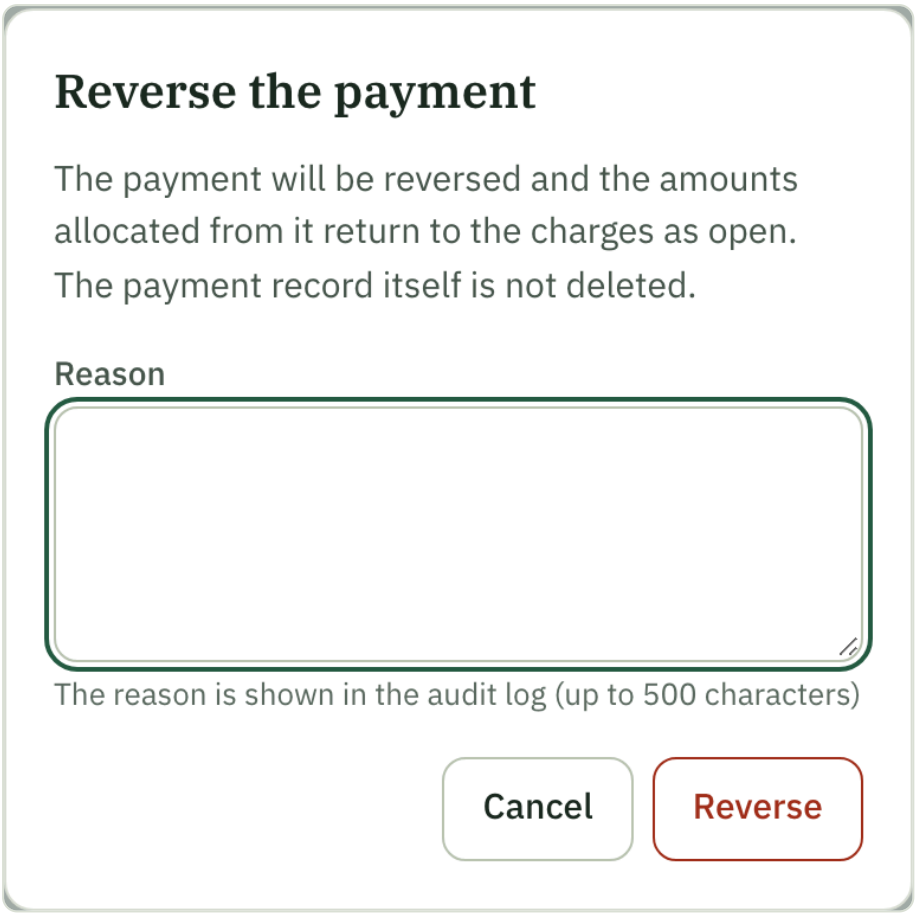
The payment card: allocation and buttons

The card shows the payment's details, the note about the payer (only to those with "View members"), the other parts of the group, and who entered the payment and when. The allocation has two parts: **"Currently allocated"** - what the payment covers now - and **"Allocation history"** - every allocation and every amount **taken back from a charge** (a release) with its period, amount, cause, author and time. When automatic allocation is off, there is a prepayment and a charge is open, the card also has the "Allocate prepayment" button.

Reallocate

"Reallocate" ("Enter payments") shows the current allocation next to a new proposal that can be changed as when entering. Saving takes back the old allocation and writes the new one in one step and is written to the audit log. This is how money is moved between charges. The dialog does not get stuck on an old preview: the table stays, wrong lines have a message and "Allocated X of Y" is shown below. While automatic allocation is on, lines left at 0.00 do not create a prepayment as long as the plot has open debts: the free amount goes to the oldest debts.

Reverse a payment



Reversing a payment with a reason

Only the chairman sees **"Reverse"**. A reason is required (up to 500 characters, for example "Wrong plot"); without it you see "Enter a reason". The system takes back the whole allocation of the payment, so the charges it paid are owed again; the payment record itself is not deleted. A payment is reversed once ("The payment has already been reversed"); a part of a group is reversed on its own and the others stay. A reversed cash payment keeps its receipt number and is shown as "Reversed". A treasurer who made a mistake asks the chairman, as for charges. A payment of an archived plot can still be reversed.

Prepayment: automatic and by hand

A plot's **prepayment** is the money paid for it that is not placed on any charge. It is never negative.

- While the **"Allocate prepayment automatically"** switch in the settings is on (the default), the prepayment goes by itself to the plot's open charges, in the same order, as soon as new charges are booked (a month, a reading, a run, an opening balance, a recalculation) or money is taken back from a charge. On the account such lines are labelled "Allocated from prepayment".
- While automatic allocation is on, prepayment never waits while the plot has open debts. When a payment is saved, the lines the treasurer chose are kept as chosen and the rest of the amount is placed at once by the standard order: the oldest debts first, then the order of charge types. The

same happens on a reallocation. Prepayment remains only when the plot has no open debt (it then waits for new charges).

- While the switch is off, nothing is placed automatically: the prepayment stays until the treasurer allocates it by hand. The **"Allocate prepayment"** button is on the plot's account and on the payment card whenever the prepayment is above 0.00 and a charge is open - a proposal over the open charges that can be changed.

Switching the setting on does not place old prepayment at once; it goes to the plot's next charges.

The plot account after payments

On the account, payments are interleaved with charges by date; next to each charge are "Charged", "Paid" and "Open". Above the list are the debt after payments per type and in total, and next to the total debt "Prepayment X" whenever the prepayment is above 0.00. The **"Payments only"** filter keeps only payments and their reversals; the debts above the list do not change.

"Charges → Months" has a **"Received"** column next to "Charged". The "Plot balances", "Debtors" and "Charges by month" reports take payments into account, and the new **"Payments"** report lists every payment with totals per source (chapter 24).

What a member sees

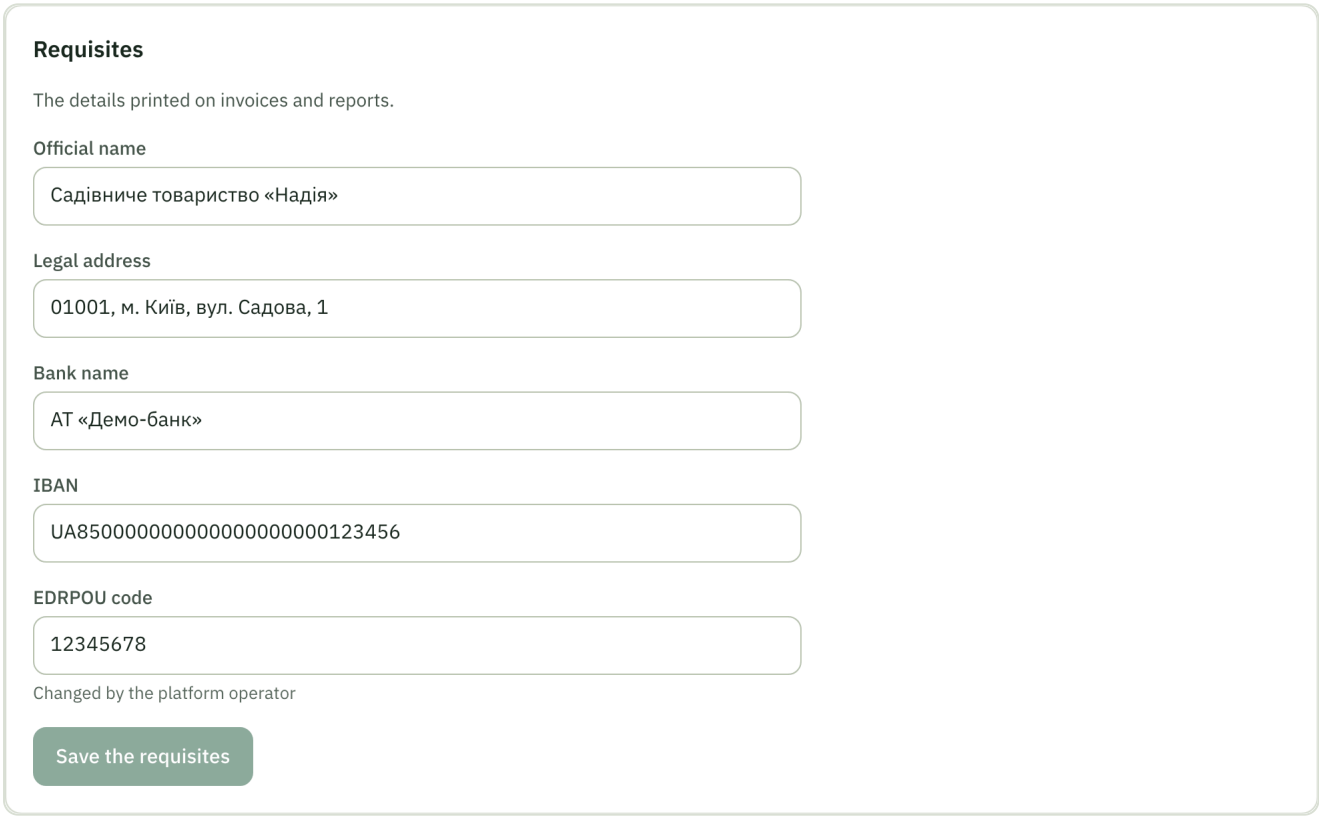
In "My charges" a member sees the payments of their own plots only - the date, the receipt or bank document number, the amount, what it paid, reversed payments with a mark, the prepayment and the debt after payments. They do not see who entered a payment or the note about the payer, nor other plots. The electrician sees no payments.

14 Invoices, certificates and receipts works

The system turns the accounts into printable documents: a monthly **invoice** for every plot with the bank details and a payment QR code, a **certificate of settlements** for a plot as of a date, and a **receipt** for a cash payment. All documents are PDF files and are always in Ukrainian - they are official documents, whatever language your screens are in. Invoices are for those with "Issue invoices", certificates for those with "Issue certificates"; both lists are seen by those with "View charges".

Before the first invoice: bank details

An invoice needs the cooperative's **IBAN** and **EDRPOU code**. The IBAN is entered by the chairman in "Settings → Requisites" (chapter 6); the EDRPOU code is set for the cooperative by the platform operator. While one of them is missing, nothing is built: the message names what is missing and links to the settings, and a warning banner is shown on the Requisites screen and on the invoice screens.



Invoices cannot be built until the IBAN and the EDRPOU code are filled in

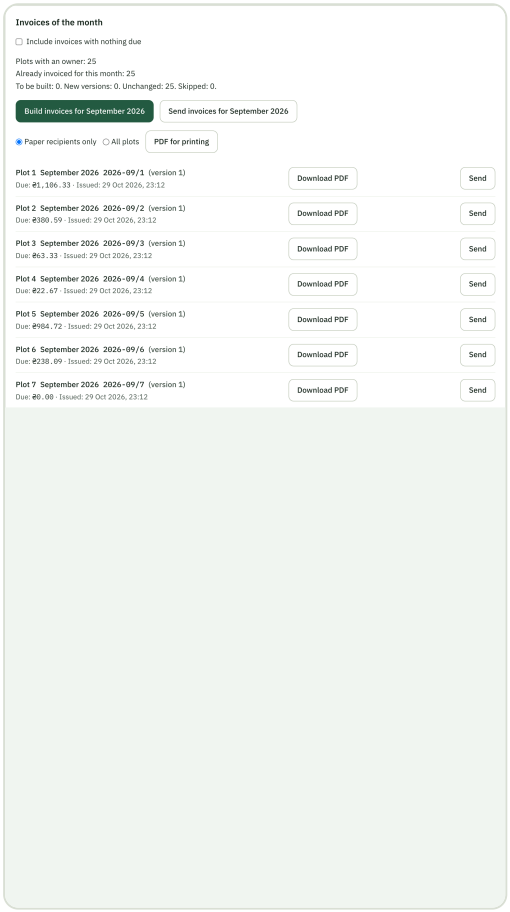
Building the month's invoices

Open "Charges → Months", choose the month and find the **"Invoices of the month"** section. An open month can be invoiced too; usually the board does it right after the month's charges.

1. **Preview** "Build invoices for October": how many plots can get an invoice (active plots that are not common land and have a current owner), how many of them already have an invoice for this

month, and what the build will do, in the words of its result: "To be built: 18. New versions: 0. Unchanged: 1. Skipped: 6." A plot that already has an invoice gets a new version only when its numbers changed; plots with nothing due are skipped. The switch **"Include invoices with nothing due"** adds the plots without a debt (their invoice has no QR code).

2. Press **"Build invoices for October"** and wait: a spinner is shown until the result appears - how many invoices were built, how many got a new version, how many stayed unchanged and how many were skipped. When nothing was booked between the preview and the build, the result matches the preview. The result takes the place of the preview until you change the month or the switch.
3. Running the build again creates nothing new while nothing has changed. If someone is already building the same month, you are told so and nothing is built twice.



The month's invoices: preview, result and the list with a PDF per row

Below is the list of the month's invoices: one row per plot with its latest version (plot, version, "Amount due", date of issue and a PDF link); a plot's older versions are folded under **"Earlier versions (N)"**. **"PDF for printing"** downloads one PDF with the latest invoices, one per A4 page, ordered by street and plot - for the members who chose paper delivery; a switch puts in all members.

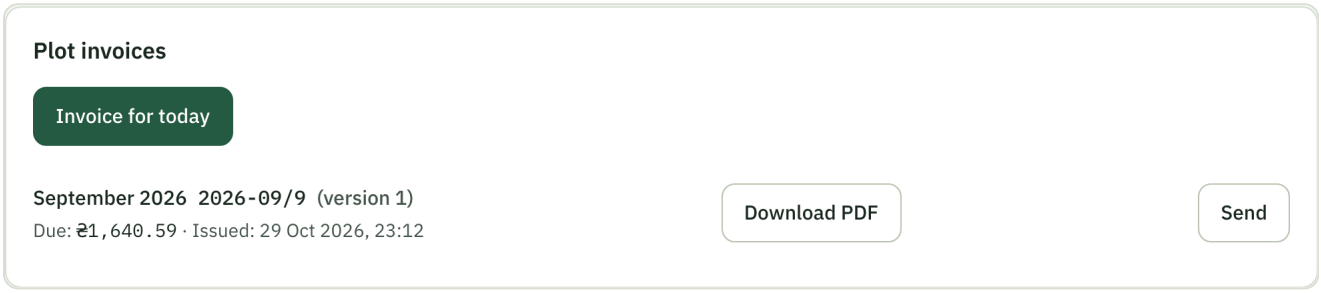
Versions

The system keeps every invoice exactly as it was built: opening it later shows the same numbers and the same owners, even after a payment or a change of owner. After a payment, a correction or a recalculation the treasurer builds the month again: a plot whose numbers changed gets a **new**

version ("version 2") under the same invoice number; a plot where nothing changed gets no new version. The older versions stay readable: the lists show the latest version, and "Earlier versions (N)" unfolds the earlier ones. A member sees only the latest version. An invoice is never deleted.

An invoice on request

On a plot card the **"Invoices"** tab lists the plot's invoices (month, version, amount due, issued) with the PDF of each: the latest version of each month, the older ones under "Earlier versions (N)". **"Invoice for today"** builds an invoice as of today for the current billing month and downloads its PDF right away (the button shows a spinner while it is prepared). A new version appears only when something changed since the latest version (a payment, a charge, the owners, the requisites); otherwise the PDF of the latest version downloads and nothing new is created.



The plot's invoices and "Invoice for today"

What is on an invoice

- The cooperative's name, EDRPOU code, address, bank and IBAN; the number "Invoice № 2026-10/9" (year-month/plot number) and, from the second version on, "(version 2)"; the date of issue.
- The plot, the street and the current owners with their shares.
- A table with one row per charge type: debt at the start of the month, charged, paid, debt at the end. Under "Charged" are the details members ask about: each electricity interval as one range of reading dates with both readings, the kWh and the price ("показники 10.08.2026 → 03.10.2026: 3 435,900 → 3 705,500 = 269,600 кВт·год × 4,32 ₴"), losses and common consumers, the basis of a fixed tariff, and for an adjustment the month it corrects.
- "Payments for October" ("Оплати за жовтень 2026"): each payment booked in the month with its date, the cash receipt number or the bank reference, and the amount, so a member can match it with their bank. A reversed payment shows as the payment and a negative "reversal" row.
- The payments received after the month but before the invoice was issued ("Received after October") - they lower the amount due.
- The totals: debt at the end of the month, prepayment and **"Amount due"**; beside them the QR code - only when the amount due is above 0.00 - with the line "Pay in the bank app" ("Оплатити в застосунку банку"): on a phone, tapping it opens the bank transfer without scanning.

Each invoice fits on one A4 page. An invoice with very many lines goes on on a second page whose header repeats the invoice number and the plot.

An invoice for a month that is not closed yet says "as of" a date. How the numbers are formed is in chapter 26.

The QR code and how to check it with a phone

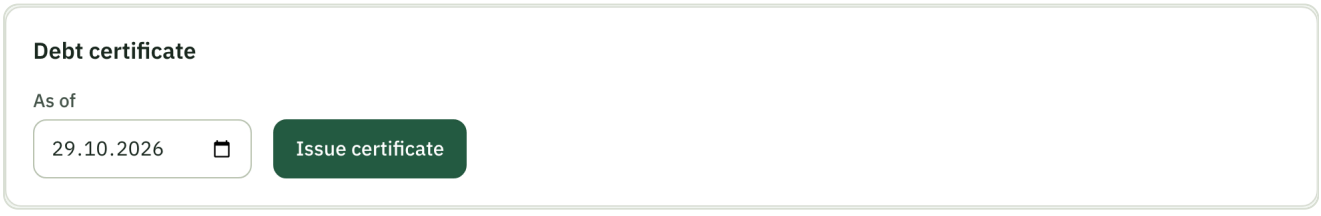
The code follows the rules of the National Bank of Ukraine. A banking app scans it and opens a filled-in transfer. To check a code before you hand invoices out:

1. Open the QR scanner in your banking app (the payments or "scan QR" section) and point the phone at the code on the screen or the paper.
2. Check the **recipient**: the cooperative's name.
3. Check the **IBAN** against the one in the settings.
4. Check the **purpose**: it must start with "Ділянка" and the number of your plot, then the month and the invoice number.
5. Check the **amount**: it equals "Amount due". The amount can be edited - a member may pay part of it.
6. The other fields (recipient, account, recipient code, purpose) are locked and cannot be changed. Close the form without paying.

A sample invoice of the demo cooperative: sample-invoice.pdf.

Debt certificates

On the plot card the **"Certificate"** tab asks for a date (today by default); press **"Issue certificate"**. The certificate "Certificate № Д-2026-1" shows the requisites, the plot with its area and current owners, the debt per charge type and in total and the prepayment as of the date - the same numbers as the "Plot balances" report. It says "No debt" when the total debt is 0.00, otherwise "The debt is ...". It carries the issuer's name and role, the date of issue and a place for a signature and a stamp. Only those with "Issue certificates" (the chairman by default) can issue. A certificate can also be issued for a plot without an owner (it then lists no owners); common land and archived plots get no certificate.

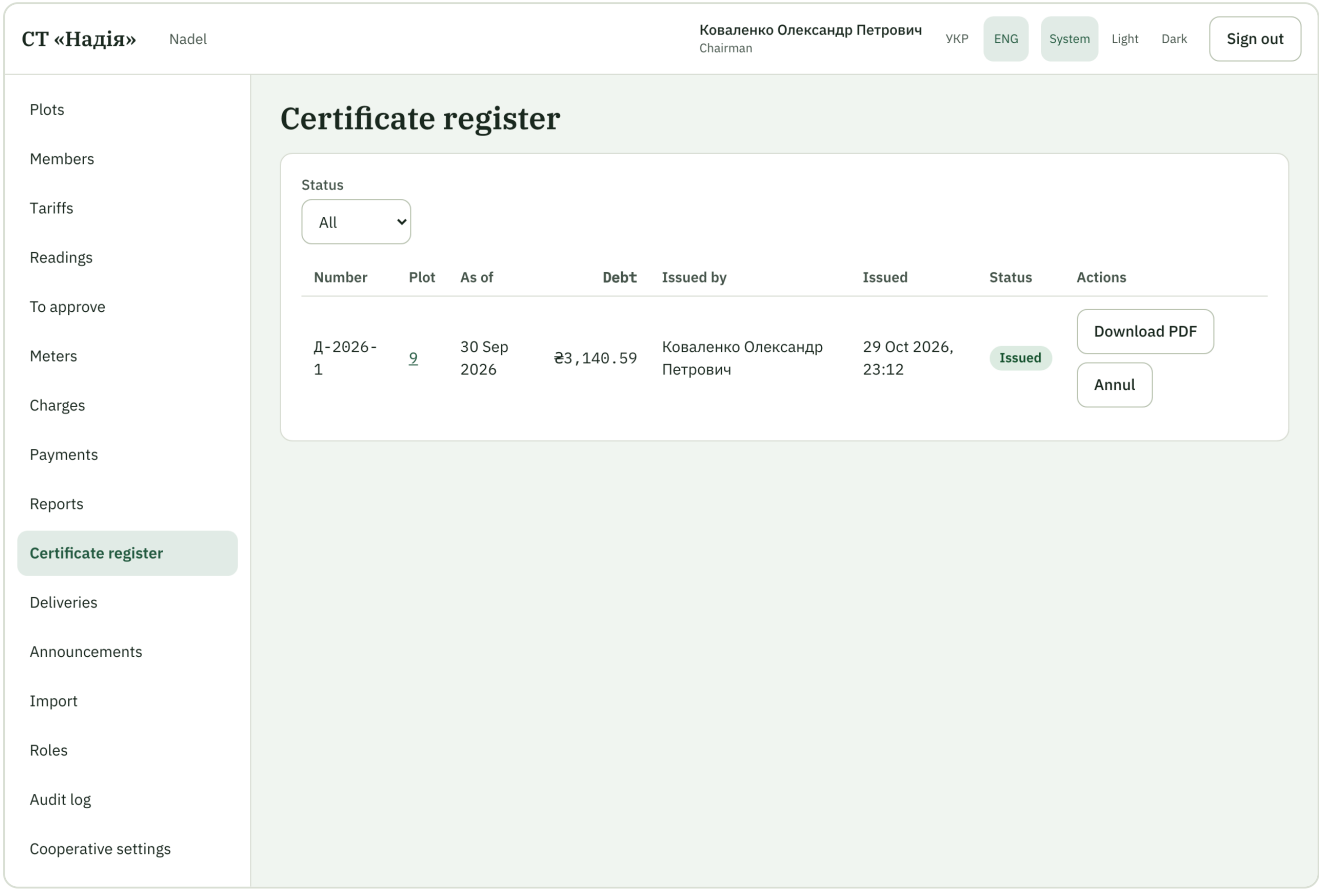


Issuing a certificate on the plot card

Numbers run in sequence per cooperative and year, with no gaps and no repeats. An issued certificate never changes. A mistake is answered with a new certificate, and the old one is **annulled** with a reason.

The register of certificates

The **"Certificate register"** menu item lists every certificate: number, plot, as-of date, debt, who and when issued it, and the status. Open its PDF, or press **"Annul"** and enter the reason (required). An annulled certificate stays in the register with its reason, its PDF is marked "ANNULLED", it cannot be annulled twice and its number is not reused.



The register of certificates

The cash receipt

After you save a cash payment, the screen with the large receipt number has **"Print receipt"**; the payment card has it too. A bank payment has no receipt. The receipt (A5 or half of A4) shows the requisites, the receipt number, the date, the plot, the payer, the amount in figures and in words, what the money was placed on ("Allocation as of" the printing date) and the name of the person who took the money, with a line for a signature. A payment split over several plots gives one receipt that lists every plot. The receipt of a reversed payment carries "REVERSED" with the date. The note about the payer is printed only for users who may see member data. It is an internal receipt, not a fiscal one.

What a member sees

In the member's account, **"Invoices"** shows the latest invoice of each of their plots first, each with "Download PDF"; older months are below. Next to a cash payment there is "Receipt". A member never sees certificates, and a document of someone else's plot is "not found" for them.



The member's invoices

AUDIT

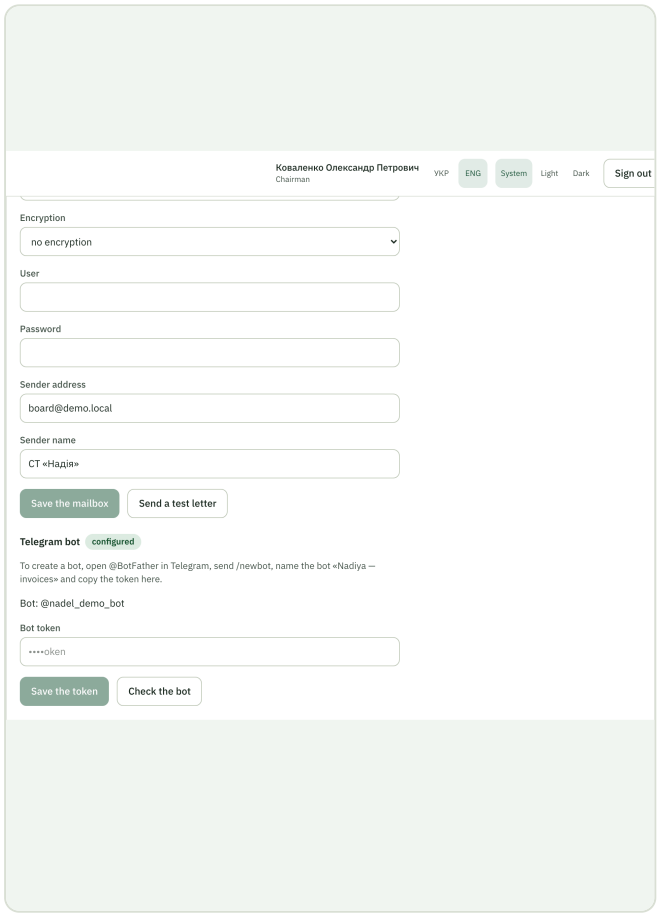
The audit log records the build of a month's invoices (month, built, skipped), a re-issued invoice, an issued certificate and an annulled certificate with its reason. Downloads are not recorded.

15 Delivery: e-mail, Telegram and the bot works

The system sends invoices, payment messages and board announcements to members through two channels that belong to the cooperative itself: **e-mail** from your own mailbox and **Telegram** through the cooperative's **own bot**. A member sees the name of the cooperative, not a third-party service. Every send is recorded with its outcome, so the treasurer knows who got the invoice and who still needs paper. Letters and messages are always in Ukrainian, like the documents. The channels are set up by those with "Manage cooperative settings" (the chairman), invoices are sent by those with "Issue invoices" (chairman, treasurer), announcements by those with "Send announcements" (the chairman).

Settings → "Delivery": the mailbox

Open "Cooperative settings" and find the **"Delivery"** panel. While a channel is not set up it says "not configured" and nothing is sent through it.



Settings → "Delivery": the mailbox, the Telegram bot and the general options

The simplest way is to create a **free mailbox** for the cooperative (for example Gmail or Ukr.net) and use an **app password**: a separate password for programs, created in the mailbox settings, which does not open the mailbox itself. Most services do not accept the ordinary mailbox password for programs. The values below are typical ones; check them in the help of your mail provider, they may change.

Provider	What to do in the mailbox	Server (SMTP)	Port and encryption	User and password
Gmail	Turn on 2-step verification, then create an "app password"	`smtp.gmail.com`	587, STARTTLS	the full mailbox address; the 16-character app password
Ukr.net	In the mailbox settings allow access from mail programs (SMTP) and create a password for them	`smtp.ukr.net`	465, SSL	the full mailbox address; the program password

1. Fill in "Mail server (SMTP)", "Port", "Encryption" (STARTTLS or SSL), "User", "Password", "Sender address" and "Sender name" (for example "Nadiya cooperative"). Press "Save the mailbox".
2. Press **"Send a test letter"**. The letter goes to your own address from your member card (without one the button says there is nowhere to send). You see either "The mail server accepted the test letter" or the server's refusal in plain words, for example about a wrong password. The button works with the saved settings: save your changes first.

After saving, the password is never shown again: the field says "saved". Leave the field empty to keep the saved password. The change history records only that the password was changed, never the value.

> **DAILY LIMITS OF FREE MAILBOXES, AND THE SENDING RATE**
>
> Free mailboxes limit the number of letters a day (in the order of hundreds; see your provider's help for the exact number) and block a mailbox that sends too fast. That is why **"Letters per second"** defaults to 1: 120 letters take a couple of minutes and the mailbox is not taken for a spammer. Raise it (up to 10) only for a paid mailbox. If the cooperative is larger than the daily limit, some letters fail with "The mail server refused the letter" - send them again the next day (see "Reading the Deliveries screen").

The Telegram bot

The chairman creates the bot in Telegram itself; it is free and takes a minute. The same short instruction is shown above the token field.

1. In Telegram open **@BotFather** (the official bot for making bots).
2. Send the command `/newbot`.
3. When asked for a name, write **"Nadiya - invoices"** (the cooperative's name; members see it).
4. Choose a unique username that ends in `bot`, for example `nadiya_invoices_bot`.
5. @BotFather answers with a **token** (a long string like `123456:ABC…`). Copy it.

6. In the settings paste the token into "Bot token" and press "Save the token".
7. Press **"Check the bot"**: the system asks Telegram and shows the bot's name (`@nadiya_invoices_bot`) or "The token is not valid".

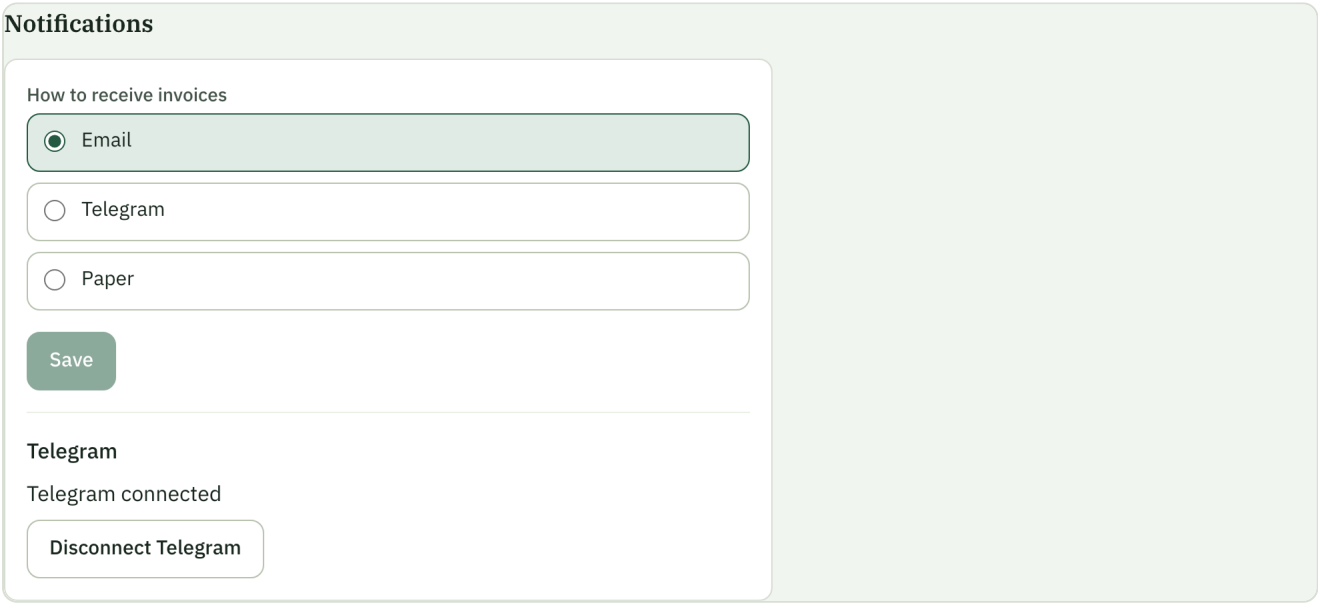
The token is the key to the bot: give it to nobody. After saving, the screen shows only its last 4 characters. If the token is changed, the cooperative's bot restarts within about a minute; if Telegram rejects the token (the bot was deleted or the token reissued), the bot stops until the token is replaced.

"Notify about payments" (on by default) decides whether members get messages about their payments and reversals (below). The switch and the letter rate are saved with "Save the options".

How a member connects Telegram

A bot cannot write first to a person by phone number or @username: the member has to open the bot once through a personal link. There are two ways.

- **From the account.** "My account" → the **"Notifications"** panel → the **"Connect Telegram"** button. The cooperative's bot opens with a one-time code (valid 30 minutes); in Telegram press "Start". The bot greets the member by name: "Вітаємо, Олено! Тепер рахунки за ділянку 9 приходитимуть сюди." ("Welcome, Olena! Invoices for plot 9 will arrive here now.") and shows the menu buttons. The account shows "Telegram connected" and a "Disconnect Telegram" button. If the cooperative has no bot, the button is not offered.
- **From a paper invoice.** Every paper copy of an invoice carries a small QR code **"Отримувати рахунки в Telegram" (Receive invoices in Telegram)** with a personal code of exactly that recipient, valid for 90 days. The member points the phone camera at it, opens the bot and presses "Start". Co-owners who get paper each have their own copy with their own code.



The member's account: "Notifications" - the preferred way to receive invoices and the Telegram link

A code works once. A used, expired or unknown link is refused by the bot with "Посилання недійсне, отримайте нове в кабінеті або в правлінні" (the link is not valid, get a new one in the account or from the board). If this Telegram account is already connected to another member of the cooperative,

the bot asks "Is it you, ...?" and moves the connection only after "Yes, it's me". The board sees only "Telegram: connected" or "not connected" on a member card, never the details of the member's account. Connecting and disconnecting are written to the change history.

In the same "Notifications" panel the member chooses the **preferred way** among those available to them: Telegram (when the cooperative has a bot), e-mail (when the member card has an address) and paper always. The board can still change it on the member form.

Who gets what, through which channel

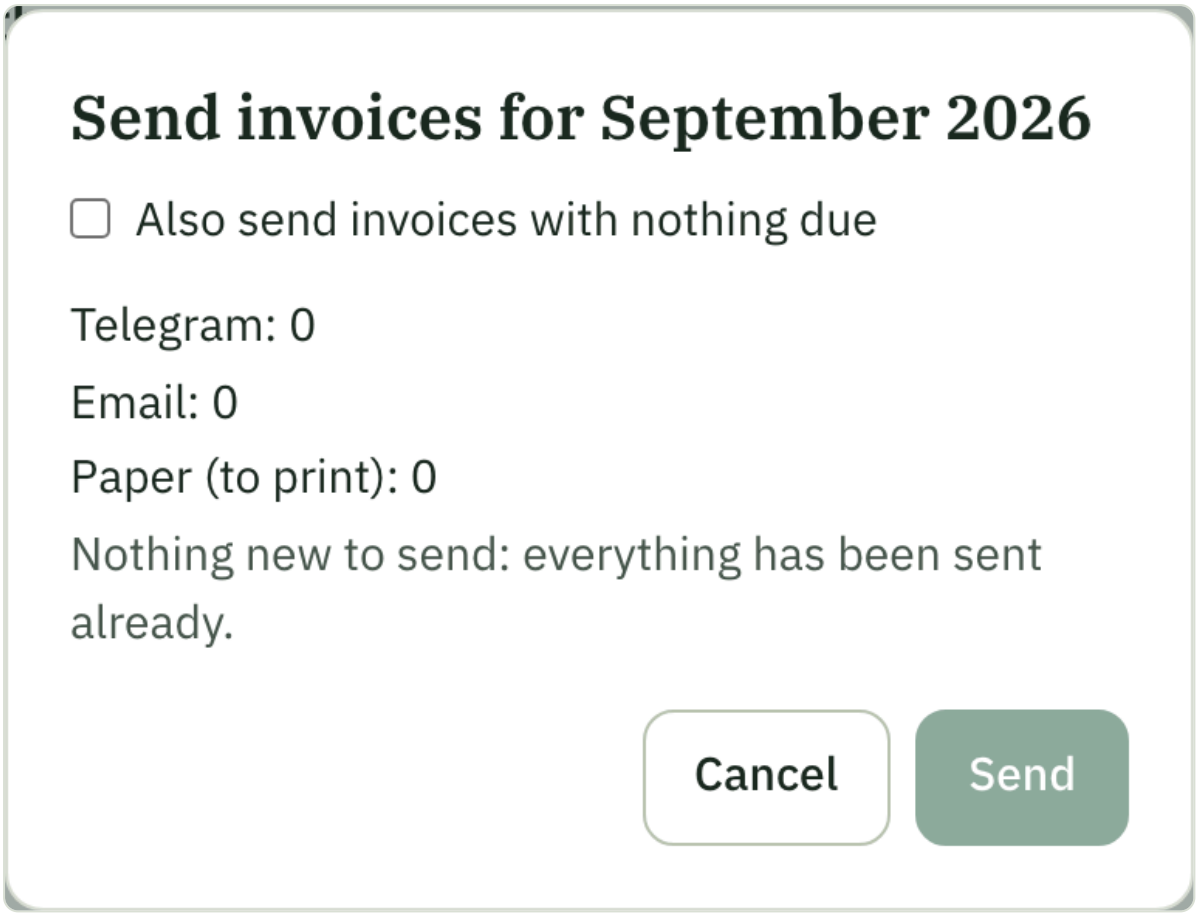
The member's preferred way and what is set up decide.

Preferred way	Situation	Gets it by
Telegram	connected, the bot is set up	Telegram
Telegram	not connected, or no bot	e-mail if the member has an address and the mailbox is set up, otherwise paper
E-mail	an address and the mailbox is set up	e-mail
E-mail	no address, or no mailbox	paper
Paper	-	paper (the print PDF)
Viber	-	paper, with the note "Viber is not supported yet" on the Deliveries screen

Co-owners. The invoice of a plot goes to each of its current owners through their own channel. A member who owns several plots gets one message per plot, each with its own PDF. Examples are in chapter 27.

Sending the month's invoices

Open "Charges → Months", choose the month and in the "Invoices of the month" section (chapter 14) press **"Send invoices for ..."**. The system sends nothing yet and first shows a **preview**: how many members get the invoice in Telegram, by e-mail and how many on paper.

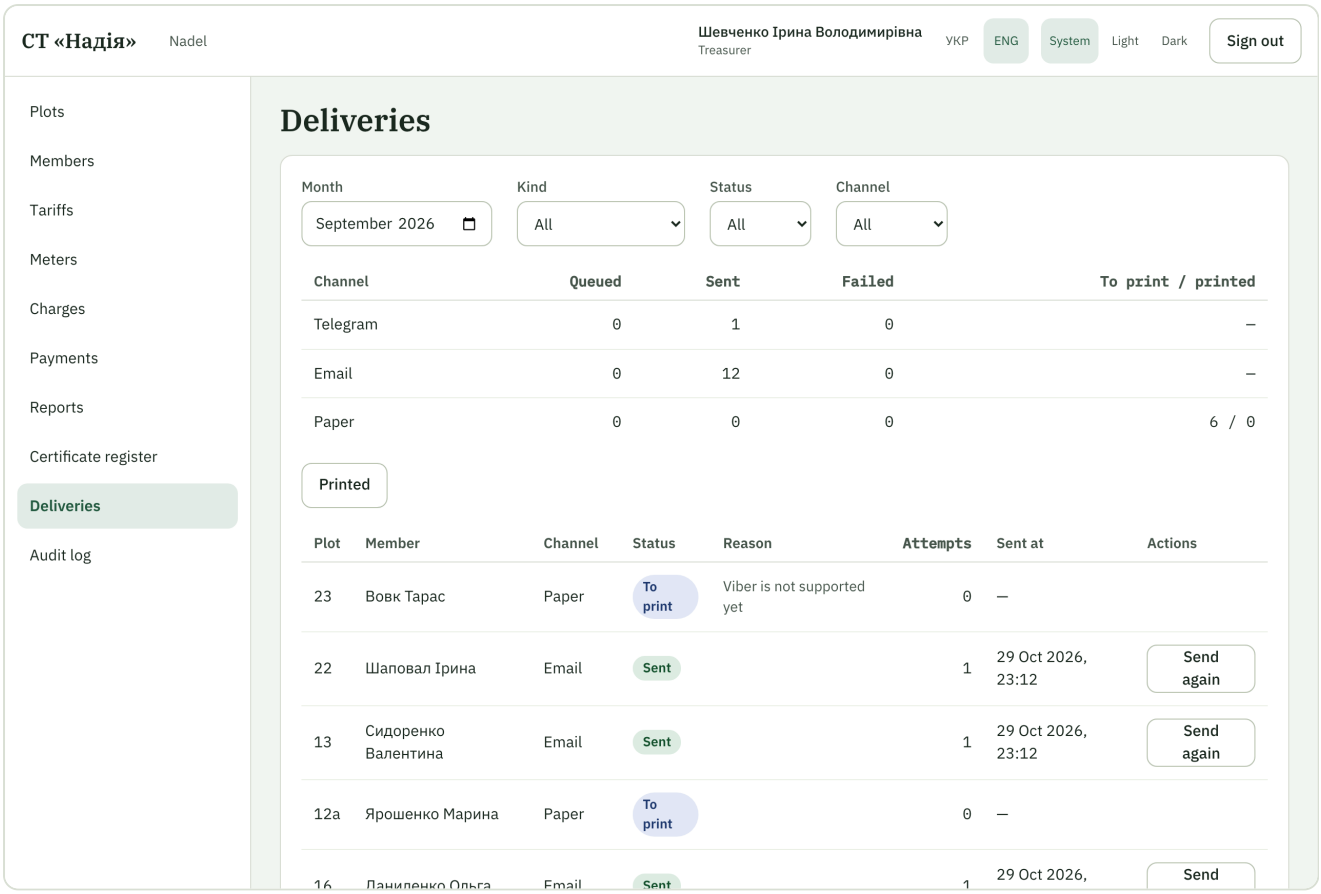


"Send invoices for the month": the preview by channel

- The checkbox **"Also send invoices with nothing due"** adds plots with nothing to pay; such a member gets "Боргу немає, дякуємо!" ("No debt, thank you!") and the PDF.
- "Send" puts the messages in the queue and returns at once; the progress is on the Deliveries screen. Telegram gets "Рахунок за жовтень 2026, ділянка 9: до сплати 413,26 ₴" (Invoice for October 2026, plot 9: 413.26 ₴ due), the PDF and a "Оплатити" (Pay) button that opens the banking app through the invoice's QR link. The letter has the same text, the PDF attached and the signature "Правління ..." (the board of ...).
- Sending the same month's invoices again **does not duplicate anything**: the same version of an invoice reaches the same member through the same channel once. A re-issued invoice with a new version is sent as new.
- One invoice can be sent to the owners of one plot with the "Send" button next to it in the month's invoice list.
- Paper invoices are marked "to print". Print the "PDF for print" (each copy has its own Telegram QR code), then press "Printed" on the Deliveries screen.

Reading the Deliveries screen

The **"Deliveries"** menu item (for those with "Issue invoices") shows what was sent for the chosen month. At the top is a summary by channel: queued, sent, failed, to print / printed. Below is the list with filters by kind (invoice, payment, payment reversal, announcement), status and channel. A row has the plot, the member, the channel, the status, the reason, the number of attempts and the time.



"Deliveries" for a month: the summary by channel, the list, failures with their reasons

Status	What it means
Queued	waits for its turn or the next attempt
Sent	the mail server or Telegram accepted the message (it does not mean the person has read it)
Failed	it did not work; the "Reason" column says why
To print / Printed	an invoice for paper, until the board presses "Printed"

The reasons of failures and what to do:

Reason	What to do
The mail server refused the letter	the mailbox refused the letter (a limit, too fast, the address not accepted); the system tries two more times. If it persists, lower "Letters per second", wait for the next day and send again
Wrong mailbox login or password	correct the app password in "Delivery" and send a test letter; then use "Send again"
Could not connect	no connection to the mail server or Telegram; usually temporary and retried; check the server address and port

Reason	What to do
The bot is blocked	the member blocked the bot; the connection is cancelled and invoices go through the fallback channel (the table above) until the member connects Telegram again
The channel is not configured	no mailbox or bot (or Telegram rejected the token); set the channel up in "Delivery"
No contact	the member has no e-mail address or Telegram chat; tell them in person or enter the address on the card

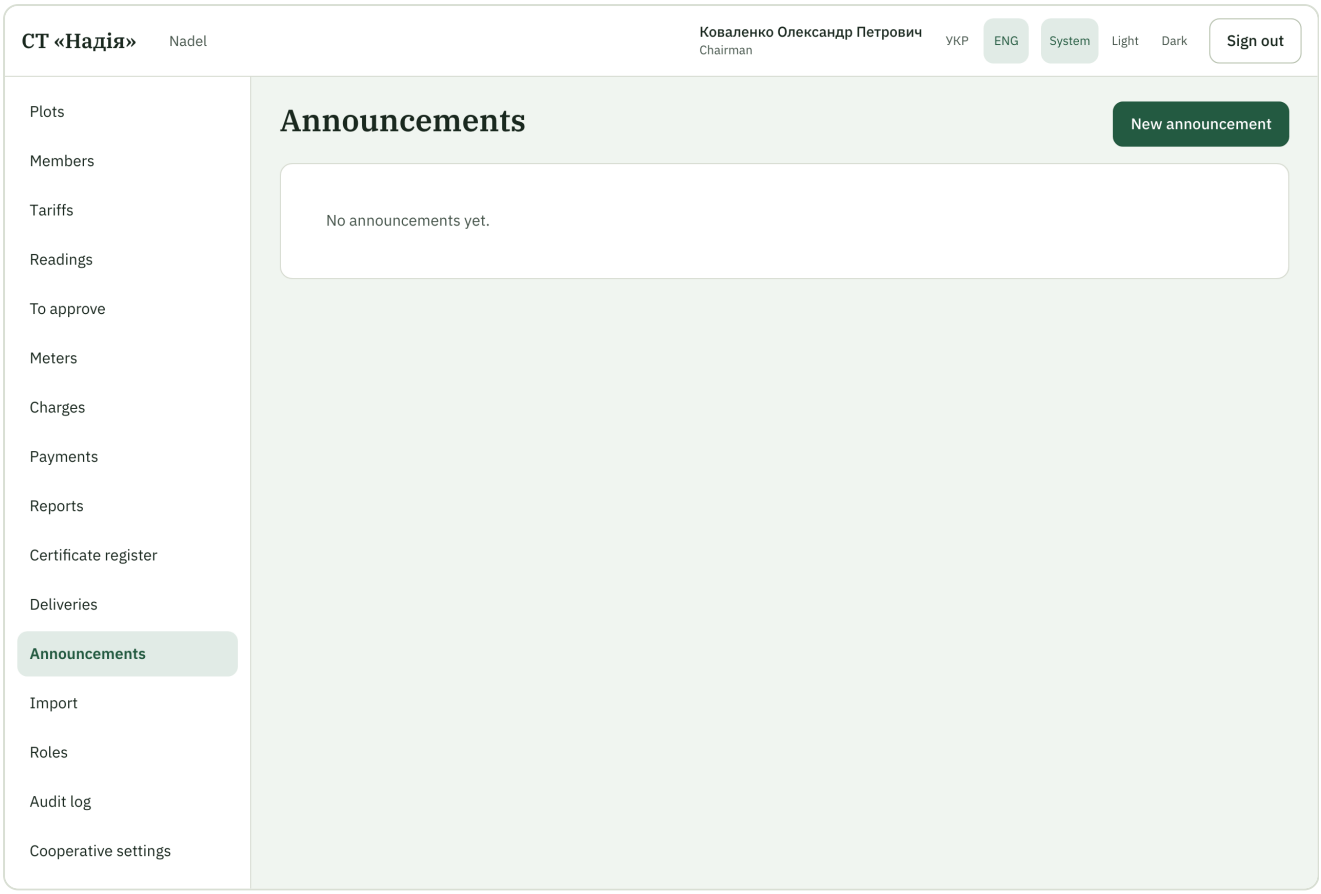
A row that is "Sent" or "Failed" has the **"Send again"** button: after fixing the cause, send the message once more. Paper rows and rows still queued do not have it. The **"Printed"** button removes the "to print" mark from all paper invoices of the chosen month (the system asks to confirm and says how many). "Sent" and "Printed" change no amounts; they only record delivery.

Payment messages

When "Notify about payments" is on, after a payment is saved every current owner of the plot who gets messages by Telegram or e-mail receives "Отримано 500,00 ₴ по ділянці 9 (28.10.2026). Борг: 1 234,56 ₴" (500.00 ₴ received for plot 9; debt 1,234.56 ₴), or "Аванс: …" (prepayment) when nothing is owed. A cash payment adds the receipt PDF. After a reversal comes "Платіж від 28.10.2026 на 500,00 ₴ сторновано. Борг: …" (the payment of … was reversed). Paper owners get no such messages. With the switch off, no messages.

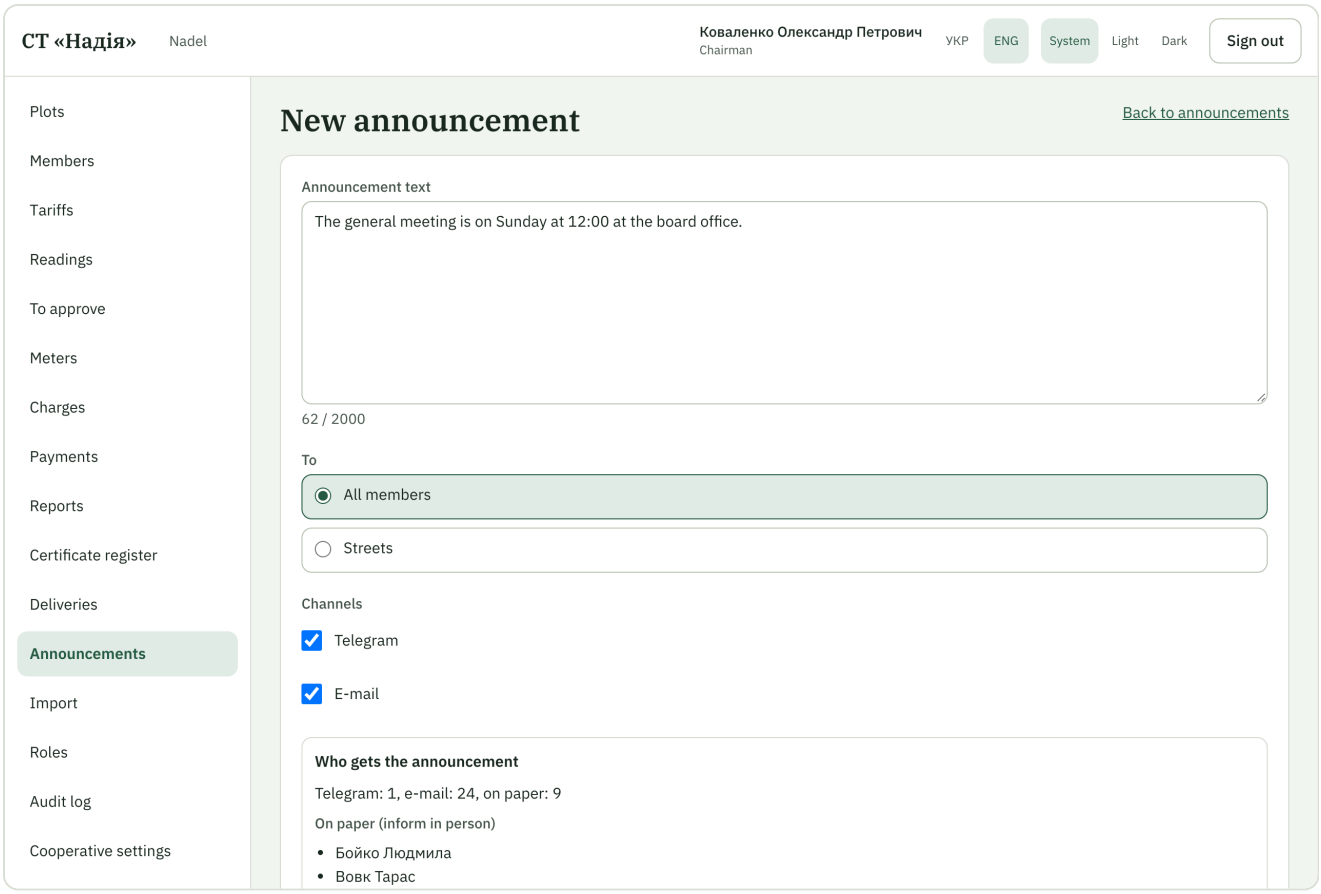
Announcements

The **"Announcements"** menu item (needs "Send announcements"; by default only the chairman) lists past announcements: date, author, to whom, channels and the outcome ("Sent: …, failed: …, queued: …, on paper: …"). Open one to see the text and each delivery.



"Announcements": past announcements with their outcome

1. "New announcement": write the text (plain text, up to 2000 characters - there is a counter under the field).
2. In "To" choose "All members" or "Streets" and tick the streets (the streets of the active plots).
3. In "Channels" choose Telegram, e-mail or both.
4. Press "Preview": how many members get the announcement in Telegram and by e-mail, and the **list of those who have none of the chosen channels** ("On paper (inform in person)") - tell them yourselves: print the list and inform them.
5. "Send". The announcement appears in the list.



A new announcement: text, recipients, channels and the preview of who stays on paper

What a member sees in the bot

A connected member has a menu of buttons under the message box. The bot shows only their own plots and nothing else. A chat that is not connected gets only the help text and an invitation to connect, no amounts. The bot speaks Ukrainian.

Button	What it does
"Борг" (Debt)	for each of the member's plots: the debt per charge type, the total and the prepayment, as in the account
"Рахунок" (Invoice)	the invoice as of today (with several plots, choose the plot first): the PDF and the "Оплатити" button. During one day the same invoice is returned
"Показання" (Readings)	choose the plot (if several), then the meter (kind and serial number), then send the number. The reading goes to the board for approval with today's date: "Показання 12 345 по лічильнику ЕЛ-9/2 передано на перевірку" (submitted for review). A wrong value is explained in the same words as in the account and the member types again
"Відключити" (Disconnect)	stop receiving invoices in Telegram (the command `/stop` does the same)

When the board approves or rejects a reading, the member gets "Показання 12 345 по лічильнику ЕЛ-9/2 прийнято" (accepted) or "... відхилено: reason" (rejected). The commands `/start` and `/help` show the menu and a short help; any other text is answered with the help. More than 20 messages a minute from one chat get "Забагато повідомлень, спробуйте за хвилину" (too many messages, try in a minute).

A short leaflet for members, "How to receive invoices in Telegram", is a separate A4 sheet in Ukrainian (`pdf/telegram-leaflet.uk.pdf`) that you can print and hand out.

CHANGE HISTORY

The history records changes of the delivery settings (which, without secrets), "Invoices sent" (the month and counts per channel), "Announcement sent" (to whom and how many), and Telegram connected or disconnected for a member. Single sends are not in the history: they are on the Deliveries screen.

Part II

How the system calculates

The rules the system calculates amounts by, and why they are what they are — with worked examples. Chapters marked "planned" describe the next capabilities as they have already been decided.

16 Core principles

Only the server counts money	The web page and the app only show amounts, debts and statuses; they never calculate them. One formula, in one place.
Kopecks	Hryvnia with two decimals. Intermediate calculations are exact, never rounded. An amount is rounded to kopecks **once**, half up, only when a charge is produced.
A rate is not money	A tariff rate has up to four decimals (1.4532 ₴ per m²) and is never rounded. A fifth decimal is rejected, not cut.
Nothing is erased	Charges, owners and readings are never edited in place. A mistake is fixed by a new record (a reversal, a new version, a new reading); the old one stays in the history.
Reports calculate nothing again	A report only selects, groups and adds up amounts already booked under these same rules, so every amount in a report can be reconciled to the kopeck with a plot account or the month screen (chapter 23).
Audit log	Every board action is recorded with old and new values.
One language	The interface shows only the selected language, never a mix.
Minimal personal data	No tax numbers, passports or dates of birth; contacts are stored encrypted; personal data never goes to the server logs.

> **WHY ROUND ONCE**
>
> A rate of 0.05 ₴ for 15 + 15 days of a 30-day month is exactly 0.05 ₴. Rounding each half separately gives 0.03 + 0.03 = 0.06 ₴, a kopeck more. Over thousands of plots and months such kopecks add up.

17 Tariffs and the expected amount

A charge type (membership fee, electricity, land tax, water, special-purpose fees) has one or more tariffs. A tariff has **versions**: each applies from its date until the day before the next version. Two versions never start on the same day.

```
units           per plot → 1 · per sotka → area m² / 100
period share    monthly 1 · quarterly 1/3 · yearly 1/12
month amount    rate × units × period share × (days the version applies in the month) /
(days in the month)
one-time fee    the whole rate in the month of its due date, 0.00 in other months
```

Worked examples

Case	Given	Result
Membership fee	50 ₴/sotka/year, plot 600 m²	50 × 6 / 12 = 25.00
Odd area	50 ₴/sotka/year, 612 m²	50 × 6.12 / 12 = 25.50
Rounded at the end	33.3333 ₴/sotka/year, 700 m²	19.444425 → 19.44
Per plot, quarterly	300 ₴ per quarter	100.00 per month
Version change mid-month	50 ₴ until the 15th, 60 ₴ from the 16th, 600 m², 30 days	12.50 + 15.00 = 27.50
Special-purpose fee	1,200 ₴, due 01.05.2026	1,200.00 in May, 0.00 otherwise
Land tax	35 ₴/sotka/year, 600 m²	17.50 per month

Which plots

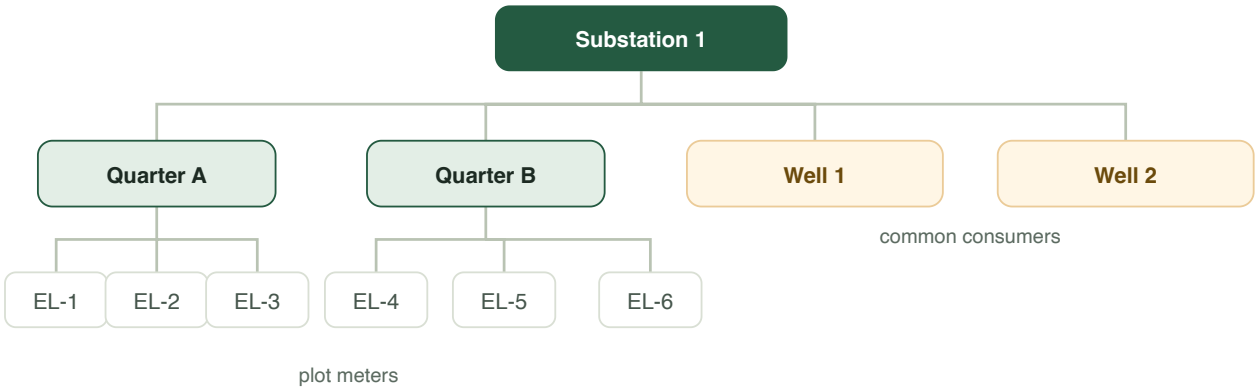
- "All plots" minus exclusions (each with a reason), or "Selected plots". Never archived plots or common land.
- Per-kWh and per-m³ tariffs only for plots with an active meter of that resource. **No meter, no charge** (confirmed by the cooperative's chairman).

Correcting a version

A version can be corrected or deleted **while no charge uses it**. After that a correction goes through recalculation (chapter 23): charges already issued to people never change silently.

18 Meters and the tree

A meter measures one resource: electricity (kWh) or water (m³). Meter values are not money: up to three decimals.



The meter tree: substation → quarters → plots; common consumers are nodes of their own. There may be any number of substations, quarters and wells.

- The chairman builds the tree; nothing about it is hard-coded. Moving a meter under another node takes effect from a date; the tree's history is kept, so a past range is calculated with the tree as it was then.
- Any number of substations, quarter meters and common consumers (wells included); every quarter meter and every well has its own list of plots.
- A plot has at most one active meter per resource.

Replacing a meter

A replacement is one action: the final reading of the old meter, the serial number and the initial reading of the new one. Consumption of a period with a replacement is the sum over both meters.

> EXAMPLE
>
> Old meter: 8,140 at the start, removed at 8,205. New one: installed at 3, reads 69 at the end. Consumption: (8,205 − 8,140) + (69 − 3) = 65 + 66 = **131 kWh**.

A counter wrapping past zero (99,999 → 00,012) is "a replacement of the meter by itself": same serial number, final 99,999, initial 0. The same rule then applies.

19 Readings and consumption

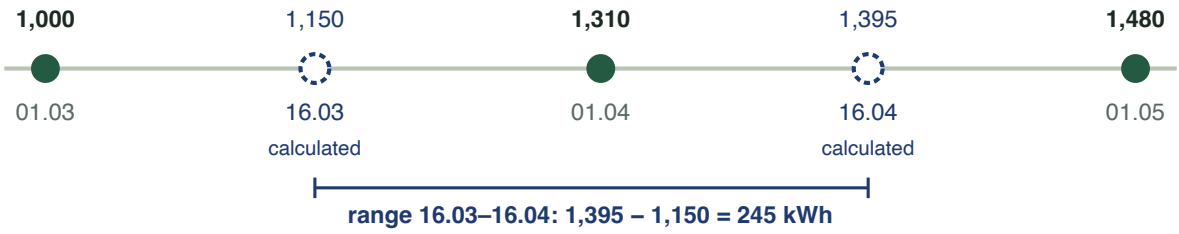
A reading is a meter's value on a date, not "the reading for a month". The electrician comes when they can: after a few days or after a few months. The person enters the date; it defaults to today, with an unobtrusive hint.

Reading statuses

Status	Meaning	Counted
Approved	Entered by the electrician or the board, imported, or a member's reading approved	yes
Pending	Submitted by a member, waits for approval	no
Rejected	With a reason the owner sees	no
Superseded	Replaced by a new reading for the same date or a correction; kept in the history	no

Intervals and the value on a date

- **Interval consumption** is the difference between a reading and the previous reading of the same meter.
- **The value on any date**: the real reading if there is one on that date; otherwise a linear interpolation by days between the neighbouring readings, marked "calculated"; with no reading after the date yet, unknown.
- **Consumption for a range "from … to …"** = the value at the end − the value at the start.



Real readings (green) and values calculated at the range boundaries (dashed)

WHERE 1,150 COMES FROM

Between 01.03 (1,000) and 01.04 (1,310) there are 31 days and 310 kWh. 16.03 is day 15: 1,000 + 310 × 15 / 31 = 1,150. Calculated values are always marked so that nobody takes them for a real reading.

Checks

- A reading lower than the previous one (or higher than the next one, when entered back-dated) is refused. A lower value is valid only as the initial reading of a new meter.
- A date in the future or outside the meter's working period is refused.
- Daily consumption three times the average of the last 180 days: the reading is accepted with the warning "Check the reading".
- A reading can be corrected or deleted while no charge uses it; after that, through recalculation.

20 Charges and closing months works

A charge is an entry in the plot's account: charge type, amount, month, the days it pays for (service period) and how the amount was obtained (rates, units, kWh, the readings and tariff versions used).

Where charges come from

Source	How it arises
Recurring tariffs	The board presses "Charge the month": preview → "Post". A rerun adds only what is missing, never duplicates.
Electricity	Every approved reading at once gives a charge for the interval since the previous reading.
Network losses	Depending on the chairman's setting: a percentage on every electricity charge, or a board run for a range (chapter 21).
Common consumers	A separate board run for a range, a separate line in the account (chapter 21).
Opening balance	A plot's debt on the day the cooperative starts in the system — by hand or with an Excel sheet.

Electricity by reading

The days of an interval run from the day after the previous reading to the day of the new one, both included. When the tariff changes inside the interval, the kWh are split by days, each part times its rate, and the total is rounded once.

> EXAMPLES
>
> Readings 112.4 (22.09) and 205.7 (04.10): 93.3 kWh × 4.32 = **403.06 ₴** for 23.09–04.10 (12 days).
>
> Readings 1,000 (01.10) and 1,300 (31.10), tariff 4.32 until 16.10 and 5.00 from 17.10: 15 days each, 150 × 4.32 + 150 × 5.00 = 648.00 + 750.00 = **1,398.00 ₴**.
>
> A meter replacement gives two charges: 65 kWh = 280.80 and 66 kWh = 285.12.

Months and closing

- A month is open or closed. Months are closed only in order; a closed month never reopens.
- Before closing the board sees a check list: is the month posted, what was not charged, which meters have not been read for long. None of it blocks closing — a person decides with everything in view.

- A charge for an already closed month (a late reading, say) goes to the earliest open month, marked "for September, charged in October".

Reversals and the "not charged" queue

- Charges are never changed or deleted — the database itself enforces it. A wrong charge is cancelled by a **reversal** (the same amount with a minus and a reason). A whole losses or common-consumer run can be reversed together.
- What the system could not charge goes to the "not charged" queue with the reason: no tariff for some days; two tariffs of one type on a plot on the same day; no node readings. Once the cause is fixed, "Retry".

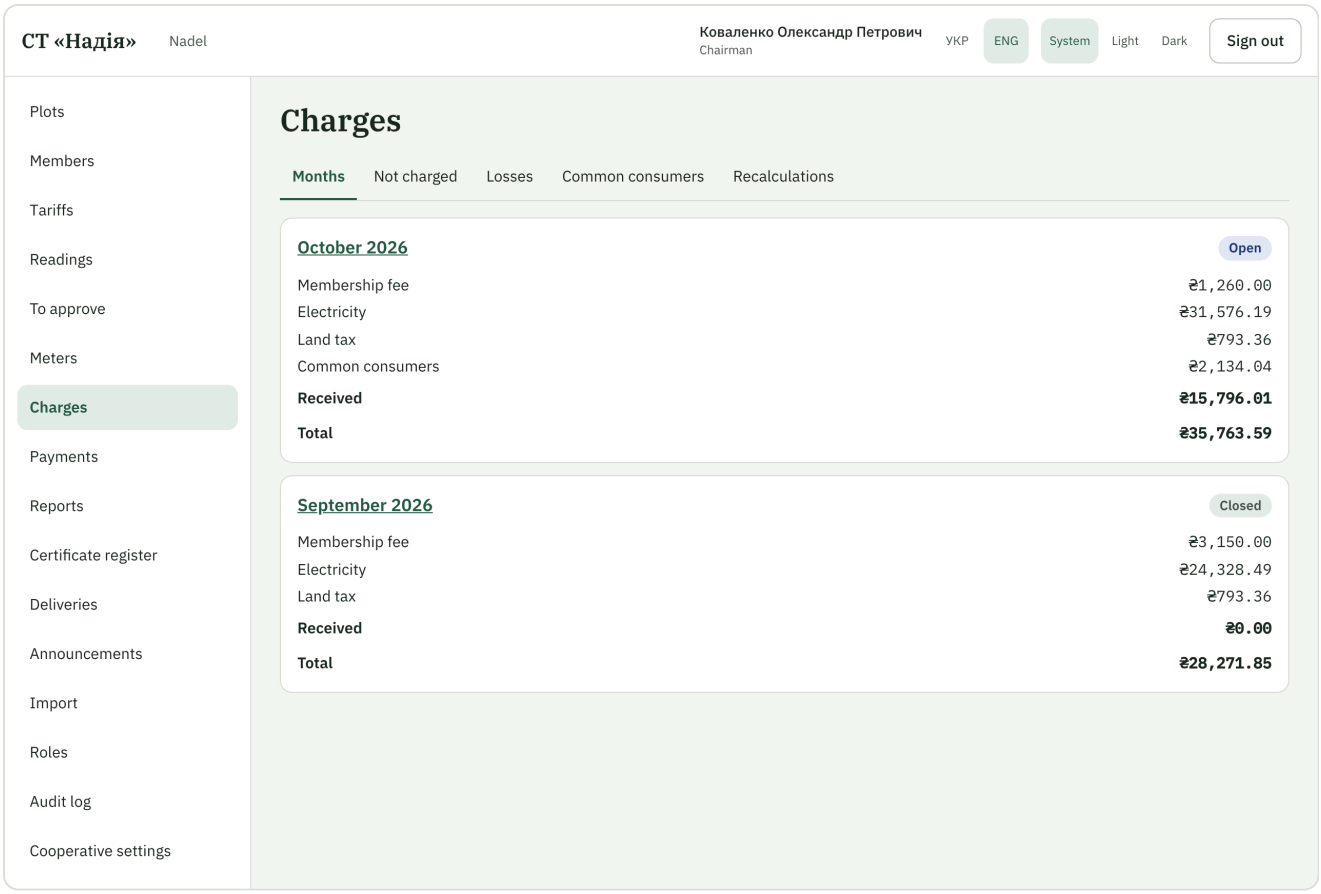
Debt

```
open amount of a charge = the charge with its adjustments (reversed: 0.00) − Σ
allocated to it
debt per charge type   = Σ open amounts of the charges of that type
plot debt              = Σ debts per charge type
prepayment             = Σ payments (net of reversals) − Σ allocated
```

A payment pays specific charges, so the debt is what is still unpaid. How a payment is allocated and what happens when a charge is reversed is in chapter 25.

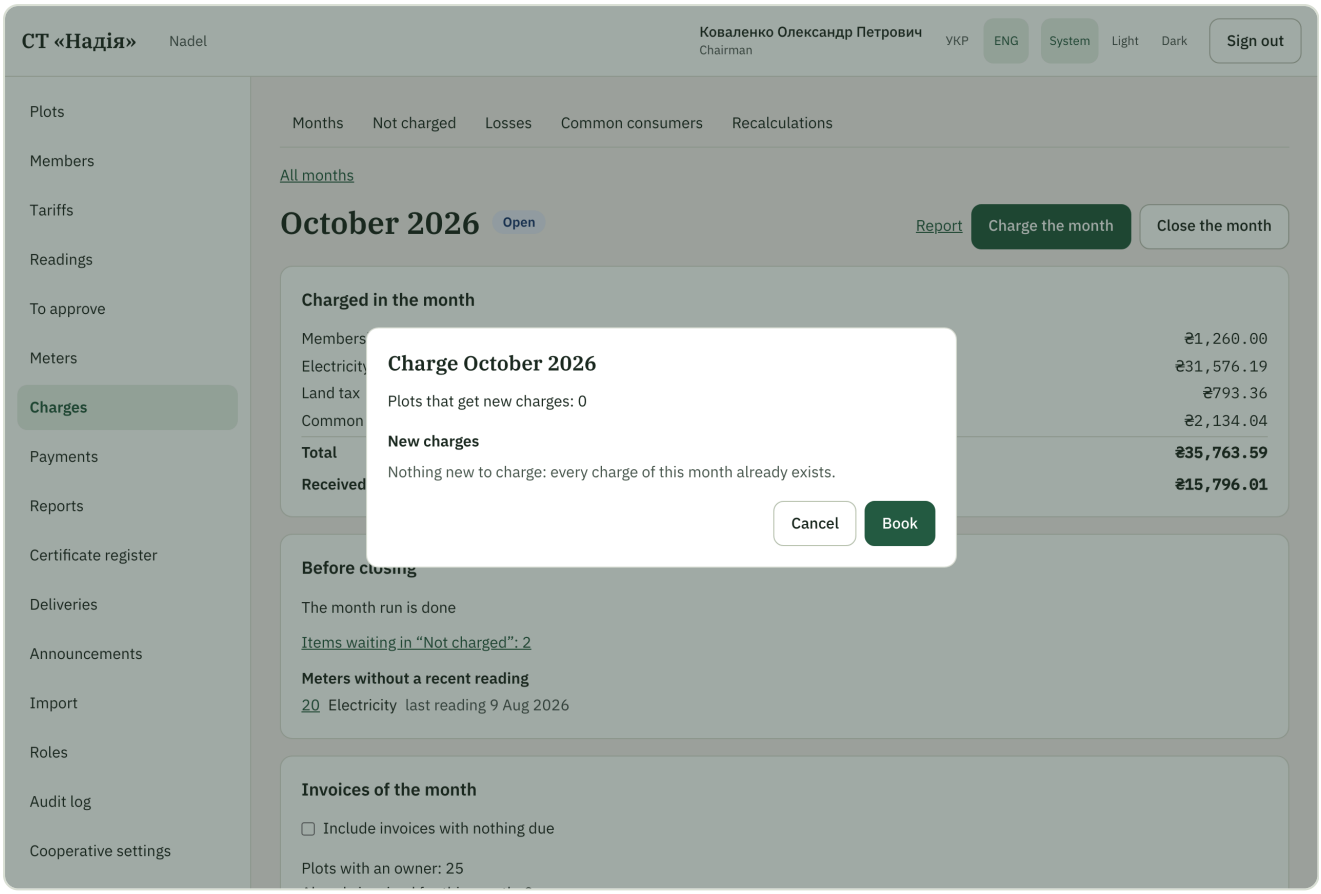
On the screen: "Charges"

The "Charges" menu item (permission "View charges") has four tabs: Months, Not charged, Losses and Common consumers.



"Charges": the months, newest first, with the total per charge type

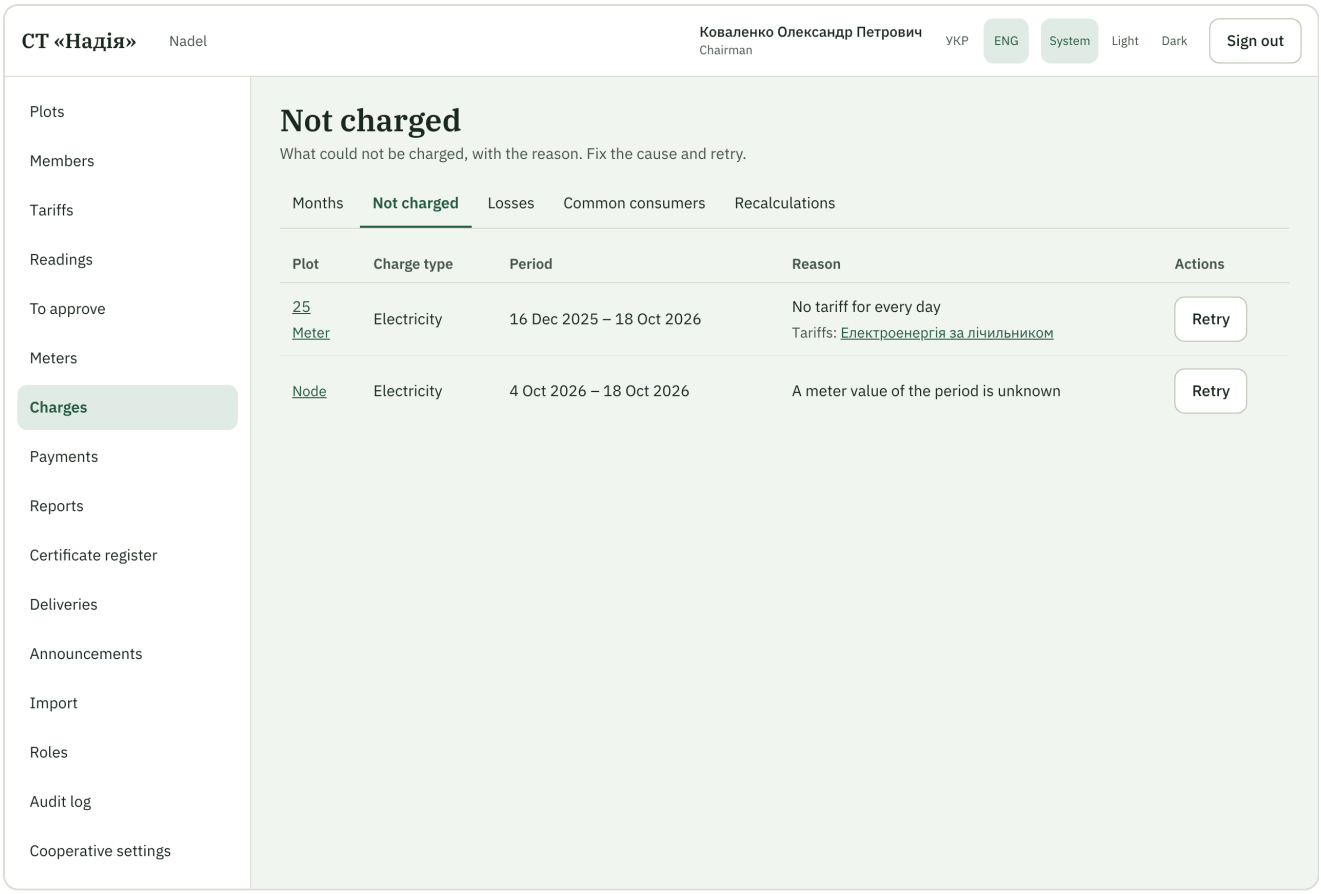
1. Open a month. The page shows what was charged per charge type and the check list ("Before closing"): is the month run done, how many items wait in "Not charged", which meters have no reading for more than 60 days.
2. "Charge the month" (permission "Run charges") first shows a **preview**: the number of plots, the new charges per charge type and the items that cannot be charged. "Book" books exactly what was shown. If anything changed meanwhile the system asks to preview again.
3. A rerun creates only what is missing: a plot or a tariff added later gets its charge, nothing is duplicated. A month after the current one cannot be charged.



The preview of the month run: nothing new, because the current month was already run

"Not charged"

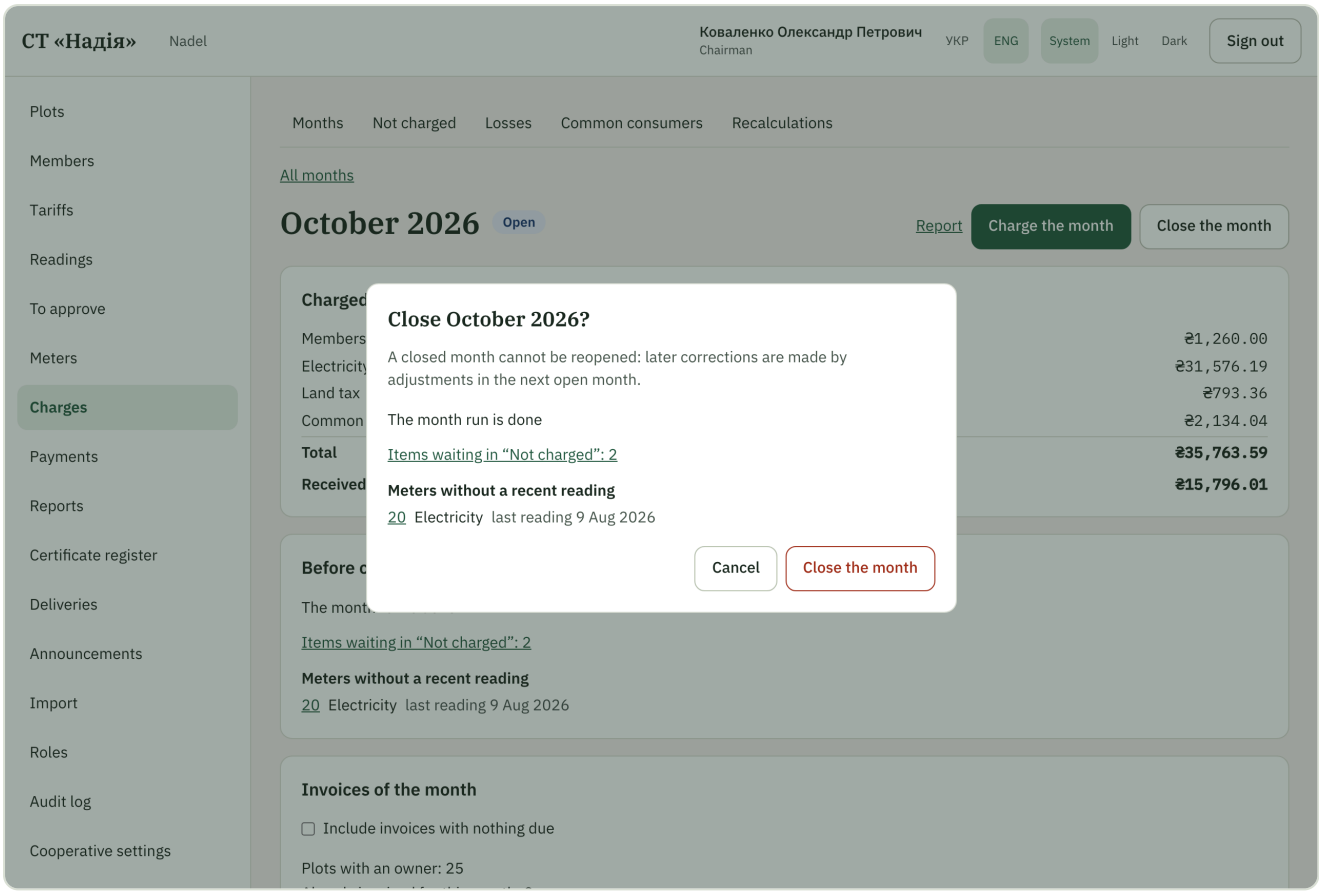
One list of everything the system could not charge, with the reason and a link to fix it: no tariff for every day, two tariffs on the same day, a plot outside every tariff, an unknown node or consumer value. "Retry" tries one item again; the next month run retries the tariff and reading items. An item disappears when the retry succeeds. Items never block closing a month.



"Not charged": the reason and "Retry" for each item

Closing a month

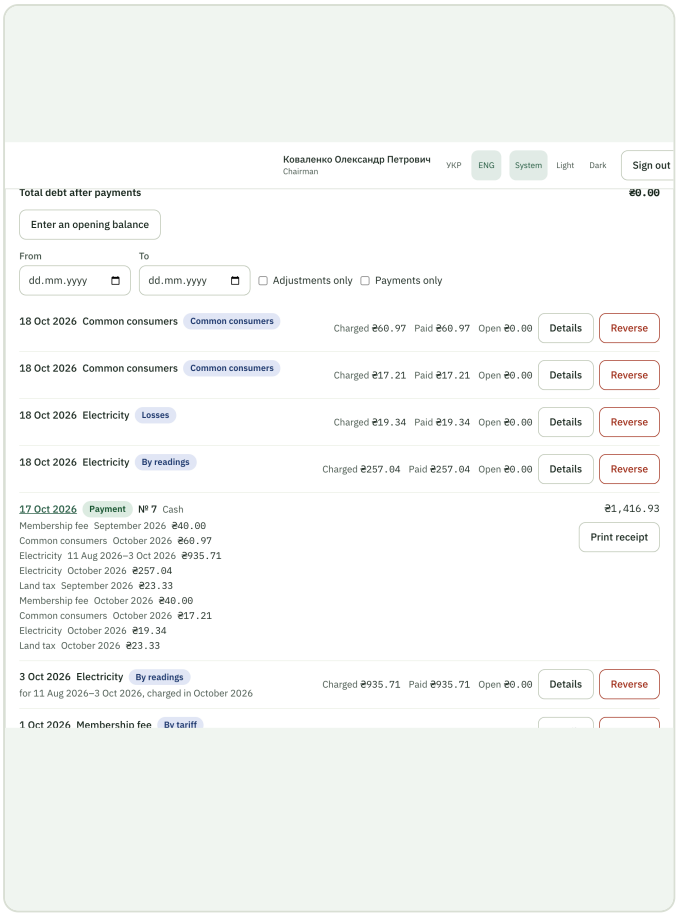
"Close the month" (permission "Close periods") shows the check list once more and asks to confirm. Only the earliest open month can be closed ("Close the earlier months first" otherwise). A closed month never changes; late entries go to the earliest open month.



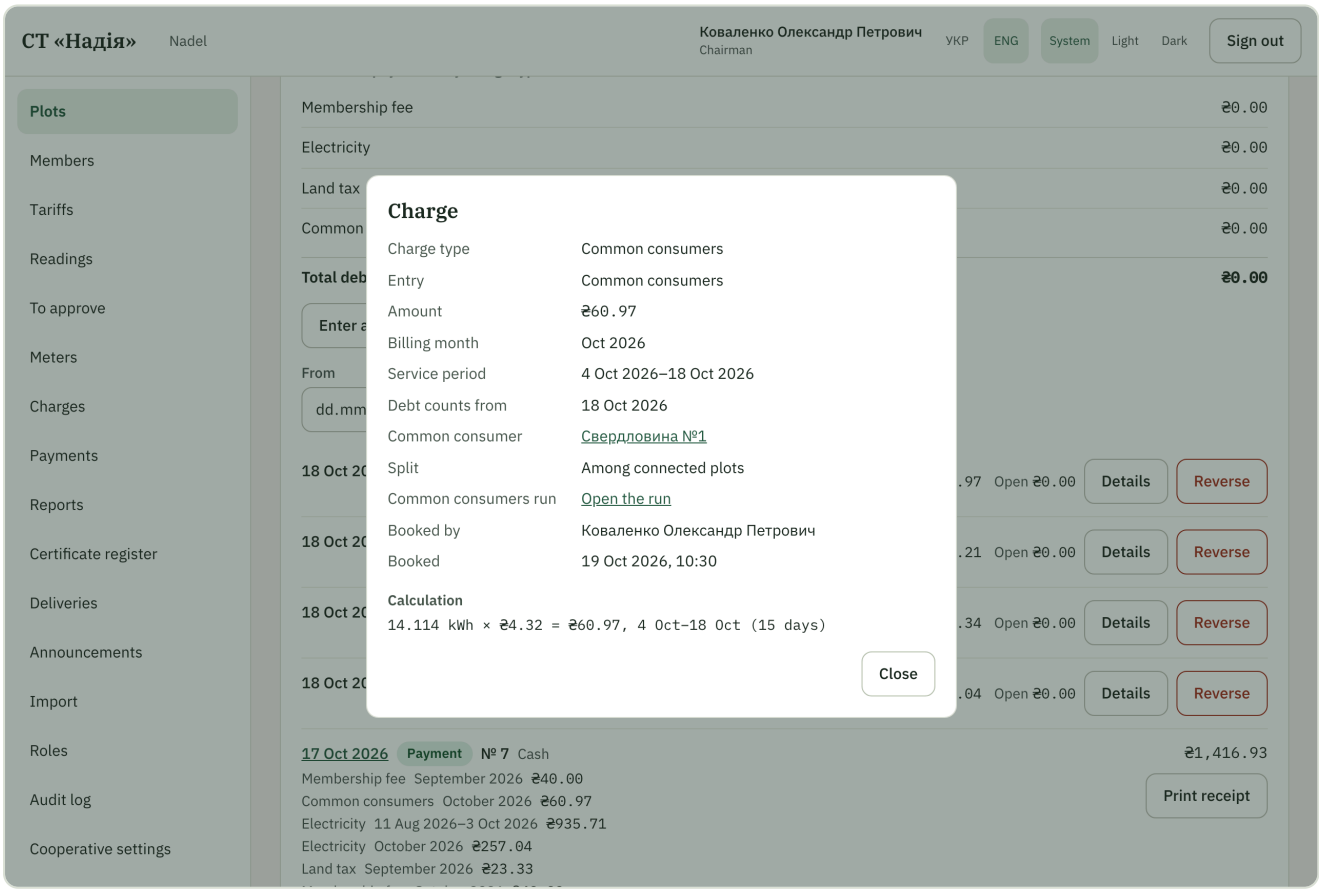
Closing: the check list stays in view, nothing blocks it

The plot account, reversals and opening balances

On the plot card the "Account" panel shows the debt per charge type and the charges, newest first, with a date filter. "Details" opens the calculation of a charge: kWh, rate, readings, days, the losses percentage. A charge booked later than the days it pays for reads "for 13.09–28.09, charged in October 2026".



The plot account



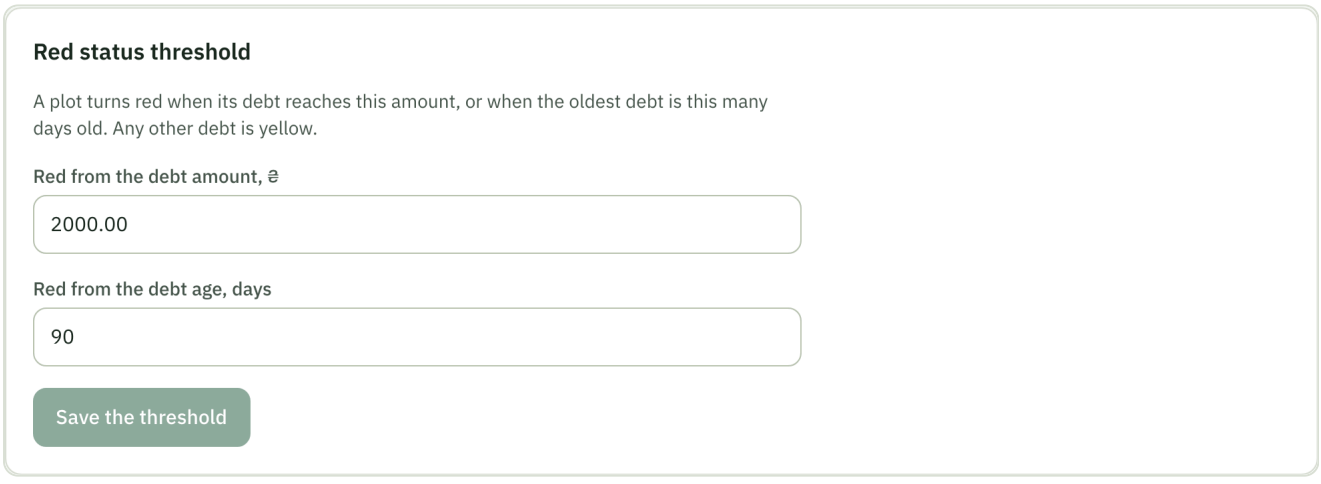
The calculation of a charge

- **Reverse** (permission "Reverse charges") asks for a reason and creates an entry for the same amount with a minus. A charge is reversed once; a reversal cannot be reversed; a charge in a closed month is corrected by an adjustment (a later phase), the button is not offered.
- **Enter an opening balance** (permission "Run charges"): charge type, amount, "debt counts from" (not later than today) and an optional reason. Many plots at once: the "Opening balances" sheet of the Excel import (chapter 7). One opening balance per plot and charge type; to change it, reverse it first.
- Reversing a whole losses or common consumers run is done on its page (below).

Plot colours on the map and in the list

Colour	When
Red	The debt is above the "red from amount" or the oldest debt is more than "red from days" old (strictly greater)
Yellow	Any other debt
Green	No debt
Grey	No owner, or common land

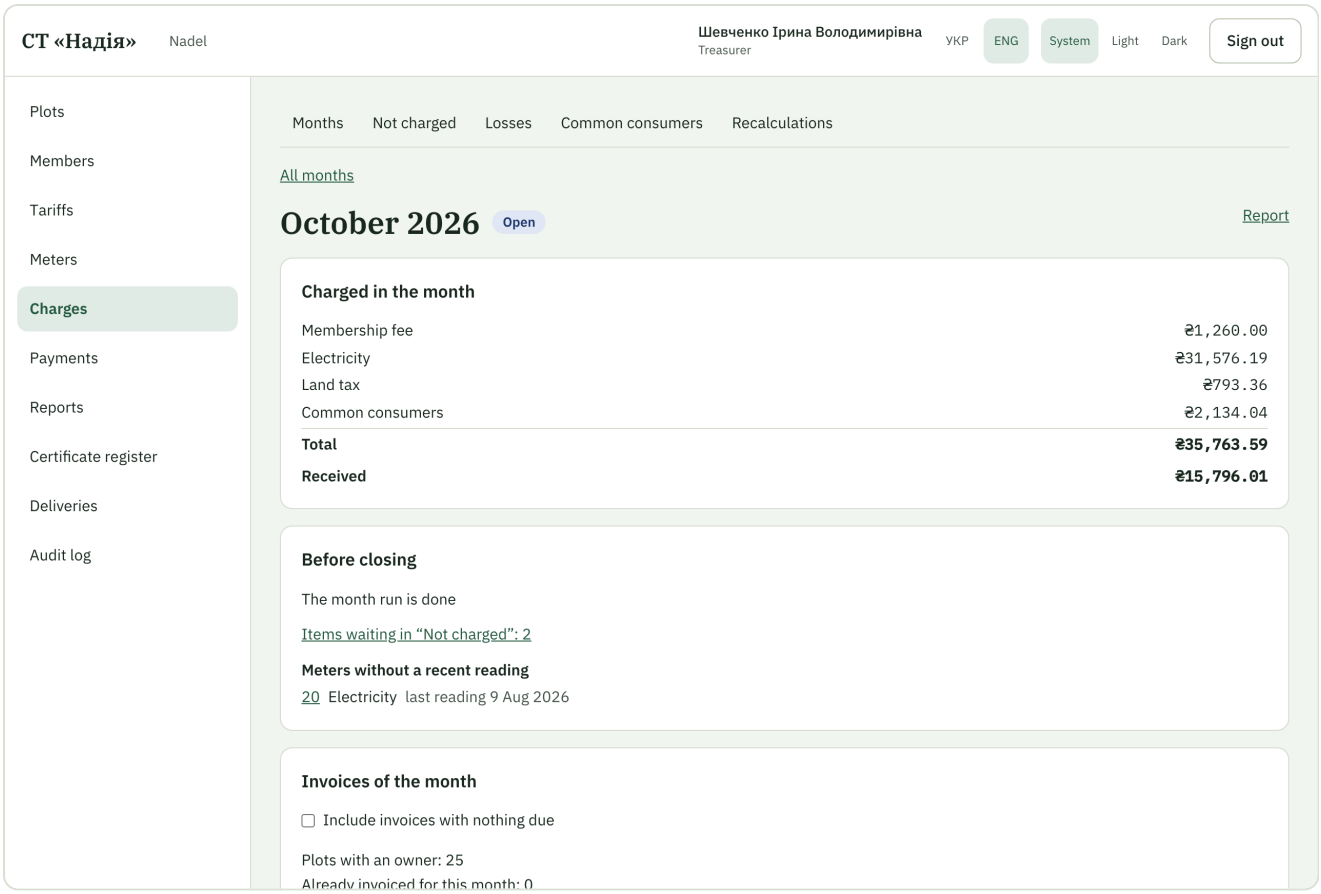
The thresholds are set in "Cooperative settings" (defaults 1,000.00 ₴ and 90 days; the demo cooperative uses 2,000.00 ₴ and 90 days).



The red status threshold in the settings

The treasurer and the member

The treasurer sees months, accounts and the lists but has no buttons. A member sees only the account of their own plots under "My charges" (chapter 12), read-only, with the calculation of every charge; other plots' amounts are never shown.



What the treasurer sees: no actions

21 Network losses and common consumers

works

Losses: two modes, the chairman chooses

- **Fixed percentage**: every electricity charge gets, for example, 5 % kWh on top. 65 kWh → losses 3.25 kWh = 14.04 ₴.
- **Proportional by node**: the board chooses a range and runs the calculation. For every node, bottom-up: losses = node consumption − the sum of its children's consumption. The losses are shared among the plots under the node in proportion to their consumption.

> EXAMPLE: A QUARTER AND THE SUBSTATION
>
> Quarter A: 1,100 kWh, plots P1 = 600 and P2 = 400 below it → losses 100: P1 60 kWh (259.20 ₴), P2 40 (172.80 ₴).
>
> Quarter B: 1,050, plot P3 = 1,000 → losses 50 → P3.
>
> Substation: 2,300; children A 1,100 + B 1,050 + well 50 = 2,200 → losses 100 shared 600 / 400 / 1,000 of 2,000: P1 30, P2 20, P3 50 kWh.

- **Negative losses** (the plots together "used" more than the node shows: a wrong reading, different reading dates, a faulty meter): nothing is charged, the board sees a warning with the numbers.
- **An unknown value** (no reading of the node or a child after the range end): that node and the nodes above it are skipped with a warning; the rest of the tree is charged.
- Two runs cannot cover the same days of the same node.

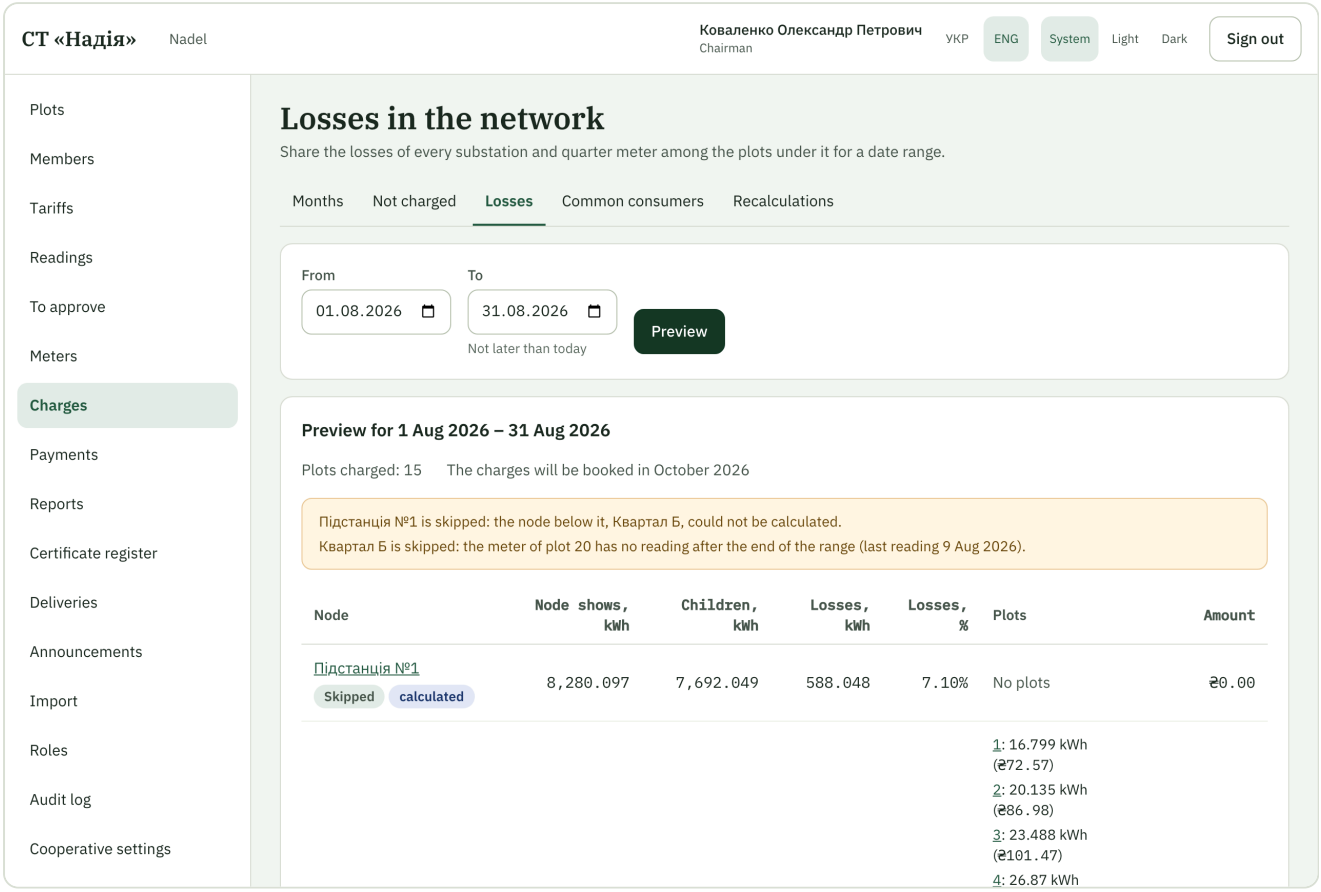
Common consumers — a separate line

Well pumps, street lighting, the office. Their consumption is a separate charge type "Common consumers" with its own per-kWh tariff and its own board run. The split is chosen per consumer:

Split	Who pays	Example
Connected plots	Plots connected to that well, for their connected days	Pump 60 kWh, three connected plots: 20 kWh each = 86.40 ₴; an unconnected plot nothing
Equally	Every active plot, with or without a meter	Lighting
Service users	The plots listed in the settings	Lighting 30 kWh, 3 plots: 10 each = 43.20 ₴
Not charged	Nobody — covered by the membership fee	Shown in reports

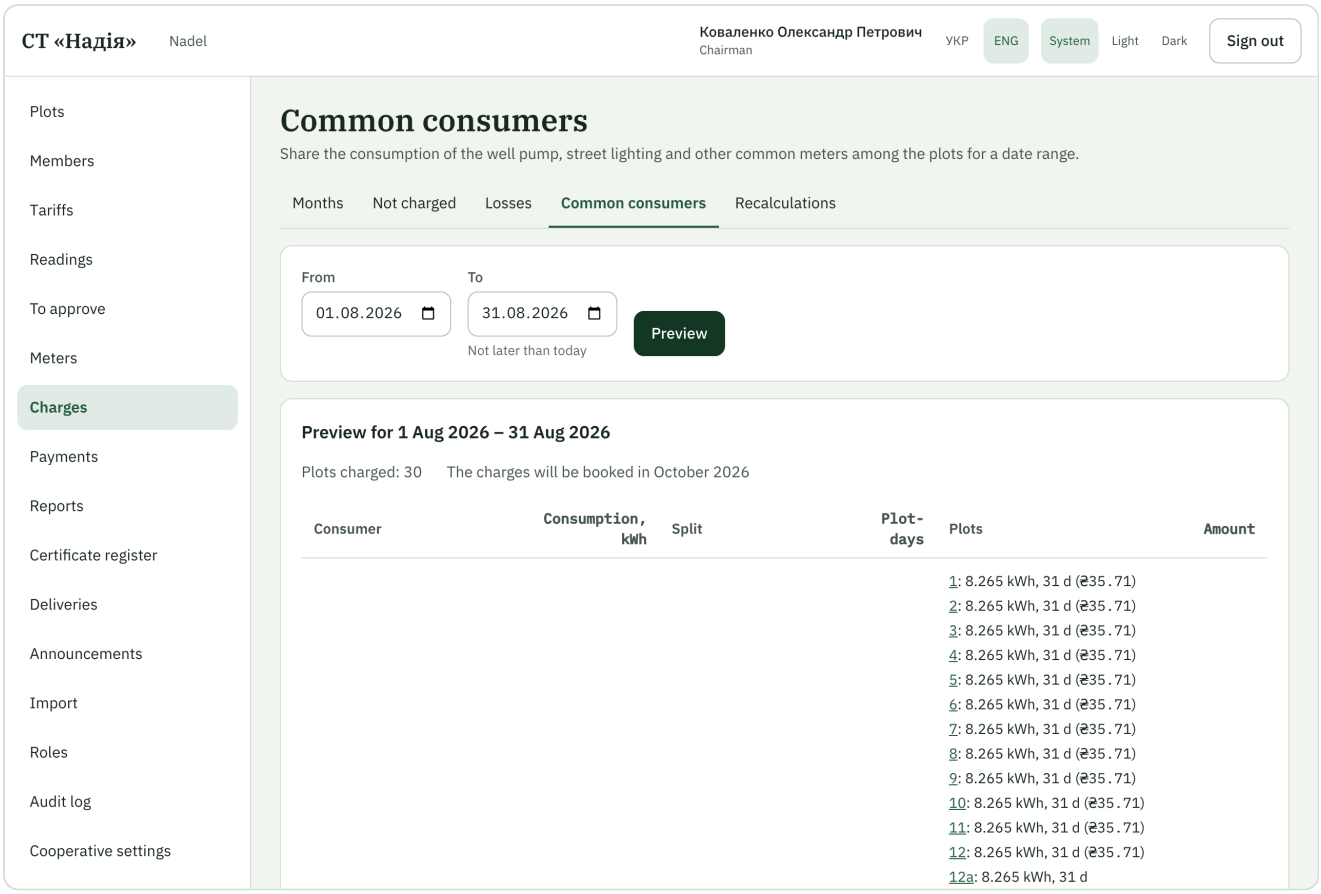
On the screen: "Losses" and "Common consumers"

Both tabs of "Charges" work the same way: pick the range "From" and "To" (not later than today), press "Preview", read the table and warnings, then "Book" (permission "Run charges"). The booking is written in the month of the "To" day, or in the earliest open month after it. The history of past runs is below, with "Details" and "Reverse the run" (permission "Reverse charges", a reason is required).

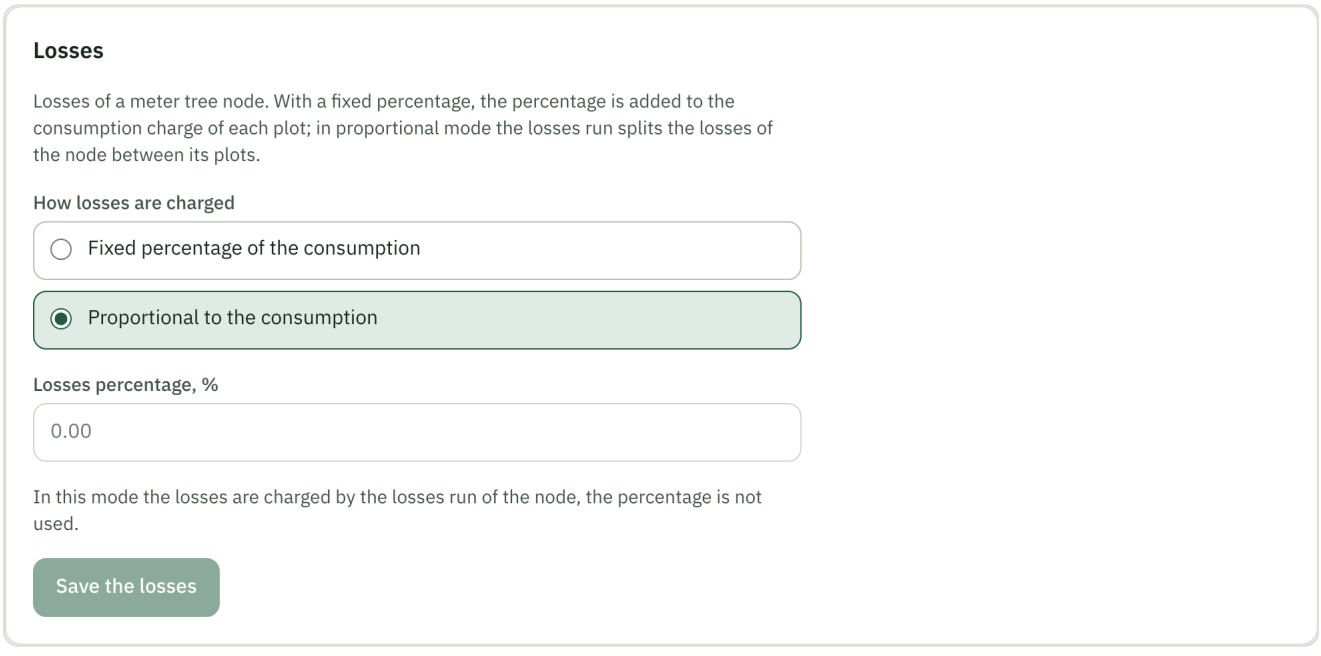


A losses preview: a node with an unknown value is skipped with a warning

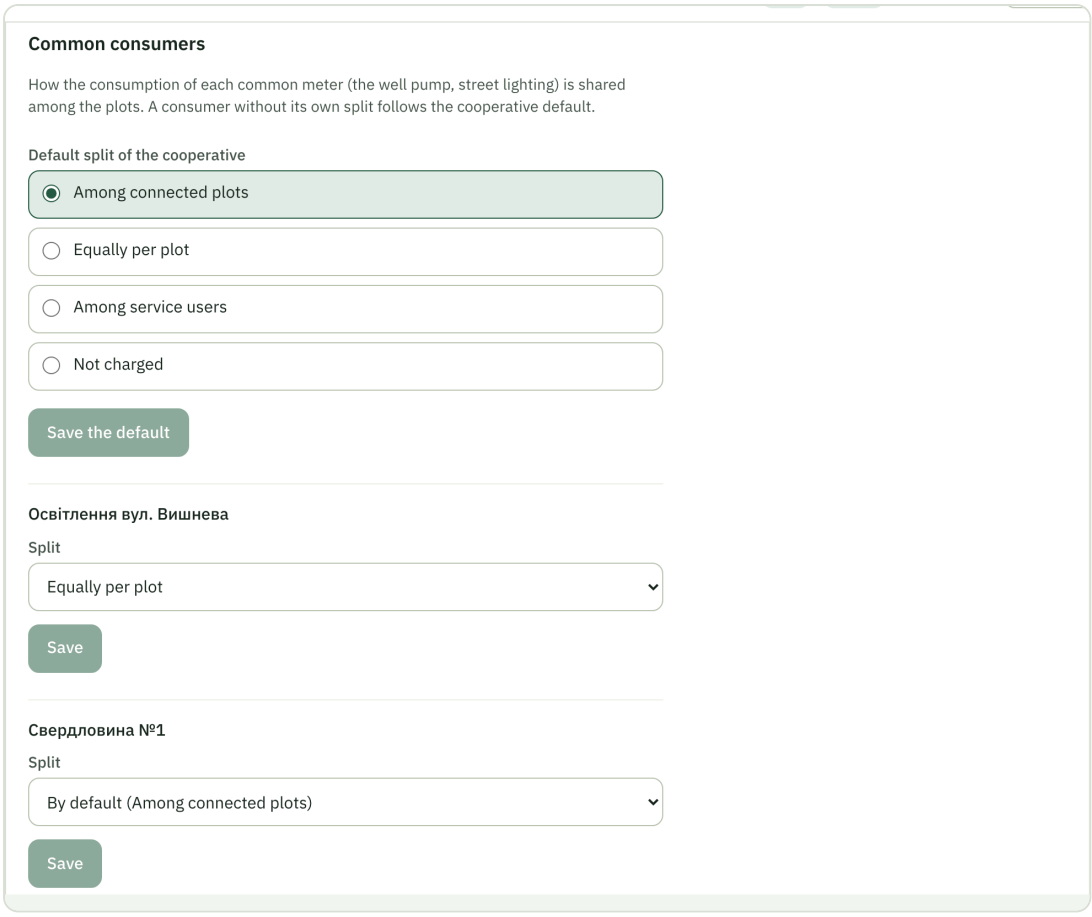
- The losses mode is chosen in the settings ("Losses"): with a fixed percentage there is nothing to run here and the page says so.
- If some nodes were already charged for these days the whole run cannot be booked; "Preview without the charged nodes" books the rest.
- A node with an unknown value goes to "Not charged"; when readings arrive, "Retry" charges it alone.
- Common consumers need a tariff of the "Common consumers" charge type; without it the preview says "No tariff" and nothing can be booked. The split of each consumer, its service users and the default are in the settings.



A common consumers preview



The losses mode in the settings



The split of each common consumer

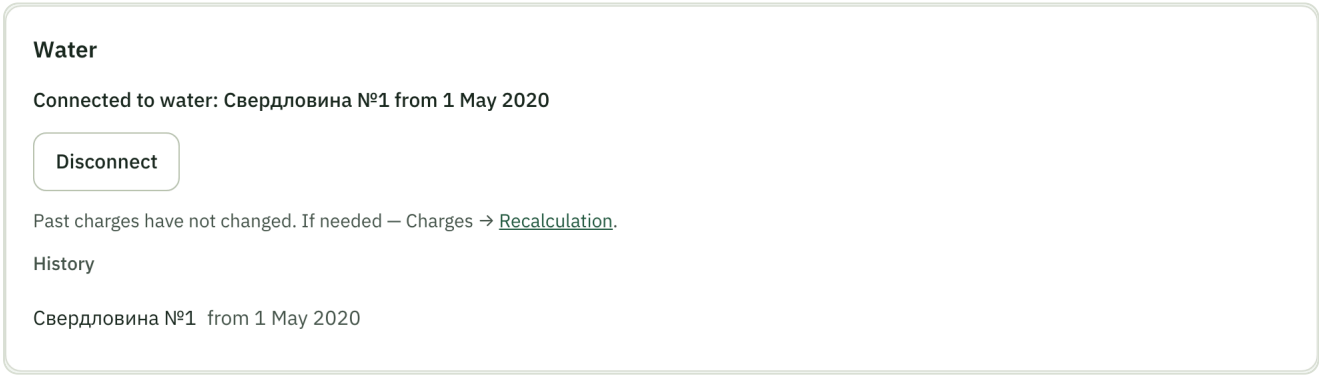
22 Water and connections works

Water works like electricity: **a plot not connected to the well pays nothing for water**.

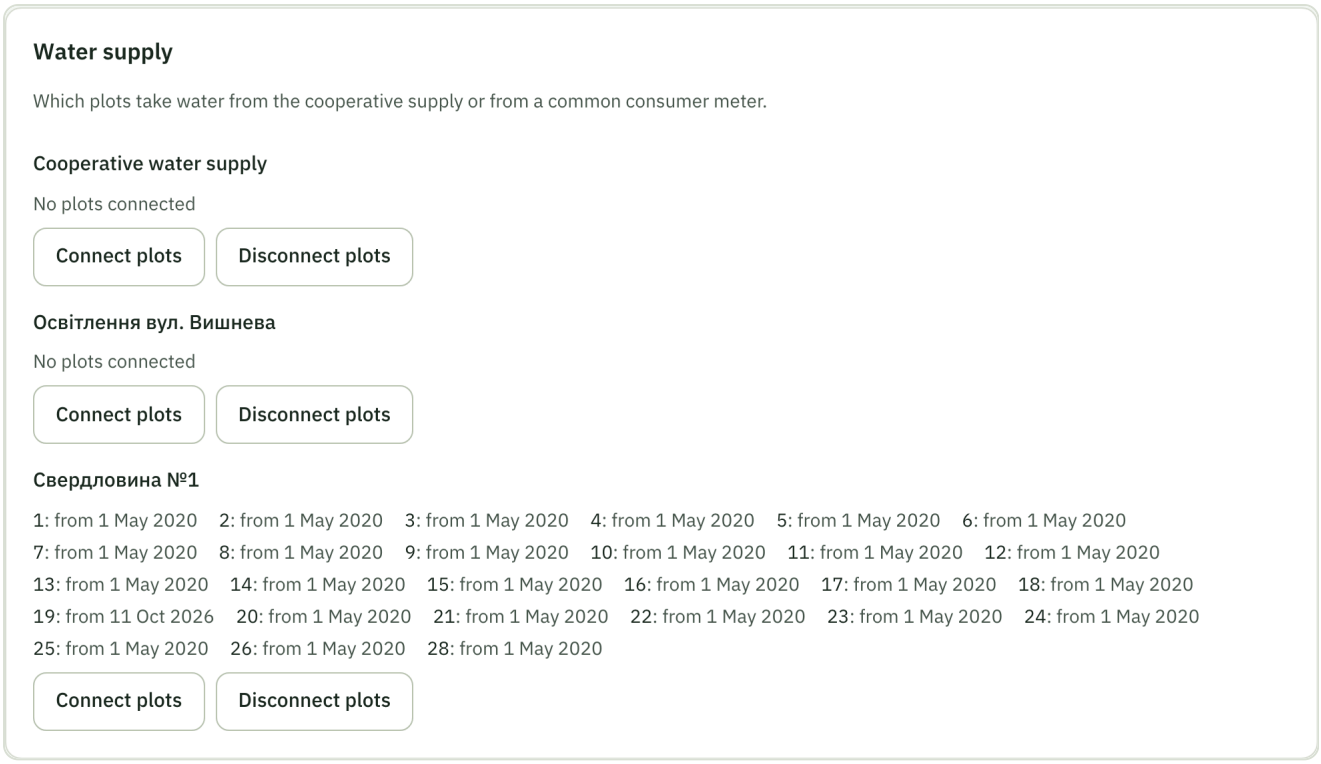
- There may be several wells: each plot is connected to a particular well, and each well's pump is shared only among its own plots.
- A plot has a history of water connections: the source (a well or "the cooperative's water supply"), from a date, until a date.
- Any water tariff — by water meter (m³) or fixed (per plot per season, say) — is charged only for the connected days. A per-m³ tariff also needs a water meter.
- A connection starting or ending mid-month splits the amount by days: 300 ₴ per year, connected from 16.10 (31 days) → 300 / 12 × 16 / 31 = **12.90 ₴**.
- The well pump's electricity is shared only among the plots connected to it, by plot-days: P1 and P2 the whole month, P3 from 16.10, pump 60 kWh → 30 + 30 + 15 = 75 plot-days → P1 24, P2 24, P3 12 kWh.
- Connections are entered on the plot card, in bulk on the well's page, or with the "Water" column of the import.

On the screen

A plot's connection is on its card, panel "Water": "Connect" (source and the first connected day) and "Disconnect" (the last connected day); the history is listed below. Many plots at once are connected or disconnected on the "Water supply" section of "Cooperative settings", per source. Without a connection a plot is never queued in "Not charged" for water and pays nothing for it.



The "Water" panel of a plot



The water sources and their plots in the settings

TWO THINGS TO KNOW

- If a well meter has been removed, its lasting connections are no longer offered in the settings: end them on each plot's card.
- A plot with a lasting water connection cannot be archived or made common land: end the connection on the plot card first.

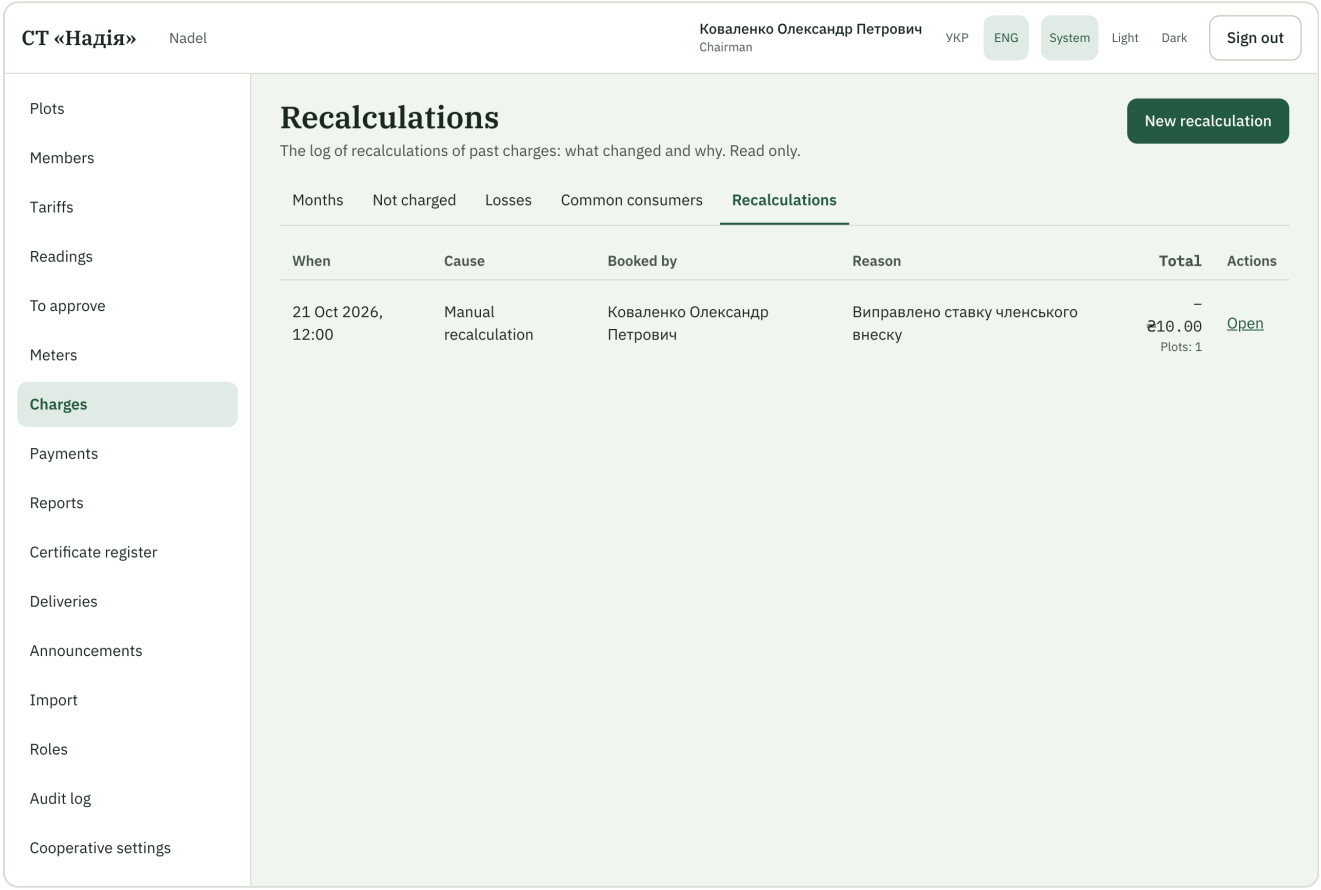
23 Corrections and recalculation works

A back-dated tariff version, a correction of a used version or of a used reading — one mechanism:

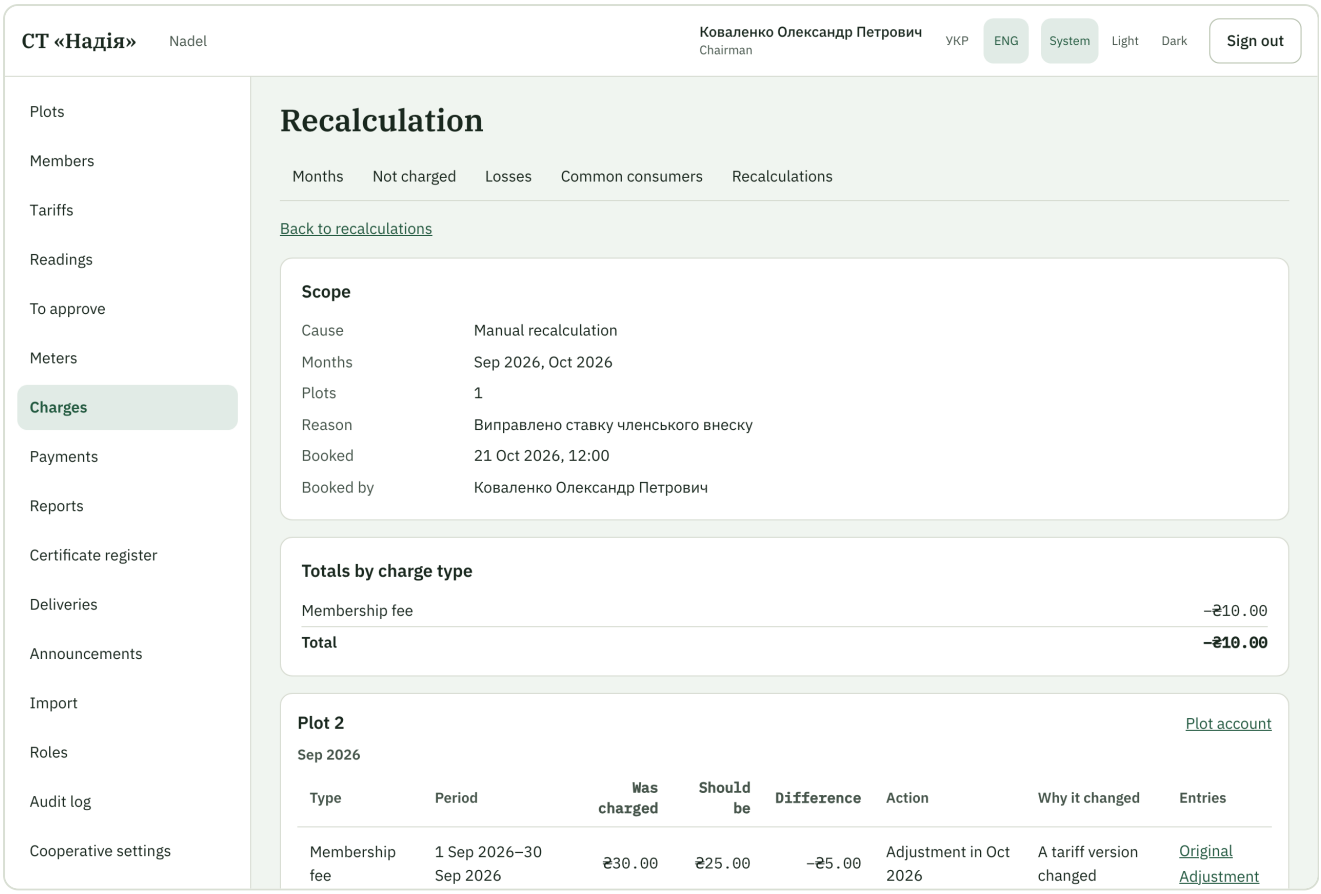
- **Open month**: the old charge is reversed and a new one created.
- **Closed month**: nothing in it changes; the current open month gets an adjustment for the difference, referring to the original month.
- A reason is required. Before posting, a preview: how many plots and how much.
- If a debt goes down after it was already paid, the excess becomes a prepayment, never a "negative debt".

On screen

Recalculations is a tab of "Charges" (viewing needs "View charges"). It is a log: when, the cause (manual, tariff version, reading), who booked it, the explanation and the "Total" difference. "Open" shows one recalculation: the scope, the totals by charge type and the lines "was charged → should be → difference", the action ("Reversal and new charge", "Adjustment") and the entries created, with a link to the plot account. A recalculation is never changed or undone: a mistake is corrected by the next recalculation.

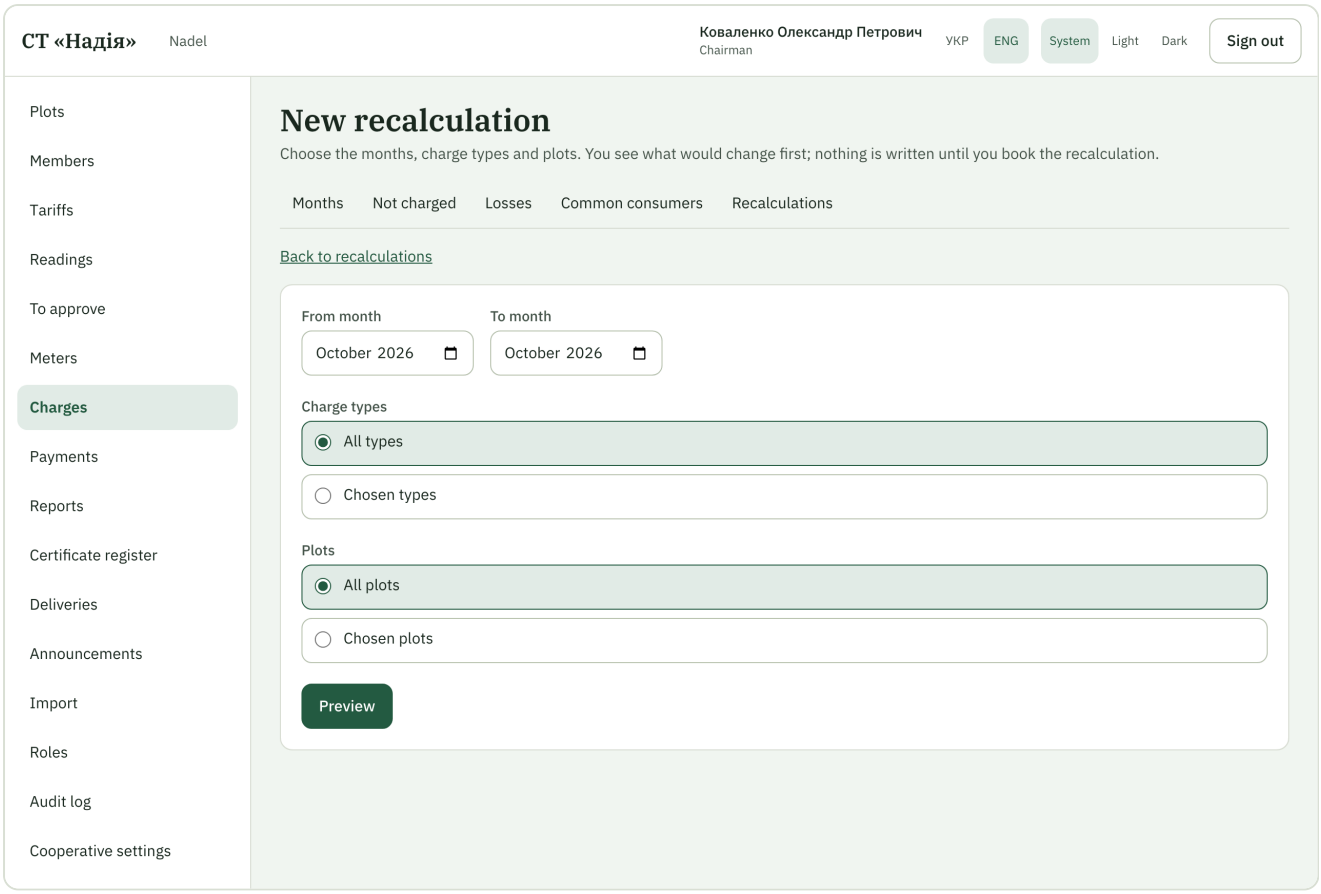


The log of recalculations

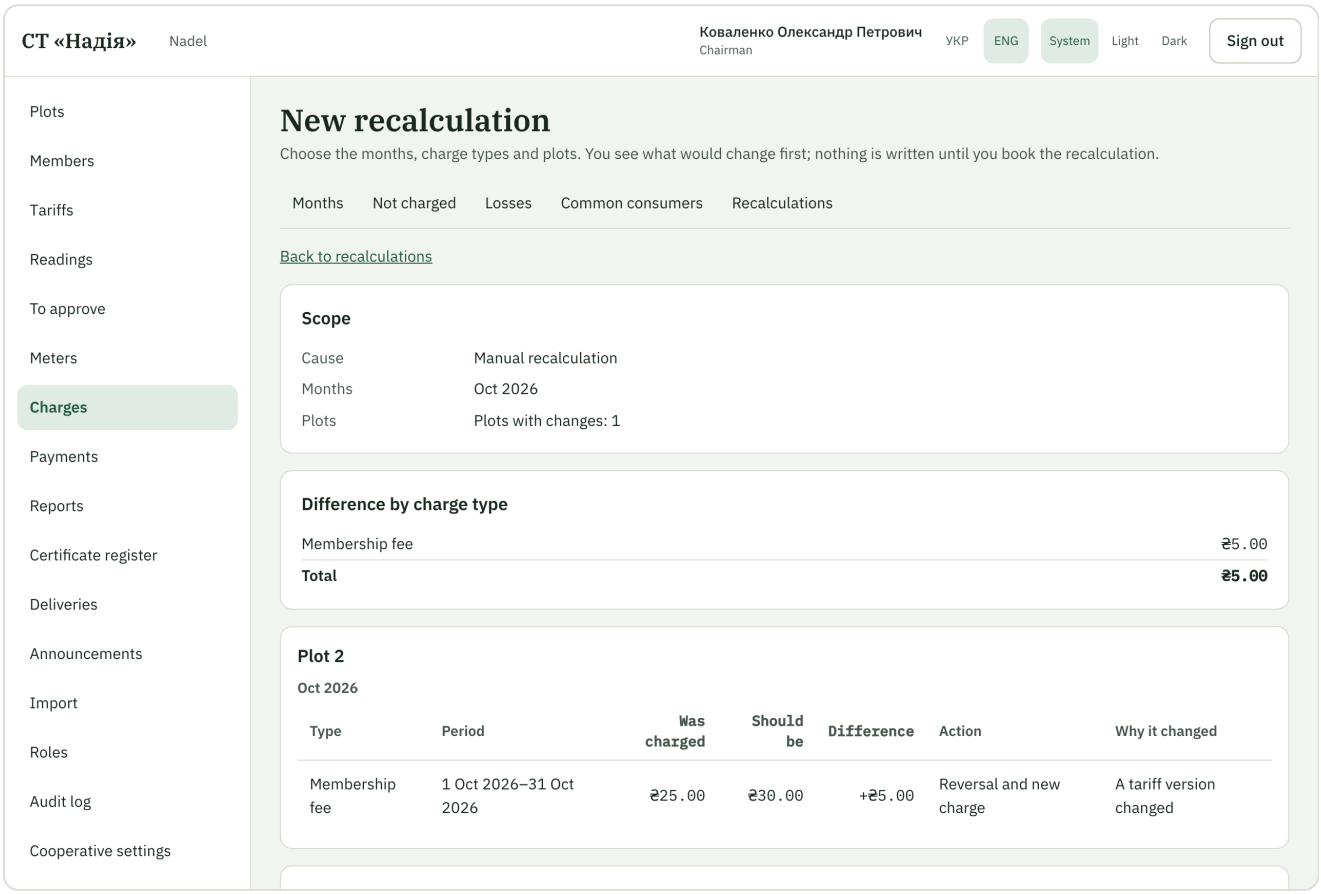


One booked recalculation: the adjustment of the closed month and the reversal of the open one

New recalculation ("Recalculate charges", the chairman only): choose "From month" and "To month" (not later than the current month, at most 24 months), the charge types and the plots ("All" or "Chosen"), and press "Preview". The preview step shows what would change: the plots, the totals by charge type and the lines with their reasons ("A tariff version changed", "The plot area changed"…). Nothing is written until you enter a **reason** (required, up to 500 characters) and press **"Book the recalculation"**; exactly the preview you saw is booked. If the charges changed in the meantime, the preview is refreshed and must be confirmed again. When the charges already match the data, the preview says "No changes" and there is nothing to book.

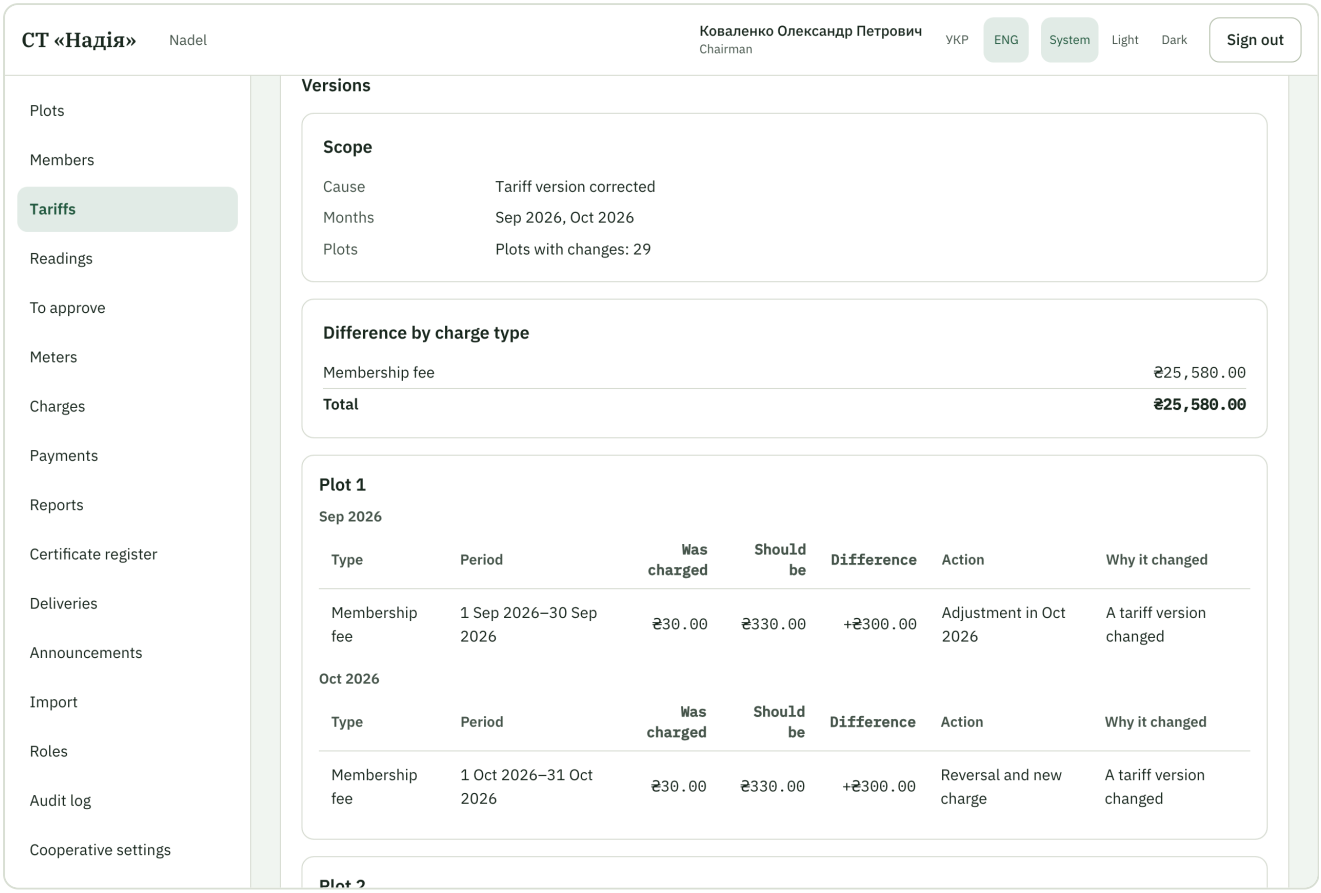


The form of a new recalculation



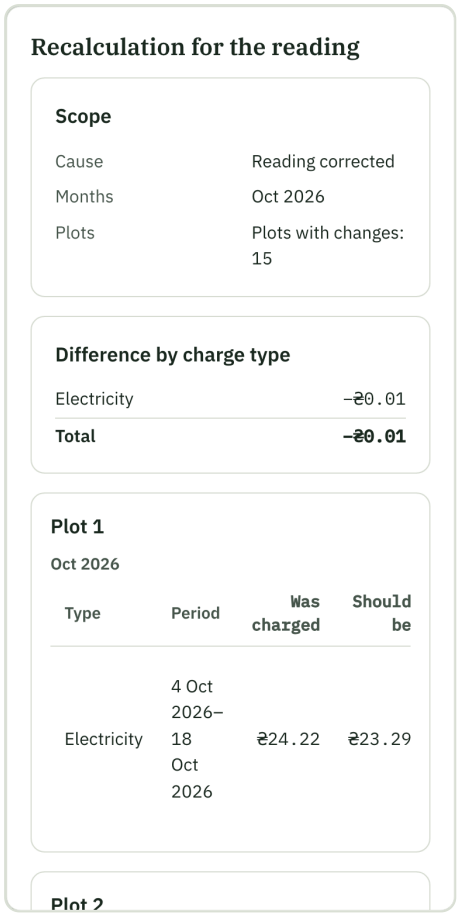
A second preview: "No changes"

Correcting a used tariff version is done on the tariff card: "Edit" or "Delete" the version. When charges already use it, the same preview step opens instead of "Version corrected": the change of the version and the recalculation are saved together or not at all. The same goes for a new back-dated version.

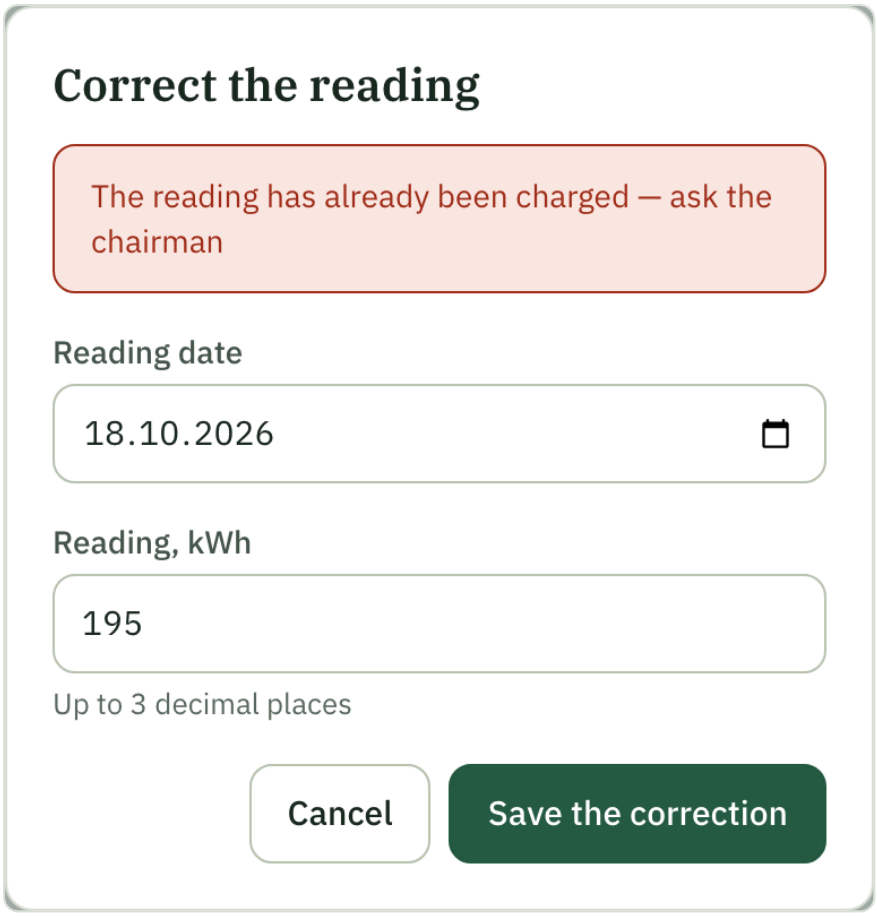


The preview step after a rate correction

Correcting a charged reading is done on the meter card: "Correct" or "Delete". The chairman gets the preview step ("Recalculation for the reading") with a reason; the electrician (may enter readings, may not recalculate) sees "The reading has already been charged — ask the chairman" and nothing changes.

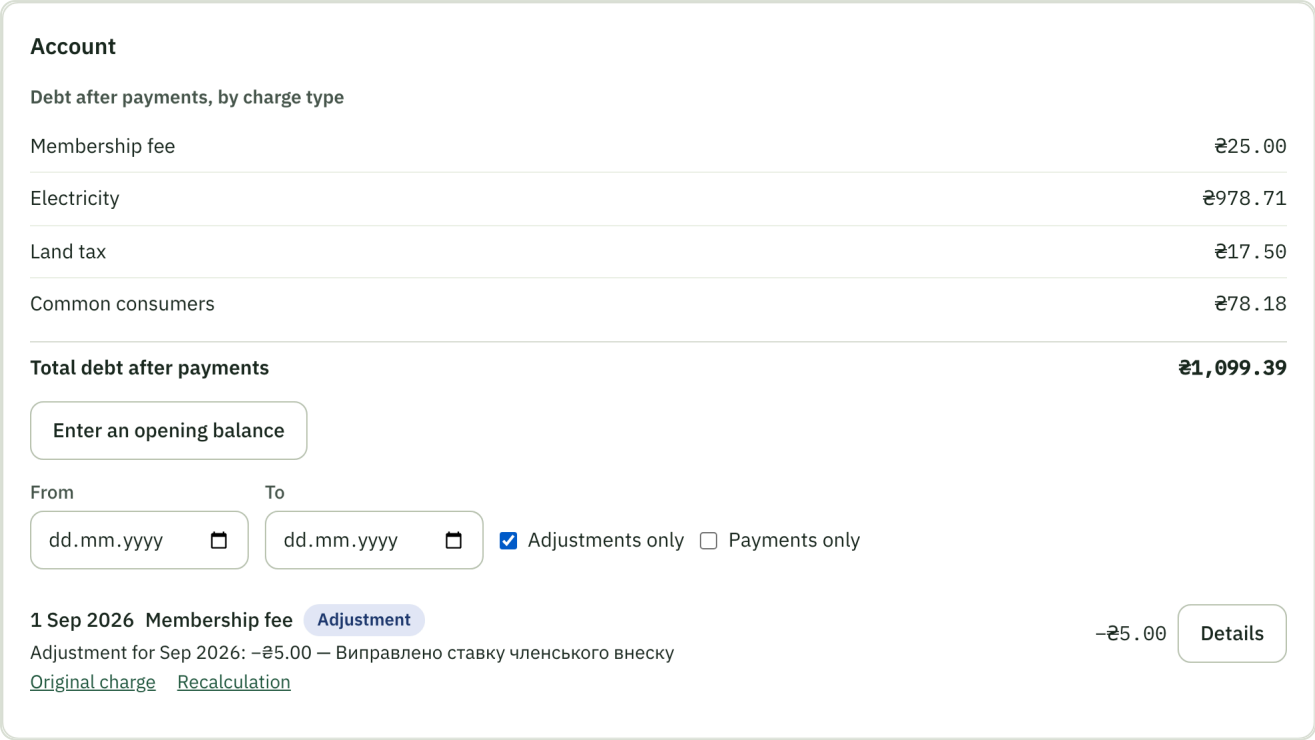


The preview step for a corrected reading



The electrician cannot correct a charged reading

On the plot account the adjustment of a closed month is a separate row "Adjustment for Sep 2026: –₴5.00" with the reason, the links "Original charge" and "Recalculation" and the badge "Adjustment". The checkbox **"Adjustments only"** keeps only such rows. A member sees the adjustments of their own plots (the reason, but not the recalculation). The demo data has such a row on plot 2.



Plot 2: the September adjustment of −₴5.00

Screens that change data affecting past charges (plot area, tariff scope, water connection, moving a meter in the tree) recalculate nothing themselves and remind you: "Past charges have not changed. If needed — Charges → Recalculation". Only those who may recalculate see the hint.

EXAMPLE

From 1 September 60 ₴/sotka/year was entered by mistake instead of 50; a 6-sotka plot. September is closed, October open.

	Was	After the correction
September (closed)	30.00	Unchanged; in October an adjustment −5.00 "for September"
October (open)	30.00	Reversal −30.00 and a new charge 25.00
Total	60.00	50.00; if 60.00 was paid, a prepayment of 10.00

KNOWN LIMIT

"To approve" has no recalculation step. If a reading is already approved and charged, correct it on the meter card, which has the preview step.

24 Reports and templates works

Reports put into one table what the system already knows: charges, payments, debts, electricity, water, corrections. Every report is shown on screen and can be downloaded as an Excel file. Besides the built-in reports, the chairman can upload their own Excel form (a template) and the system fills it in with the cooperative's data.

REPORTS INTRODUCE NO NEW RULES

A report calculates nothing "its own way": it only selects, groups and adds up the amounts already booked under the rules of chapters 16-23 and 25. So any amount in a report can be reconciled to the kopeck with a plot account or the billing months screen. Amounts in reports are always **effective** amounts: charges plus adjustments minus reversals.

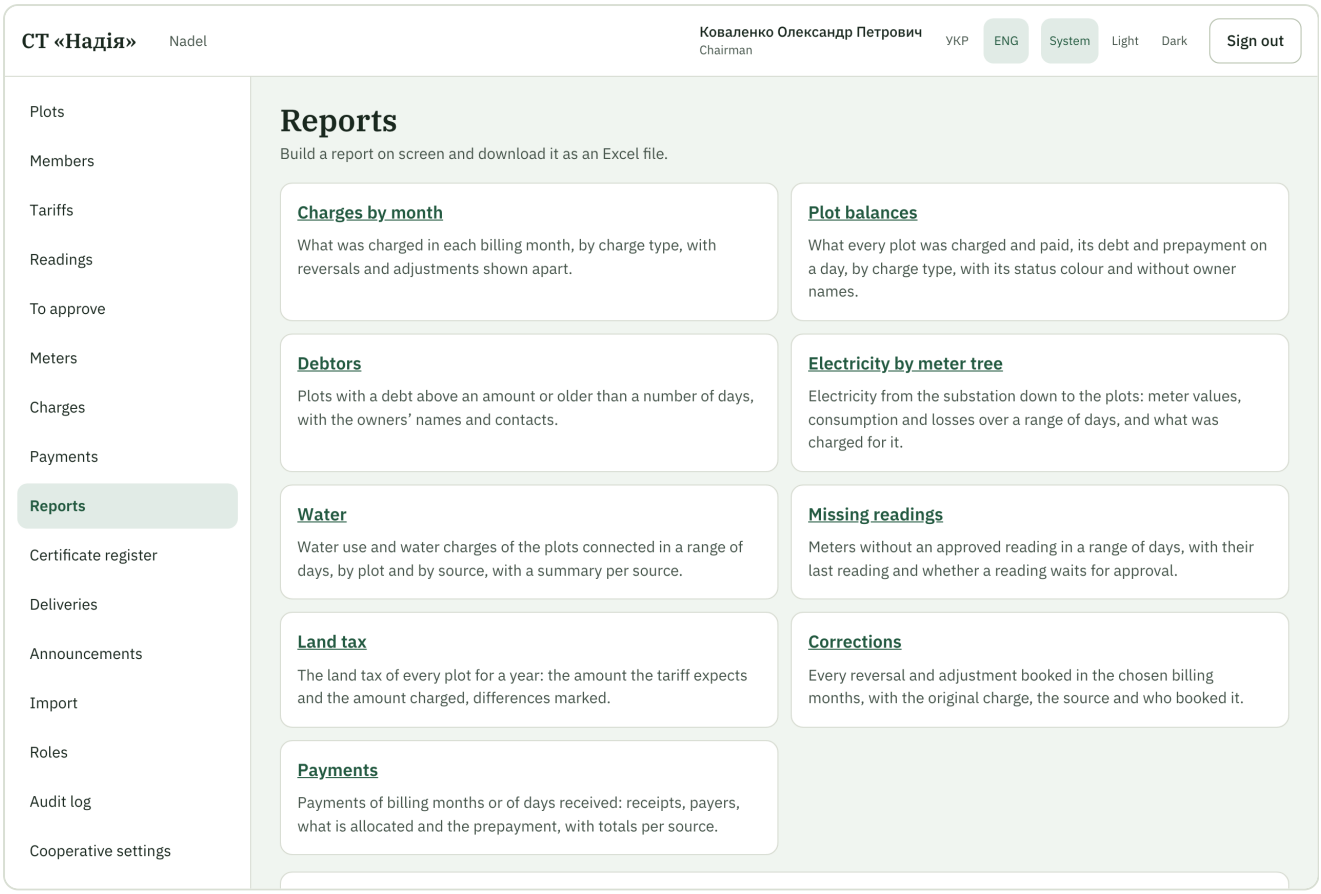
Who sees reports

What	Permission	Who has it by default
Open and download reports, generate files from templates	View reports	Chairman, treasurer
The "Debtors" report (this permission is needed as well)	View members	Chairman, treasurer
The "Payments" report (this permission is needed as well; the note about the payer is shown only to someone who also has "View members")	View payments	Chairman, treasurer
Upload, replace, download and archive templates	Manage report templates	Chairman only

A member and the electrician have no "Reports" item in the menu, and a direct address shows a message that they have no access. You can give your own role these permissions in "Roles" (chapter 3). A role with "View reports" but without "View members" sees every report except "Debtors".

The "Reports" screen

Menu → **Reports**. At the top are the cards of the reports available to you, below them the templates. A card has the report's name and one sentence about it; a click opens the report.



The "Reports" section: report cards and templates

- **Filters** are at the top of a report page; press **"Show"** to refresh the table. Under the filters it is written what the period counts by: *billing months* (the month a charge belongs to in the ledger), *service days* (the days a service was provided) or *reading dates*. They differ: an adjustment booked in March for January stands in March in "billing months" and in January in "service days".
- **Preview**: the screen shows the first 200 rows and the totals; the line "Showing first … of … rows. The Excel file has all of them" tells how many rows the report has in all.
- **"Download Excel"** gives a file with all rows and the same totals. It has one sheet per report (some reports have two), a header block (cooperative, report name, filters, date and time in Kyiv time, who made it) and a frozen header row; numbers are stored as numbers and dates as dates. The file is in the interface language. The file name is the report code and the period in Latin letters, for example `land-tax_2026-01-01_2026-12-31.xlsx`.
- **The limit**: a report with more than 20 000 rows on a sheet is not produced; you see "The report has more than 20 000 rows on a sheet. Narrow the filters and try again".
- A value that is not known (for example a reading at the start of the period) is shown **empty** with the reason, never as zero. "Plot balances" and "Charges by month" have "Paid" columns (the allocation of payments), "Plot balances" also "Prepayment"; the separate "Payments" report lists the payments themselves.
- Every report that is made is recorded in the audit log: who, which report, the filters and the row count; the content is not recorded.

The eight reports

Charges by month	What was charged in each month, by charge type. Filters: "From month", "To month", charge types, street. A row is a month and a type: "Plots", "Charged", "Opening balances", "Reversals", "Adjustments" and "Effective total" (the amount you see on the month screen). The second sheet, **"By service period"**, groups the same money by the month of the service.
Plot balances	The debt of every active plot by charge type as of a date ("As of", today by default), with the oldest unpaid charge, days overdue and status (red, yellow, green, grey). There are no owner names. For a past date the status is worked out with today's thresholds; the header says so.
Debtors	Plots with a debt from an amount ("Debt from", by default the cooperative's red threshold) *or* older than a number of days ("Older than, days"). Unlike the other reports, this one has the **names, phones and e-mails of the owners**: personal data. So the report needs the "View members" permission as well, and making it is recorded in the audit log. Do not pass the file on to outsiders.
Electricity by meter tree	Electricity from the substation down to the plots for the range "From"-"To" (the previous month by default, service days). The **"Consumption"** sheet: one tree node per row (indented by level), readings at the start and end with their type ("Real", "Calculated", "Unknown"), consumption, "Children, kWh" (the sum of the consumption of the nodes below it), **losses in kWh and %** (the node's consumption minus the sum below it; only for substations and quarter meters whose values are known at both ends) and "Plots fed". When a value is unknown, everything that depends on it in the row is **empty** with the reason, not zero. The **"Charged"** sheet shows what was booked in the ledger for the range: plots' consumption (kWh, price, amount), losses (a fixed percent or runs per node), common-consumer runs with their split, plots and plot-days; adjustments and reversals are separate rows with the original service period. The totals are "Consumption of plots", "Losses charged" and "Common consumers charged". The header shows the current losses mode; a run booked under another mode shows what it booked.
Water	Plots connected to water on the chosen days ("From", "To", "Water source": "Cooperative supply" or "Well"). A row has the source, "Connected days", the meter, cubic metres for connected days only, and the charges by kind: "By meter", "Fixed fee", "Well pump share", with a separate column for reversals and adjustments. The "By source" sheet totals plots, plot-days, cubic metres and amount. Plots that were not connected are not listed.
Missing readings	Meters with no *approved* reading in the range; filters: dates, resource (electricity, water), meter kind (plot meters by default), street. A row has the resource, kind, plot or name, serial number, last reading (date and value), days since it and the mark "Reading waiting for approval" (such a meter stays in the list). Not listed: a water meter of a plot that was not connected

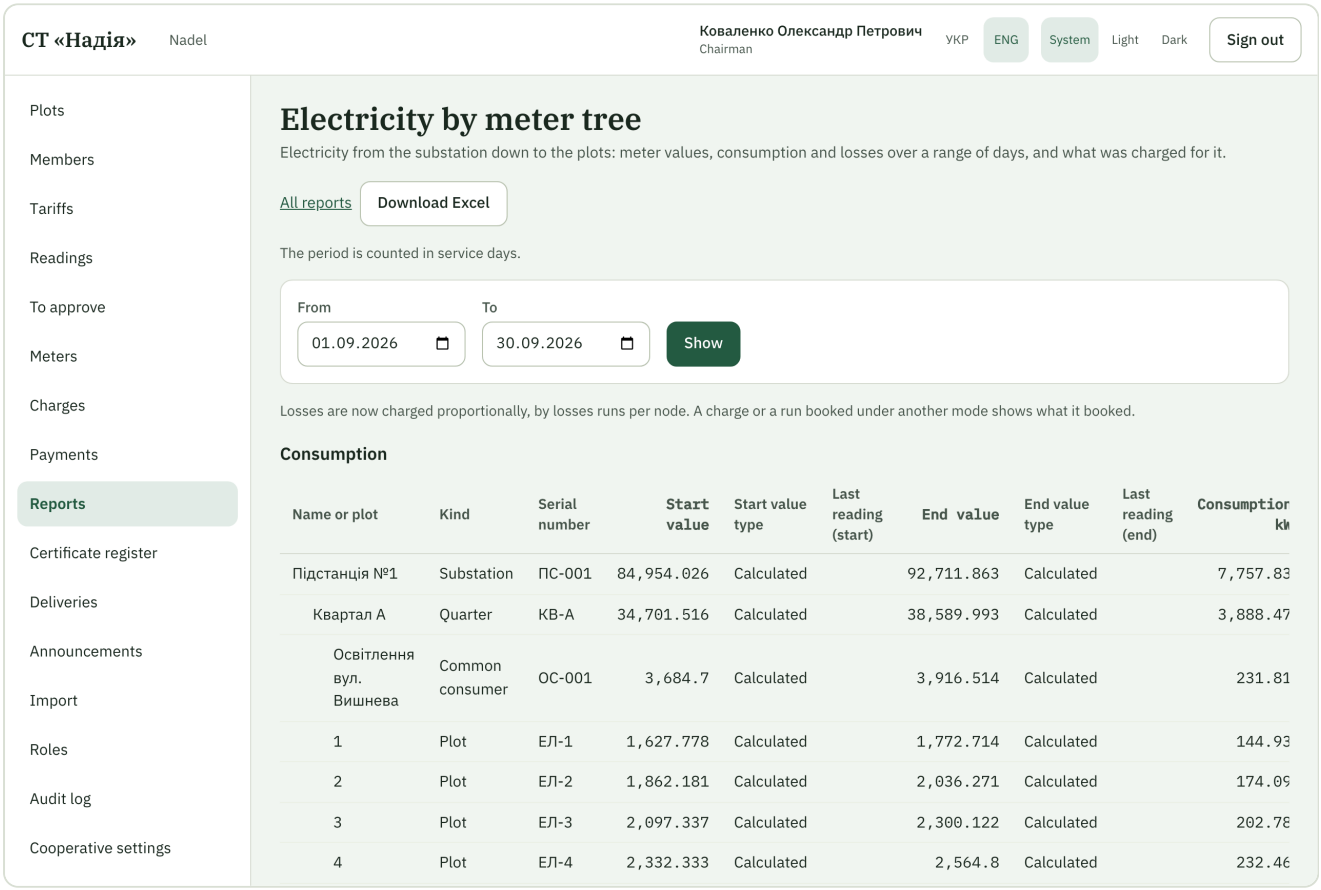
on any day of the range; a meter installed after the range or removed before it; and also a meter installed or removed *inside* the range, which is treated as having had a point to read. Under the table is how many open "Not charged" items there are for these meters, with a link to the list.

Land tax

The filter is "Year". A row is a plot the tax applies to: area in m² and sotkas, rate per sotka, "Expected" (the amount the system expects for the year) and "Charged for the year". If two rates applied in the year, the plot has one row per rate with its days. A plot whose expected and charged amounts differ is marked in "Differs", and a hint under the table says that a recalculation may be due (the report books nothing itself). Example: the rate is 35.00 ₴ per sotka per year and the plot is 600 m² = 6 sotkas → 6 × 35.00 = **210.00 ₴** a year (17.50 ₴ a month).

Corrections

Every reversal and adjustment of the chosen billing months (filters: months, charge types, "Reversal" or "Adjustment"). A row: booked on, month, plot, type, the original charge (period and amount), the amount, the source ("Manual reversal", "Recalculation", "Run reversal"), the recalculation cause, the reason and who booked it. It is a ready list of changes of the past for a general meeting. When there are no corrections, the report says so.



The report "Electricity by meter tree": readings, consumption, losses

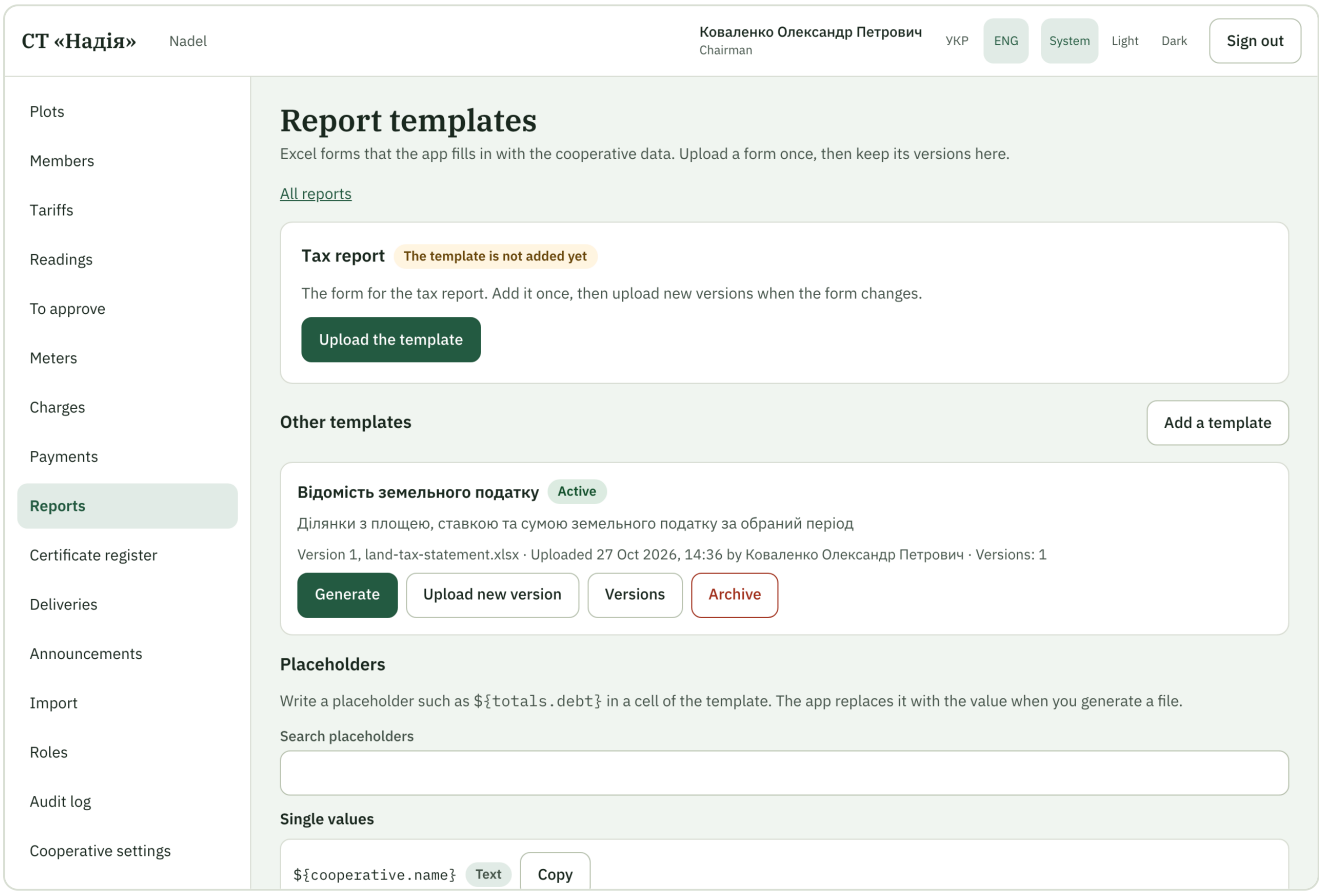
THE SAME PUMP SHARE IN TWO REPORTS

The electricity of a well pump is split among its plots as a common-consumer row (chapter 22). That row is on the "Charged" sheet of the electricity report *and* in the "Well pump share" column of the "Water" report. It is the same amount seen from two sides: do not add the two reports together.

Reports are easy to open from other screens: on a billing month the "Report" link opens "Charges by month" for exactly that month, and a plot account has a link to the balances report for that plot's street.

Templates: your own Excel form

If you need your own form (a statement for the meeting, a report for the tax office), prepare it in Excel as usual, and where a value from the system should stand, write a **placeholder** such as `${name.of.field}`. The system replaces placeholders with data and keeps your formats, merged cells, column widths and formulas. The full list of placeholders is on the **"Report templates"** page, in the "Placeholders" block: type, description, example, a search box and a "Copy" button.



The "Report templates" page: the tax report, other templates, placeholders

- **A placeholder alone in a cell** (`${rows.plots.landTax.expected}` and nothing else) is filled with a number or a date, not text, so formulas over it work. Give the cell a date format for a date. **A**

placeholder inside text ("Land tax for ${period.year}") is filled as text in Ukrainian number and date format.

- **Single values**: `${cooperative.name}`, `${cooperative.officialName}`, `${cooperative.iban}` (requisites from the settings), `${period.from}`, `${period.to}`, `${period.year}` (the year of the last day of the period), `${totals.debt}`, `${totals.landTax.expected}` and other totals.
- **A repeating row**: when a row holds placeholders of one set — `rows.plots.*` (plots), `rows.months.*` (billing months), `rows.meters.*` (electricity meters) or `rows.water.*` (water) — the row is repeated once per row of the set. **One repeating row per sheet**; two sets in one row or two repeating rows on a sheet are refused.
- Names and contacts of people are not available in templates: only data of the cooperative, plots and meters, and amounts.

A SMALL EXAMPLE: THE LAND TAX STATEMENT

The template is an Excel table (rate 35.00 ₴ per sotka per year; plots of 600, 800 and 1000 m²). What you write in the template:

Row	A	B	C
1	`${cooperative.officialName}`		
2	Land tax for `${period.year}`		
4	Plot	Area, m²	Tax for the year, ₴
5	`${rows.plots.number}`	`${rows.plots.area}`	`${rows.plots.landTax.expected}`
6	Total		`=SUM(C5:C5)`

What you get for the period 01.01-31.12.2026 (row 5 is repeated for every plot):

Plot	Area, m²	Tax for the year, ₴
1	600	210.00
2	800	280.00
3	1000	350.00
Total		840.00

An area of 600 m² is 6 sotkas, 6 × 35.00 = 210.00; likewise 8 × 35.00 = 280.00 and 10 × 35.00 = 350.00. The formula `=SUM(C5:C5)` under the repeating row was turned by the system into `=SUM(C5:C7)`: it covers all three copies, so "Total" equals the sum of the rows. The title in row 1 is the full name of the cooperative from the requisites, and row 2 reads "Land tax for 2026".

What the system does with a template

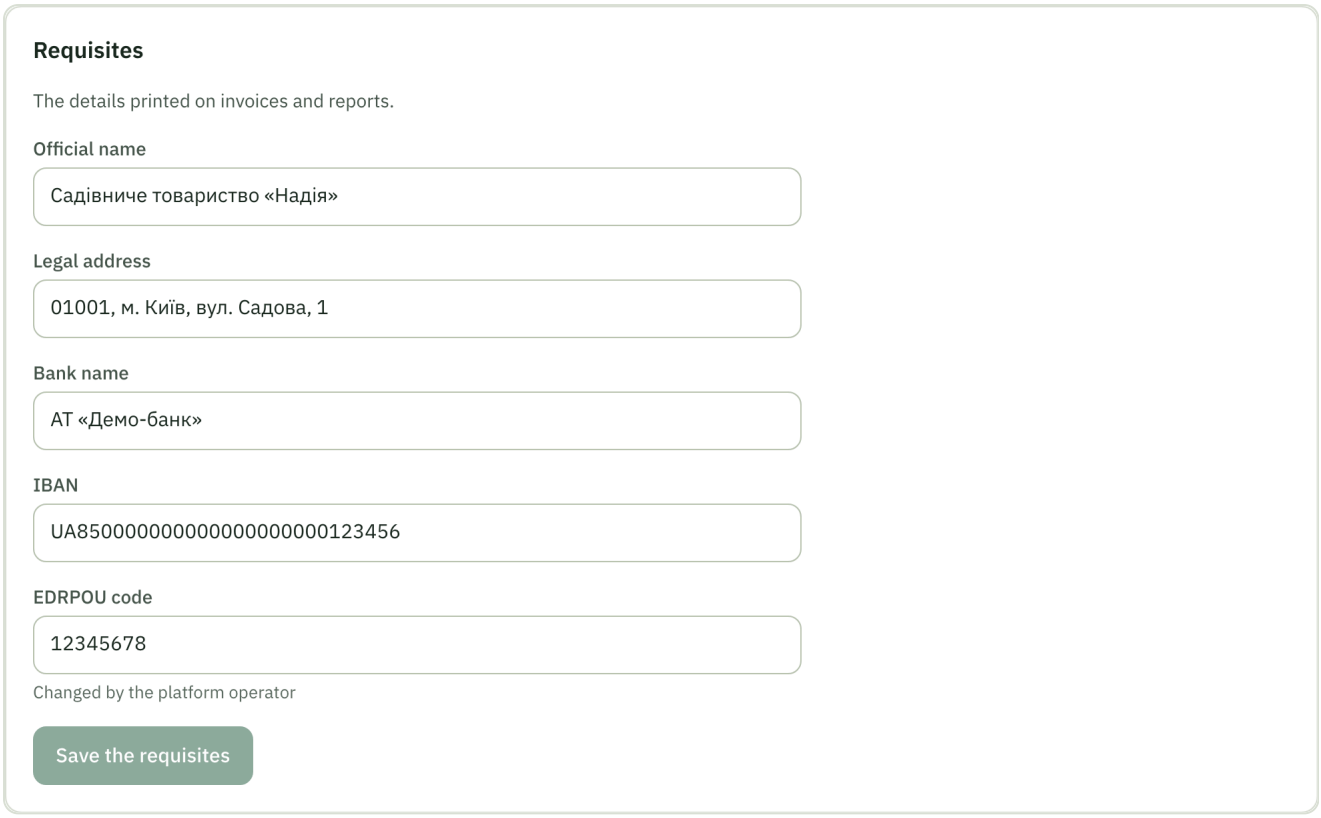
- **An unknown placeholder** (a typo, or a field that is not in the catalogue) **does not disappear**: it stays in the cell as written, and a **"Warnings"** sheet at the end of the file names the cell and the reason. The same goes for a placeholder whose value is unknown (no reading, a requisite not entered): the cell is empty, not zero, and it is listed in the warnings.
- The system does **not evaluate** formulas: it writes the placeholders and leaves the formulas to you. Excel or LibreOffice recalculates them when you open the file; a viewer that does not calculate formulas shows them empty.
- A valid file is an ordinary `.xlsx` workbook **up to 2 MB**. Files with macros (`.xlsm`), active content or embedded objects, damaged files and files that are "too large when unpacked" are refused and nothing is stored. Links to other files are removed from a template with a warning. Charts that refer to the repeating row are not shifted (a warning).

Versions, the archive and the tax report

- **Add a template**: a name, a description, a file. Uploading a new file into an existing template ("Upload new version") makes **version 2, 3…**; the latest version is always the one used, and older ones can be downloaded under "Versions" ("Download"). The generated file's name contains the template's name; the file properties and the audit log record the template and its version.
- **Archive**: the template is no longer offered for generation and takes no new versions, its name becomes free, and its versions stay downloadable.
- **Tax report** is a separate card with the status "The template is not added yet" until the chairman uploads the form with "Upload the template". The official form will be added as an ordinary template, without changing the program; when the form changes, upload a new version. Fields the official form may be missing will be added by a later feature.
- **"Generate"** next to an active template opens the "Generate a file" window: "From" and "To" (both days are included, the period is up to one year). "Generate and download" makes the file and downloads it. The system **does not keep** generated files. If there are warnings you see "The file was downloaded. Warnings in the file: …" with the first of them; the full list is on the "Warnings" sheet. The treasurer generates from the "Reports" section, the chairman from the templates page.

Cooperative requisites

Settings has a **"Requisites"** block (changing it needs the "Manage cooperative settings" permission): "Official name", "Legal address", "Bank name", "IBAN" and "EDRPOU code". The IBAN must be a valid Ukrainian IBAN (otherwise "The IBAN is not valid" and nothing is saved). The EDRPOU code is read-only — "Changed by the platform operator". These are public data of the legal entity, not personal data; changes are recorded in the audit log. These are the data that go into `${cooperative.officialName}`, `${cooperative.iban}` and so on.



The "Requisites" block in the settings

25 Payments and allocation works

A payment is a row in the ledger that erases nothing: a mistake is corrected by a reversal, and the allocation to charges changes only by new rows. How to work with the screens is chapter 13; here are the rules and the examples.

The allocation order	The oldest debt first (by the day it counts from); for the same day, in the order of the cooperative's charge types (membership, electricity, land tax - as on the charge types page); then by the start of the service period and the booking order. Each charge receives at most its open amount; the rest is prepayment if the plot has no open debt (while automatic allocation is on, the free remainder goes at once to the oldest debts).
The treasurer's change	The proposal can be changed before saving: the lines may not add up to more than the payment, and a line may not exceed the charge's open amount. With automatic allocation on, the unallocated remainder goes to the oldest debts; with it off, or when no debt is open, it becomes prepayment.
Taking money back from a charge	When money is taken off a charge (it is reversed, an adjustment lowers it, or the payment is reversed), this is a separate "release" record. The released sum becomes the plot's prepayment and, if automatic allocation is on, goes to the other open charges.
Reversing a payment	The chairman only, with a reason. Counted in the payment's own month while it is open, otherwise in the earliest open month. A payment is reversed once; a part of a group on its own.
A payment of a closed month	Counted in the earliest open month and keeps its date received; the closed month does not change.
Splitting between plots	In proportion to the debts, kopecks rounded down and the leftover kopecks to the largest remainders (a tie goes to the plot listed first). This is the only place of rounding in allocation; the parts always add up to the payment.
One at a time	Every change of a plot (payment, allocation, release) first locks the plot's ledger, so two treasurers cannot allocate the same money twice; at the end the database checks it too and refuses to allocate more than a payment holds or a charge is worth.

Worked examples

Order of charge types: membership, electricity, land tax.

> EXAMPLE 1. A PAYMENT SMALLER THAN THE DEBT
>
> Debts: January membership 600.00, February electricity 250.00. Payment 700.00. The oldest first: membership 600.00, then electricity 100.00. Electricity has 150.00 open, prepayment 0.00. The plot's debt is 150.00.

EXAMPLE 2. A PAYMENT LARGER THAN THE DEBT

The same debts, payment 1,000.00: membership 600.00 and electricity 250.00 are paid in full, prepayment 150.00. There is no debt, so the plot is green although there is a prepayment.

EXAMPLE 4. PREPAYMENT GOES TO A NEW CHARGE

Example 2, then a land tax charge of 120.00 is booked. Automatic allocation is on: 120.00 goes from the prepayment to the land tax and the prepayment becomes 30.00. With the switch off, the land tax would stay a debt of 120.00 and the prepayment 150.00 until the treasurer presses "Allocate prepayment".

EXAMPLE 19. A PAYMENT WITH A REMAINDER WHILE DEBTS ARE OPEN

Debts: membership 600.00 (January), electricity 250.00 (February). Payment 700.00; the treasurer allocates only the electricity 250.00. Automatic allocation is on: the remaining 450.00 is placed at once on the membership, 150.00 of membership stays open, the prepayment is 0.00.

EXAMPLE 20. THE SAME WITH AUTOMATIC ALLOCATION OFF

The same case with the switch off: electricity 250.00, the remaining 450.00 stays as prepayment and the membership stays open at 600.00. The treasurer can press "Allocate prepayment".

EXAMPLE 6. A CHARGE IS REVERSED

Example 1 (payment 700.00: membership 600.00 and electricity 100.00), then the chairman reverses the January membership. 600.00 is taken back from it - that is a prepayment of 600.00. Automatic allocation moves 150.00 to the open electricity; the prepayment stays 450.00 and there is no debt.

EXAMPLE 12. ONE PAYMENT FOR TWO PLOTS

A member has plot 7 with a debt of 300.00 and plot 15 with a debt of 100.00 and pays 200.00. The parts in proportion to the debts: 200 × 300 / 400 = 150.00 for plot 7 and 50.00 for plot 15. These are two payments of one group; cash has one receipt number. If the member paid 500.00, plot 7 would get 300.00 and another 100.00 as prepayment (the excess goes to the first plot), and plot 15 would get 100.00.

Debt and colour after payments

- **The plot's debt** is the sum of the open amounts of its charges, that is charged minus allocated. A payment lowers the debt only by what is allocated to charges.

- **The oldest unpaid** is the oldest debt day among charges with an open amount above zero; a charge paid in full does not count, a partly paid one counts with its open part.
- **The colour** uses the same thresholds as before, but over the debt after payments: red if the debt exceeds the threshold or the oldest unpaid day is older than the threshold in days; yellow for any other debt; green when there is no debt, **also when there is a prepayment**.
- The colour is refreshed after every payment, reversal, allocation and release - within seconds. Example: a plot with a debt of 1,200.00 (red with a threshold of 1,000.00) after a payment of 900.00 has a debt of 300.00 and is yellow.
- An adjustment upward (for example +50.00 to a paid membership) leaves 50.00 open, so the plot is no longer green. An adjustment downward (–100.00 to a paid membership) takes 100.00 back from the latest allocation and it becomes prepayment, not a "negative debt".

26 Invoice numbers works

An invoice does not recalculate anything: it reads the same ledger as the plot account and the reports, and arranges it by billing month. How to work with the screens is chapter 14; here are the rules and the examples.

The booking-month rule — Every row of the ledger belongs to the billing month it is *booked* in. For an invoice of October, "Charged" is everything booked in October (regular charges, adjustments and reversals), and "Paid" is the payments placed in October. "Debt at the start" is what was open before October, "debt at the end" what is open after it.

A closed month never changes — A correction made after October was closed is booked in the month it is made in. So the October invoice and its re-issues keep the same numbers, and the correction shows up in the invoice of the later month, with the month it corrects.

Amount due — Debt at the end of the month, minus the money received after the month and before the issue, never below 0.00. Those payments are listed under the table as "Received after October". When the amount due is 0.00 there is no QR code; a prepayment is shown as "Prepayment".

The invoice number — Year-month/plot number, for example `2026-10/9`. A re-issue keeps the number and raises the version.

Worked examples

Plot 9, charge types in the order Membership, Electricity. The invoice for October 2026, issued on 05.11.2026.

#	Situation	The October invoice
1	No debt before. October: membership 75.00, electricity 338.26. No payments	Start 0.00; charged 413.26; paid 0.00; end 413.26; due 413.26; QR amount 413.26
2	Electricity debt from September 1,135.08. In October a payment of 1,500.00 went to 1,135.08 (September electricity) + 75.00 (membership) + 289.92 (October electricity)	Start 1,135.08; charged 413.26; paid 1,500.00; end 48.34 (electricity); due 48.34
3	Example 1 and a payment of 413.26 on 03.11	End of October 413.26; "Received after October" 413.26; due 0.00; no QR code
4	Overpaid: no debt at the end, prepayment 100.00	Due 0.00; "Prepayment 100.00"; no QR code

#	Situation	The October invoice
5	October was closed on 03.11. On 06.11 a recalculation lowers an October charge by 20.00 (booked in November)	The October invoice does not change, and a re-issue gives the same numbers. The November invoice shows the -20.00 adjustment "for October 2026"

27 Which channel, and how sending is retried

works

Delivery calculates nothing and changes no amounts: it only delivers documents that are already built. How to work with the screens is in chapter 15; here are the rules for choosing a channel and for retries, with examples.

The channel is decided for each owner separately — For every *current* owner of a plot, the owner's preferred way (Telegram, e-mail, paper or Viber) is taken and checked for being possible: Telegram - the member is connected and the bot exists; e-mail - there is an address and a mailbox is set up. If not, the fallback of the table in chapter 15 applies: from Telegram to e-mail, then paper; from e-mail to paper; Viber is always paper.

One message per plot and owner — A member who owns two plots gets two messages, each with its own PDF. Two people who own a plot together get one message each, each through their own channel.

The same is never sent twice — For one invoice version, one member and one channel there is exactly one delivery record. Pressing "Send invoices" again adds nothing; a new invoice version is sent as a new message.

Worked examples

October 2026 invoices; the cooperative's mailbox and bot are set up.

No.	Situation	What happens
1	Plot 9, one owner Olena: prefers Telegram, connected	Olena gets the invoice in Telegram: the text, the PDF and the "Оплатити" (Pay) button
2	Plot 7 is co-owned. Ivan: prefers Telegram, has not connected yet, has an e-mail address. Maria: prefers paper	Ivan gets a letter with the PDF; Maria is on the "to print" list, her copy is printed with its own QR code for connecting Telegram
3	Peter: prefers Viber (not supported yet). Hanna: prefers Telegram, has not connected, no e-mail address	Both are "to print"; Peter has the note "Viber is not supported yet" on the Deliveries screen. When Hanna connects Telegram, the next invoices go there
4	A member owns plots 9 and 10, prefers e-mail	Two letters: one for plot 9, one for plot 10, each with its own PDF
5	The invoices are sent already and the treasurer presses "Send invoices" again	Nothing new: the system says everything has been sent

Sending attempts

Sending runs in the background at a limited pace: at most one message a second to one member, up to 20 messages a second per bot in Telegram, and e-mail at your "Letters per second". Every message has **three attempts**: now, after 1 minute and after 5 minutes. After the third failure the status is "Failed" with the reason of the last attempt.

Reason	Retried?	What it means
The mail server refused the letter	yes, up to three attempts	the mailbox did not accept the letter: the daily limit, too fast, an invalid recipient address
Could not connect	yes, up to three attempts	no connection to the mail server or Telegram
Wrong mailbox login or password	no, "Failed" at once	the password does not fit; a retry changes nothing until it is corrected
The bot is blocked	no	the member blocked the bot; the connection is cancelled
No contact	no	no e-mail address or connected chat
The channel is not configured	no	no mailbox or bot, or Telegram rejects the token

If Telegram asks to wait (too many requests), the message is simply moved to the time Telegram names, and that pause does *not* use up an attempt. A message that became "Failed" does not disappear: it stays on the Deliveries screen, and after the cause is fixed it is sent with "Send again".

26 What comes next planned

What	What it gives the board
Bank and cash book	The cash book, handing cash over to the bank and reconciliation; later, bank statement import with matching payments to plots.
Viber and reminders to debtors	Sending invoices, messages and announcements by e-mail and Telegram already works (chapter 15); Viber and automatic reminders to debtors come later.
Mobile app and map	An app for members and the electrician with photo readings; the cooperative map coloured by debt.

This guide is updated with every new capability of the system.